VMware Cloud Foundation: Build, Manage and Secure [V9.1]

Lab Manual

vmware by Broadcom

VMware Cloud Foundation: Build, Manage and Secure [V9.1]

Lab Manual - GA

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Typographical Conventions

The following typographical conventions are used in this course.

Conventions	Usage and Examples

`Monospace`	Identifies command names, command options, parameters, code fragments, error messages, filenames, folder names, directory names, and path names: • Run the `esxtop` command. • ... found in the `/var/log/messages` file.
`Monospace Bold`	Identifies user inputs: • Enter **`ipconfig /release`**.
Boldface	Identifies user interface controls: • Click the **Configuration** tab.
Italic	Identifies book titles: • *vSphere Virtual Machine Administration*
< >	Indicates placeholder variables: • <ESXi_host_name> • ... the `Settings/<Your_Name>.txt` file

Contents

Lab 1 Case Study 1
Lab 2 Accessing the Lab Environment 3
Task 1: Access the Student Desktop 3
Task 2: Log In to vCenter, SDDC Manager, and VCF Operations 3
Task 3: Review the Lab Network Topology 5
Lab 3 (Simulation) Deploying VMware Cloud Foundation with VCF Installer 6
Task 1: Prepare for the Lab 7
Task 2: Plan VMware Cloud Foundation Deployment 7
Task 3: Deploy VMware Cloud Foundation 8
Lab 4 Managing the VCF Environment with the User Interface 15
Task 1: Manage VCF Compute with the vSphere Client 15
Task 2: Manage VCF Instances with SDDC Manager 18
Task 3: Manage VCF Instances with VCF Operations 20
Task 4: Manage VCF Networking with NSX 22
Lab 5 Verifying the VCF Operation Integration with the Management Domain 24
Task 1: Prepare for the Lab 24
Task 2: Verify the VCF Operation Integration with vCenter, vSAN, and NSX 24
Lab 6 (Simulation) Deploying Log Management 26
Task 1: Prepare for the Lab 26
Task 2: Deploying Log Management 27
Task 3: Using Log Management UI 28
Task 4: Shutting down VCF Log Management 28

Lab 7 Configuring VCF Single Sign-On for the Management Domain 29
Task 1: Prepare for the Lab 29
Task 2: Configuring Authentik 30
Task 3: Enable Single Sign-On 39
Task 4: Configure SCIM in Authentik 50
Task 5: Enable Single Sign-On on Management Domain vCenter 57
Task 6: Verify the Integration of VCF Instances with SSO 65
Task 7: Verify VCF Automation and VCF Operation Integration with VCF SSO 68
Lab 8 Assigning Permissions to VCF SSO Users 71
Task 1: Prepare for the Lab 71
Task 2: Import an SSO User and Assign Permissions to VCF Operations 72
Task 3: Add VCF SSO Users to vCenter 77
Task 4: Add VCF SSO Users to NSX Manager 78
Task 5: Verify VCF SSO Access to VCF Fleet 79
Lab 9 Managing Passwords 81
Task 1: Prepare for the Lab 81
Task 2: Rotate the ESX Host Passwords 82
Task 3: Schedule Password Rotation for the vCenter Root Account 82
Task 4: Re-Create a Primary Password File 83
Task 5: Use the API to Create a Primary Password List 83
Task 6: Managing Passwords from VCF Operations 84
Lab 10 (Simulation) Commissioning Hosts 85
Task 1: Prepare for the Lab 86
Task 2: Configure a New Network Pool 86
Task 3: Commission a Host 87
Task 4: Bulk-Commission Additional Hosts 89
Lab 11 (Simulation) Creating Workload Domain 92
Task 1: Prepare for the Lab 92
Task 2: Create a Workload Domain 93
Task 3: Retrieve Passwords and Log in to vCenter and NSX Manager 96
Lab 12 Creating Workload Domain without vSphere Cluster 98
Task 1: Prepare for the Lab 98
Task 2: Create a Workload Domain (Deploy Infrastructure only) 99

Task 3: Retrieve Passwords and Log in to vCenter and NSX Manager 101
Lab 13 (Simulation) Creating a Workload Domain with the Supervisor Cluster 104
Task 1: Prepare for the Lab 105
Task 2: Create a Workload Domain with Supervisor Cluster 105
Task 3: Retrieve Passwords 109
Task 4: Silence vSAN Alerts 110
Lab 14 Verifying the Workload Domain Components Integration with VCF Operations 112
Task 1: Prepare for the Lab 112
Task 2: Verify the Workload Domain Components Integration with VCF Operations 112
Lab 15 (Simulation) Assigning a License to the Workload Domain vCenter 114
Task 1: Prepare for the Lab 115
Task 2: Assign a License to the Workload Domain vCenter 115
Lab 16 (Simulation) Adding an ESX Host to the Workload Domain 116
Task 1: Prepare for the Lab 117
Task 2: Commission Hosts 117
Task 3: Add a Host to the Workload Domain 118
Lab 17 Creating vCenter Group 120
Task 1: Prepare for the Lab 120
Task 2: Create a vCenter Group with vCenter Instances 121
Task 3: Verify vCenter Linking 121
Lab 18 Configuring Distributed Transit Gateway 122
Task 1: Prepare for the Lab 122
Task 2: Configure Default Transit Gateway for External Connectivity 123
Task 3: Create an IP Address Block 124
Task 4: Configure VPC Connectivity Profiles 125
Task 5: Verify the Default Transit Gateway details in vCenter 126
Lab 19 Creating Virtual Private Cloud (Distributed) 127
Task 1: Prepare for the Lab 127
Task 2: Create Virtual Private Cloud (Distributed) 128
Task 3: Create Private VPC Subnets in VPC 128

Task 4: Create Private Transit Gateway Subnets in VPC 129
Task 5: Create Public Subnets in VPC 129
Task 6: Deploy Virtual Machines in VPC 130
Task 7: Test Connectivity Between the Subnets within the VPC 132
Task 8: Review the VPC in NSX Manager 134
Lab 20 Creating VNA Cluster and Update Connectivity Profile 135
Task 1: Prepare for the Lab 135
Task 2: Create VNA Cluster 135
Task 3: Update Connectivity Profile with VNA Cluster details. 137
Lab 21 (Simulation) Creating Edge Cluster and Tier-0 Gateway using vCenter 138
Task 1: Prepare for the Lab 139
Task 2: Create Edge Cluster and Setup Tier-0 Gateway 139
Task 3: Verify the NSX Edge Cluster in NSX Manager 145
Lab 22 Configuring Centralized Transit Gateway 146
Task 1: Prepare for the Lab 146
Task 2: Verify External Connection 147
Task 3: Create and verify the IP Address Block details 147
Task 4: Configure VPC Connectivity Profiles 148
Task 5: Configure Default Transit Gateway 149
Lab 23 Creating Centralized Virtual Private Cloud 151
Task 1: Prepare for the Lab 151
Task 2: Create a Virtual Private Cloud (Centralized) 152
Task 3: Create Private VPC Subnets 152
Task 4: Create Private Transit Gateway Subnets 153
Task 5: Create Public Subnets 154
Task 6: Deploy Virtual Machines in VPC 154
Task 7: Test Connectivity between the Subnets within the VPC 156
Task 8: Review the VPC in NSX Manager 158
Lab 24 Reviewing the Workload Domain vSAN Cluster Details 159
Task 1: Prepare for the Lab 159
Task 2: Review the vSAN Cluster Details in the vSphere Client 160
Task 3: Verify the vSAN Cluster Configuration Details Using ESXCLI Commands 160

Task 4: Review Storage Details in VCF Operations 162
Lab 25 Analyzing the Placement of VM Objects and Components in the vSAN Cluster 163
Task 1: Prepare for the Lab 163
Task 2: Deploy a Virtual Machine 163
Task 3: Verify the VM Object and Component Placement 164
Lab 26 Running vSAN Tests 165
Task 1: Prepare for the Lab 165
Task 2: Run vSAN Proactive Tests 165
Task 3: Run a Data Migration Precheck 166
Lab 27 Creating and Applying a vSAN Storage Policy 167
Task 1: Prepare for the Lab 167
Task 2: Verify the VM Storage Policy Applied to a vSAN Data Store 167
Task 3: Create a VM Storage Policy 168
Task 4: Create a Virtual Machine with a Custom VM Storage Policy 170
Task 5: Check Compliance with the VM Storage Policy 170
Lab 28 Implementing Certificates 172
Task 1: Prepare for the Lab 172
Task 2: Configure the Certificate Authority in SDDC Manager 173
Task 3: Rotate the Certificate for SDDC Manager 174
Task 4: Manage Certificates from VCF Operations 176
Lab 29 Configuring Backup 178
Task 1: Prepare for the Lab 178
Task 2: Configure Backups for VCF Instances 178
Task 3: Configure SFTP Settings for VCF Fleet Management 180
Task 4: Create Backup Schedule for VCF Management 181
Lab 30 (Optional) Using APIs 183
Task 1: Prepare for the Lab 183
Task 2: Use the API to Determine vSAN Storage Usage 184
Task 3: Use the API to Obtain NSX Cluster Details 184
Lab 31 (Optional) Using SoS Commands 187
Task 1: Use SoS Commands to Verify Cluster Health 187

Answer Key 190

Lab 1 Case Study

Rainpole Learning Organization

Rainpole, a prominent education services provider, is embarking on a strategic initiative to enhance its learning platform. This expansion aims to bolster the organization's future growth by establishing a robust, agile, and scalable private cloud infrastructure. This infrastructure is crucial for supporting their learning lab environment and delivering hands-on training for cloud service providers.

Currently, Rainpole's infrastructure relies on traditional vSphere, which has reached its operational limits. This legacy system lacks the necessary flexibility and automation to facilitate rapid innovation. Consequently, Rainpole is evaluating VMware Cloud Foundation as a potential solution for its next-generation data center. VMware Cloud Foundation offers the promise of a highly available, agile, and secure environment, essential for their evolving needs.

Problem Statement

The lack of flexibility and modern cloud capabilities hinders the delivery of advanced, hands-on training for cloud service providers. To overcome these limitations and enable rapid innovation, Rainpole needs to deploy VMware Cloud Foundation to establish a highly available, secure, and future-ready private cloud environment.

Proof of Concept

As a proof of concept, the CIO has decided to build a sample private cloud using VCF to set up the infrastructure and use VCF Automation on top to build the private cloud. The cloud serves as a playground environment for the learning team to run virtual machines as a classroom teaching aid as well as for learning content development. VCF Operations will be used to monitor and troubleshoot the environment.

To deploy a VMware Cloud Foundation (VCF) 9.1 environment, the proof of concept would begin deploying the VCF Installer virtual appliance, which helps to orchestrate the initial bring-up. In the VCF Installer UI, you provide the deployment parameter, reflecting the enhanced networking, storage (including new NVMe tiering capabilities), and security constructs in VCF 9.1 for vCenter, NSX, and SDDC Manager.

The VCF Installer then automates the deployment of the management domain onto a minimum of four qualified ESX hosts with vSAN, leveraging the integrated components in VCF 9.1. For configuring workload domains, SDDC Manager is used to provision additional clusters, ensuring they benefit from the VCF 9.1 advancements for improved storage efficiency. The PoC would highlight SDDC Manager's enhanced life cycle management for VCF 9.1, demonstrating streamlined patching and upgrades for the entire stack. Finally, integrated solutions for operational insights and automation would showcase the VCF 9.1 capabilities for a modern, self-service private cloud, emphasizing its new features for performance, compliance, and developer enablement.

Learning Goals

This lab environment and the accompanying document are meant to serve as a learning and teaching tool to gain expertise on deploying, configuring, and managing VMware Cloud Foundation 9.1. When working together, these features provide an industry-leading environment to build, automate, consume, and operate your own private cloud.

Participants acquire the following primary VCF skills from these lab exercises:

- Deploying VMware Cloud Foundation with VCF Installer.
- Managing the VCF environment with the vSphere Client, NSX UI, and VCF Operations UI.
- Configuring single sign-on for user authentication and authorization.
- Configuring Open LDAP as an Identity Source.
- Commissioning hosts and creating workload domains.
- Creating vCenter groups for linking vCenter systems.
- Managing licenses and certificates with VCF Operations.
- Creating Virtual Private Clouds (VPCs).
- Creating NSX Edge cluster and Tier-0 gateways.
- Running SoS commands, APIs, and vSAN Tests for VCF troubleshooting.

Lab 2 Accessing the Lab Environment

Objective and Tasks

Prepare for all labs by reviewing the lab environment and its configuration:

1. Access the Student Desktop
2. Log In to vCenter, SDDC Manager, and VCF Operations
3. Review the Lab Network Topology

Task 1: Access the Student Desktop

You access and manage the lab environment from the student desktop.

The system assigned to you serves as an end-user terminal.

1. Verify that you are successfully logged in to the student desktop.
2. If you are not logged in already, log in to your student desktop.
    - User name: holuser
    - Password: VMware123!VMware123!

Task 2: Log In to vCenter, SDDC Manager, and VCF Operations

You log in to vCenter and SDDC Manager to familiarize yourself with the lab environment.

1. From your student desktop, log in to vCenter using the vSphere Client.
    a. Open Firefox.
    b. Click **Region A > vc-mgmt-a Client** on the bookmarks bar.

c. If the `Your connection is not private` message appears, click **Advanced** and click **Proceed to vc-mgmt-a.site-a.vcf.lab (unsafe)**.

d. Log in to the vSphere Client:

- User name: administrator@vsphere.local
- Password: VMware123!VMware123!

When the vSphere Client opens, the **Hosts and Clusters** inventory appears in the navigation pane on the left. This pane is also called the navigator.

2. Review the vCenter inventory for vc-mgmt-a.site-a.vcf.lab in the left navigation pane.
3. Expand **vc-mgmt-a.site-a.vcf.lab** vCenter and the objects beneath it to become familiar with all the items in the inventory.

   The license-a, NSX Manager (nsx-mgmt-01a & nsx-wld-01a), VCF Operations(ops-a), license-a, ops Collector-01a, SDDC Manager (sddcmanager-a), vCenters (vc-mgmt-a & vc-wld-01a), vsp-xxxx, web01 and web-02a VMs are running under the cluster-mgmt-01a cluster.

4. Launch SDDC Manager in a new browser tab.
   a. On the Firefox bookmarks bar, click **Region A** and right-click **SDDC Manager** and select **Open in new tab**.
   b. Click the newly opened **Privacy error** tab.
   c. If the `Your connection is not private message` appears, click **Advanced** and click **Proceed to sddcmanager-01a.vcf.lab (unsafe)**.
   d. Click the **SDDC Manager** browser tab.

      SDDC Manager is integrated with vCenter Single Sign-On. So it authenticates automatically when you log in to the vSphere Client.

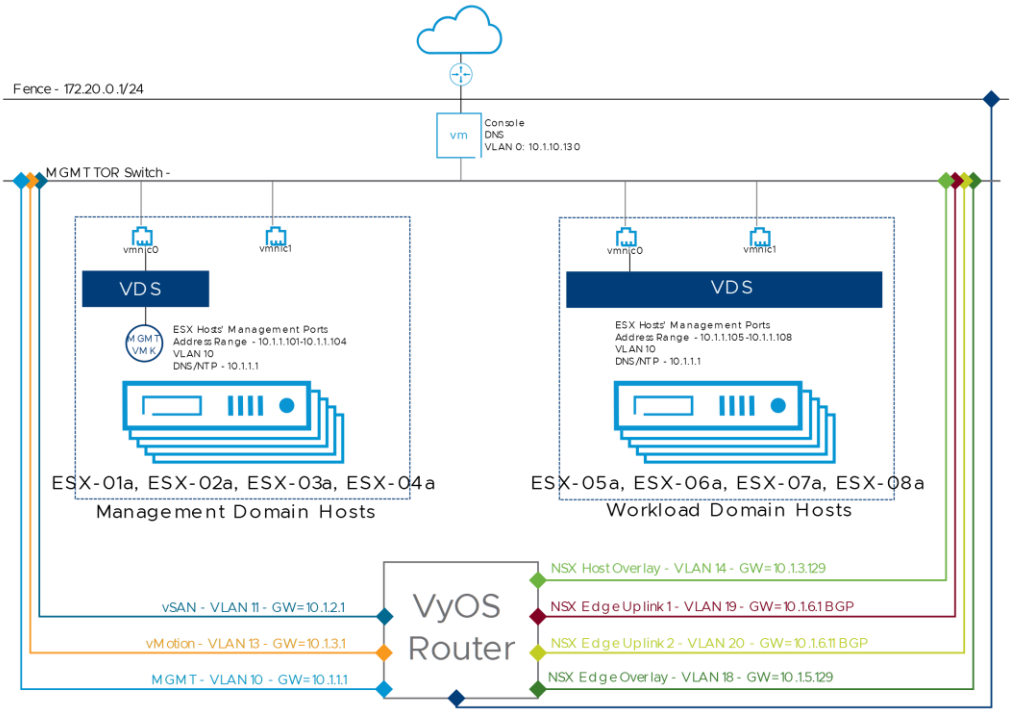
   e. In the top-right corner of the right pane, click the question mark (?) icon to open the drop-down menu.
   f. Select **About** to verify the VMware Cloud Foundation software version.
   g. Click **CLOSE**.
5. Launch VMware Cloud Foundation Operations in a new browser tab.
   a. On the Firefox bookmarks bar, Click **Region A** > right-click **VMware Cloud Foundation Operations** and select **Open in new tab**.
   b. If the `Your connection is not private message` appears, click **Advanced** and click **Proceed to ops-a.vcf.lab (unsafe)**.
   c. Log in to the VCF Operations:

- User name: admin
- Password: VMware123!VMware123!

When VCF Operations opens, the **Home** page appears in the navigation pane on the left.

Task 3: Review the Lab Network Topology

1. You use the network topology diagram to deploy, configure, and manage the VMware Cloud Foundation lab environment.



During the lab, you can refer to the diagram for any clarification.

Lab 3 (Simulation) Deploying VMware Cloud Foundation with VCF Installer

Objective and Tasks

Deploy VMware Cloud Foundation using VCF Installer:

1. Prepare for the Lab
2. Plan VMware Cloud Foundation Deployment
3. Deploy VMware Cloud Foundation
4. Login to vCenter and SDDC Manager.

In the web browser in your local desktop, go to https://core-vmware.bravais.com/s/21891/Deploying_VMware_Cloud_Foundation_with_VCF_Installer to open the simulation.

Perform the following steps in the simulated environment. Refer to these steps as you work through the simulation.

IMPORTANT

Do not refresh, navigate away from, or minimize the browser tab hosting the simulation. Such actions might pause the simulation and prevent it from progressing.

NOTE

If you're stuck anywhere in the simulation, press the **TAB** key to reveal the next clickable area.

Task 1: Prepare for the Lab

You configure the deployment parameter workbook to enable image-based installation of VMware Cloud Foundation using VCF Installer.

1. Log in to VMware Cloud Foundation Installer.
   a. From the bookmarks bar, select **HOL Admin** > **VCF Installer**.
   b. Log in to VCF Installer.
      - User name: admin@local
      - Password: VMware123!VMware123!
2. Under **Get Started**, click **DEPOT SETTINGS AND BINARY MANAGEMENT**.

   The VMware Cloud Foundation bring-up wizard starts.
3. Under **Binary Management**, verify that the binaries for vSphere Foundation and VMware Cloud Foundation are downloaded.
   a. Verify that the download status for all components is Success.
4. Click < **RETURN HOME.**

Task 2: Plan VMware Cloud Foundation Deployment

You deploy VMware Cloud Foundation with VMware Cloud Foundation Installer.

1. On the VMware Cloud Foundation Installer, click **DEPLOYMENT WIZARD** and select **VMware Cloud Foundation**.
2. On the **About the Deployment Wizard**, click **NEXT.**
3. On the **Deployment Paths** wizard, validate that **Deploy a new VCF fleet** is selected and click **NEXT.**
4. Use the **Plan** wizard to plan the resources needed for the VCF Fleet.
   a. On the **Existing Components** wizard, click **NEXT**.

      Do not select any of the options in this wizard, as you are deploying VMware Cloud Foundation in a greenfield deployment.

      ---

   b. On the **Size Options** wizard, select **Simple** from the Deployment model drop-down menu.
   c. Click **NEXT**.

d. On the Network Options, click **CUSTOMIZE** and select the options.

Options	Value
VM management network	Use ESX management network
VCF management network	Use VM management network
Transit gateway type	Centralized Connectivity

e. Click **NEXT** on the Network Options.

f. On the **Storage** wizard, select the options:

Option	Value
Select Storage type	vSAN
vSAN Architecture	vSAN ESA
Activate Data-In-Transit encryption	Verify that the checkbox is cleared.

g. Click **NEXT** on the Storage wizard.

5. On the **Review Prerequisites** page, review the deployment information.

   a. Click **Show details** link, under **Resource requirements** to review the resource requirement details for the various components.

   b. Click **CLOSE** to close the Resource Requirements Details window.

   c. Click **Show details** link, under **VLAN requirements** to review the VLAN requirement details for the various VLANs required for the deployment.

   d. Click **CLOSE** to close the VLAN Requirements Details window.

   e. Click **Show details** link, under **FQDNs and IP address pools** to view the FQDN and IP addresses details needed for the deployment.

   f. Click **CLOSE** to close the FQDN and IP Addresses Details window.

   g. Click **NEXT** to proceed to the Prepare step.

Task 3: Deploy VMware Cloud Foundation

You deploy VMware Cloud Foundation with VMware Cloud Foundation Installer.

1. Use the Prepare wizard to deploy VCF Fleet.

a. On the **General Information** wizard, select options.

Option	Value
Version	9.1.0.0
VCF Instance Name	instance-a
Management Domain name	mgmt-a
Customer Experience Improvement Program (CEIP)	Select **Opt-in**
DNS Servers	10.1.1.1
NTP Servers	10.1.1.1
Default hostname DNS suffix	site-a.vcf.lab

b. Click **NEXT**.

2. On the **Hosts** page, enter the hosts information to add the hosts for the deployment.
   a. In the **enter FQDN** text box, enter `esx-01a.site-a.vcf.lab` .
   b. Enter `VMware123!VMware123!` in the **Root password** text box.
   c. Select **Use this password for all additional hosts** check box under the Root Password text box.
   d. In the **enter FQDN** text box, enter `esx-02a.site-a.vcf.lab`
   e. In the **enter FQDN** text box, enter `esx-03a.site-a.vcf.lab`
   f. Click **+ ADD Host**.
   g. In the **enter FQDN** text box, enter `esx-04a.site-a.vcf.lab`
   h. Click **CONFIRM ALL FINGERPRINTS**.
   i. Click **NEXT**.
   j. On the Resource Requirements Warning, click **YES, PROCEED** to continue.
3. On the **Networks** wizard, configure the options for ESX Management Network, VCF Managements Services IP Range, VCF Automation IPV4 Range, vMotion Network, vSAN Network and NSX Overlay Network.
   a. ESX Management Network:

Option	Value
VLAN ID	10

IPv4 Gateway (CIDR notation)	10.1.1.1/24
VCF Management Services IP Range: IPv4 Address Range: From*	10.1.1.141
VCF Management Services IP Range: IPv4 Address Range: To*	10.1.1.160
VCF Automation IPv4 Range: IP Address Range: From *	10.1.1.71
VCF Automation IPV4 Range: IP Address Range To	10.1.1.74

b. vMotion Network:

Option	Value
VLAN ID	13
MTU	8000
IPV4 Gateway (CIDR notation)	10.1.3.1/25
IP Address Range From *	10.1.3.101
IPv4 Address Range To	10.1.3.121

c. vSAN Network:

Option	Value
VLAN ID	11
MTU	8000
IPV4 Gateway (CIDR notation)	10.1.2.1/25
IP Address Range From *	10.1.2.101
IPv4 Address Range To	10.1.2.121

d. NSX Overlay Network:

Option	Value
VLAN ID	14
IPV4 Gateway (CIDR notation)	10.1.3.129/25

IP assignment mode	IP Pool
IP Address Range From	10.1.3.201
IP Address Range To	10.1.3.217

i. Click **NEXT**.

4. On the **VCF management services**, enter the configuration details for all VCF management components.

Option	Value
VCF Operations Primary node FQDN	ops-a.site-a.vcf.lab
Cloud Proxy FQDN	opsproxy-01a.site-a.vcf.lab
License Server FQDN	license-a.site-a.vcf.lab
Fleet components FQDN	opslcm-a.site-a.vcf.lab
Instance components FQDN	shared-01a.site-a.vcf.lab
Identity Broker FQDN	vidb-01a.site-a.vcf.lab
VCF services runtime FQDN	vsp-01a.site-a.vcf.lab
VCF Automation FQDN	auto-a.site-a.vcf.lab
VCF services runtime FQDN	auto-platform-a.site-a.vcf.lab

a. Click **NEXT**.

5. On the **vCenter** page, enter the configuration details for the vCenter.

Option	Value
vCenter FQDN	vc-mgmt-a.site-a.vcf.lab
Datacenter name	dc-a
Cluster name	cluster-mgmt-01a
SSO Domain name	vsphere.local

a. Click **NEXT**.

6. On the **Storage** wizard, validate the storage type.

Option	Value
Select Storage type	vSAN
vSAN architecture	vSAN ESA
vSAN Datastore Name	cluster-mgmt-01a-ds-vsan01

a. Click **NEXT**.

7. On the **Distributed Switch** wizard, click **CHANGE** from Selected Profile,

a. Click **SELECT** from the Custom Switch Configuration,

b. Verify the following parameters and configure network traffic for the cluster-mgmt-01a-vds distributed switch and verify that the parameters match those shown in the tables.

Option	Value
Distributed Switch Name	cluster-mgmt-01a-vds
MTU	Enter 8000
Number of Uplinks	2
uplink1	vmnic0
uplink2	vmnic1
Configure network traffic for this Distributed Switch	Turn on the **ALL** toggle to enable the networks: ESX Management vMotion vSAN NSX

i. Verify the parameters under Network Traffic: ESX Management.

Option	Value
PortGroup Name	cluster-mgmt-01a-vds01-pg-esx-mgmt
Load Balancing	Route Based On Physical NIC Load
uplink1	Active

uplink2	Active

c. Verify the parameters under Network Traffic: VM Management.

Option	Value
PortGroup Name	cluster-mgmt-01a-vds01-pg-vm-mgmt
Load Balancing	Route Based on Physical NIC Load
uplink1	Active
uplink2	Active

d. Verify the following parameters under Network Traffic: vSAN.

Option	Value
PortGroup Name	cluster-mgmt-01a-vds01-pg-vsan
Load Balancing	Route Based On Physical NIC Load
uplink1	Active
uplink2	Active

e. Verify the following parameters under Network Traffic: NSX.

Option	Value
Apply the default virtual switch mode configured in NSX Manager	Select the checkbox.
Load Balancing	Use explicit failover order.
uplink1	Active
uplink2	Active

f. Click **NEXT**.

8. On the **NSX Manager** wizard, enter the configuration details for the NSX Manager.

Option	Value
Cluster FQDN	nsx-mgmt-a.site-a.vcf.lab

Appliance 1 FQDN	nsx-mgmt-01a.site-a.vcf.lab

9. On the SDDC Manager, validate the configuration details for the SDDC Manager appliance.

Option	Value
SDDC Manager FQDN	sddcmanager-a.site-a.vcf.lab
Administrator Password	VMware123!VMware123!

   a. Click **NEXT**.

   b. On the **Certificate Thumbprint** window, select the **Confirm Thumbprint checkbox** and click **CONFIRM**.

10. On the **Review** page, review the deployment information and click **NEXT**.
11. On the **Validate & Deploy** page, wait for the validations to run.

    a. Click **DEPLOY.**

12. Expand **Tasks** to monitor the deployment status and wait for the deployment status to be 100% completed.
13. On the **Congratulations! Your deployment completed successfully!** page, click **REVIEW PASSWORDS** to review the deployment passwords.

    a. On the Review the passwords, review the passwords used or generated during the deployment for future use and click **CANCEL** to close the window.

    b. Click **OPEN VCF OPERATIONS UI.**

14. Log in to VCF Operations with Local Account as the login method.
    - User name: admin
    - Password: VMware123!VMware123!

    a. Click the **User** drop-down menu and click **Logout** to log out of Operations UI.

    b. Close the **VMware Cloud Foundations** tab in Chrome.

Lab 4 Managing the VCF Environment with the User Interface

Objective and Tasks

Manage the VCF environment with the vSphere Client, NSX Manager, VCF Operations, and SDDC Manager:

1. Manage VCF Compute with the vSphere Client
2. Manage VCF Instances with SDDC Manager
3. Manage VCF Instance with VCF Operations
4. Manage VCF Networking with NSX

Task 1: Manage VCF Compute with the vSphere Client

You use the vSphere Client to familiarize yourself to configure and manage the Compute resources in the VMware Cloud Foundation private cloud environment.

1. From your student desktop, log in to vCenter using the vSphere Client.
   a. Open Firefox and click **Region A > vc-mgmt-a Client** on the bookmarks bar.
   b. Log in to the vSphere Client:
      - User name: administrator@vsphere.local
      - Password: VMware123!VMware123!

      When the vSphere Client opens, the **Hosts and Clusters** inventory appears in the navigation pane on the left.
2. From the Hosts and Clusters view, verify the vCenter details.
   a. Click **vc-mgmt-a.site-a.vcf.lab** and click the **Summary** tab to verify the vCenter details.

Option	Value
Release	9.1.0.0
Version	9.1.0.0
Build	vCenter Build Number

3. From the Hosts and Clusters view, verify the ESX details.
   a. Click **esx-01a.site-a.vcf.lab** and click **Summary** to verify the host details.

Option	Value
Hypervisor	VMware ESXi 9.1.0.0.XXXXXX
Model	VMware20.1
Logical Processors	32
NICs	4

4. From the Hosts and Clusters view, verify the ESX cluster details.
   a. Click the **cluster-mgmt-01a** cluster and click the **VMs** tab to verify the VMs running on the cluster.
5. Navigate to the Storage view.
   a. From the Hosts and Clusters view, click the **Storage** view.
   b. Expand **dc-a**.
   c. Click the **vsan-mgmt-01a** data store to view the summary of the data store.

      You can verify the Capacity and Usage details of the vSAN data store.
6. Navigate to the Networking view.
   a. From the Datastore view, click the **Networking** view.
   b. Click the **dc-a** data center to show the network details.
   c. Expand **vds01-mgmt-01a**.
   d. Click the **Networks** tab to verify the distributed portgroup details.

Name	VLAN ID
cluster-mgmt-01a-dvsvlan19	VLAN access:19

mgmt-vds01-mgmt-01a	VLAN access:10
vmmgmt-vds01-mgmt-01a	VLAN access:10
vmotion-vds-mgmt-01a	VLAN access:13

e. Click **vds01-mgmt-01a** and click the **Hosts** tab to verify the hosts connected to the distributed switch.

Name	State
esx-01a.site-a.vcf.lab	Connected
esx-02a.site-a.vcf.lab	Connected
esx-03a.site-a.vcf.lab	Connected
esx-04a.site-a.vcf.lab	Connected

7. Click **vc-mgmt-a.site.vcf.lab** from the inventory and click the **Networks** tab to view the tabs available under Networks.
   a. Click **Distributed Switches** to list the switches configured in the vCenter system.
   b. Click the **Distributed Port Groups** tab to list the portgroups configured in the vCenter system.
   c. Click the **Uplink Port Groups** tab to list the uplink port groups configured in the vCenter system.
   d. Click the **VPCs** tab to view the VPCs configured in the vCenter system.
   e. Click **VPC Subnets** to view the VPC subnets in the vCenter system.
   f. Click **Transit Gateways** to view the default Transit Gateway configured in vCenter.
8. Verify the Client plug-ins installed in the vSphere Client after the VCF deployment.
   a. From the **vSphere Client** menu, click **Administration**.
   b. Click **Solutions > Client Plugins** to verify the plug-ins.

Client Plug-In Name	Status	Vendor
NSX Manager NETWORKING	Deployed	Broadcom

NSX Manager VPC	Deployed	Broadcom
SDDC Manager	Deployed	Broadcom Inc
VMware Cloud Foundation Operations	Deployed	Broadcom Inc
VMware vCenter Server Lifecycle Manager	Deployed	Broadcom Inc

c. Click **Solutions > vCenter Server Extensions** to verify the extensions installed in the vSphere Client.

Extension Name	vCenter Server	Vendor
NSX Manager	vc-mgmt-a.site-a.vcf.lab	Broadcom Inc
NSX Manager NETWORKING	vc-mgmt-a.site-a.vcf.lab	Broadcom Inc
NSX Manager VCP	vc-mgmt-a.site-a.vcf.lab	Broadcom Inc
NSX Manager VPC	vc-mgmt-a.site-a.vcf.lab	Broadcom Inc
vService Manager	vc-mgmt-a.site-a.vcf.lab	VMware Inc
vSphere ESX Agent Manager	vc-mgmt-a.site-a.vcf.lab	VMware Inc

9. Click **Solutions > Foundation Load Balancers** to view the foundation load balancer.

   Foundation load balancers are not configured at this stage of the lab.

10. Click **Licensing > Version 9+ Licenses** to view the primary license name, primary license product, and expiration date.

Task 2: Manage VCF Instances with SDDC Manager

You use the SDDC Manager UI to familiarize yourself to manage the VMware Cloud Foundation private cloud environment.

CAUTION

The SDDC Manager UI is being deprecated and will be removed in a future release. Use the VCF Operations console to perform all Day-N activities.

1. Launch SDDC Manager in a new browser tab.

a. On the Firefox bookmarks bar, click **Region A** > right-click **SDDC Manager** and select **Open in new tab**.

b. If the `Your connection is not private message` appears, click **Advanced** and click **Proceed to sddcmanager-a.site-a.vcf.lab (unsafe)**.

2. Click **Close** on the Tasks bar to close the tasks and to maximize the view of the home page.

3. Read the cautionary message at the top of the **SDDC Manager** page.

   `The SDDC Manager UI is being deprecated and will be removed in a future release. Please use VCF Operations console to perform all Day-N activities GO TO VCF OPERATIONS CONSOLE.`

   `Backups of SDDC Manager and NSX-T Managers are configured with the default settings. Register an external SFTP server for backups`

   a. Click **X** to close the messages.

4. From the navigation pane, click **Inventory > Workload Domains** to view the capacity utilization across domains.

   Use the vertical scroll bar to view details about domains.

5. From the navigation pane, click **Inventory > Hosts** to view all hosts available in the VMware Cloud Foundation inventory.

   You can navigate to the **ASSIGNED HOSTS**, and **UNASSIGNED HOSTS** tabs to view the host details.

6. From the navigation pane, click **Lifecycle Management** and click **SDDC Manager** to view the current SDDC Manager version.

7. From the navigation pane, click **Lifecycle Management > Release Versions** to view the Release Bill of Materials (BOM).

8. From the navigation pane, click **Administration > Network Settings** to view the network pool, DNS configuration, and NTP configuration information.

9. From the navigation pane, click **Administration > Single Sign On** to view the Users, Groups and Identity Provider details.

10. From the navigation pane, click **Security > Password Management** to manage the passwords for ESX, vCenter, NSX Manager, and SDDC Manager.

    If you see `"Password rotate for resource: vc-mgmt-a.site-a.vcf.lab, user:svc-sddcmanager-a-vc-mgmt-a-4622@vsphere.local and credential type: SSO)"` click **Cancel**.

11. From the navigation pane, click **Developer Center** to view the available APIs in the API Explorer.
12. Navigate to the top-right corner of the SDDC Manager UI, click the **administrator@vsphere.local** user name, and click on **Log Out**.
13. Close the **SDDC Manager** tab in the Firefox browser.

Task 3: Manage VCF Instances with VCF Operations

You use the VCF Operations UI to familiarize yourself to configure and manage the VMware Cloud Foundation private cloud environment.

1. Launch VMware Cloud Foundation Operations in a new browser tab.
    a. On the Firefox bookmarks bar, click **Region A >** right-click **VMware Cloud Foundation Operations** and select **Open in new tab**.
    b. Log in to VCF Operations:
        - User name: admin
        - Password: VMware123!VMware123!

        When the **VMware Cloud Foundation Operations** page opens, the **Home** screen appears with the **VCF Overview** and vCenter details. under **Operate** tab.
2. You use the Operate tab to manage the following tasks.
    a. From the navigation pane, click **Inventory** to view the inventory details.

        You can view the inventory in DETAILED VIEW and BASIC VIEW.
    b. Click Inventory **BASIC VIEW** to switch the view to basic view.
    c. Click Inventory **DETAILED VIEW** to switch the view to detailed view.
    d. From the navigation pane, click **Alerts** to view the **Alerts, Administrative Alerts** and **Intelligent Alerts**.
    e. From the navigation pane, click **Dashboards** to view the **Dashboards, Views and Reports**.
    f. From the navigation pane, click **Findings** to view the **ACTIVE FINDINGS, HISTORICAL FINDINGS** and **FINDINGS CATALOG.**
    g. From the navigation page, click **VCF Health** to view VCF Health.

        VCF Health has two views: VCF VIEW and COMPONENT VIEW.

        VCF VIEW is the default for VCF Health.

        i. Click the **COMPONENT VIEW** tab to find the component-level VCF Health.

h. From the navigation pane, click **Workloads > Applications** to view the VCF applications registered with VCF Operations.

i. From the navigation pane, click **Administration > SDDC Manager** to view the SDDC Manager network settings, storage settings, and backup settings.

j. From the navigation pane, click **Administration > Integrations.**

   i. Expand **VMware cloud Foundation.**

   ii. Verify that the **Status** is **Collecting**.

   iii. Expand **instance-a** to verify the details.

   iv. Expand **mgmt-a** to verify that the status is **Collecting**.

   v. Expand **wld01-a** to verify that the status is **Collecting**.

3. From the VMware Cloud Foundation Operations home page, click **Manage** tab.

   a. From the navigation pane, Click **Fleet Management > Certificates** to view the certificates for VCF Management, VCF instances and Standalone vCenter Instances.

      i. Expand **VCF Instances > instance-a > mgmt-a** to view the Management domain's VCF instance certificates.

      ii. Click **wld01-a** to view the workload domains VCF instance certificates.

   b. From the navigation pane, click **Fleet Management > Passwords** to manage the VCF Management and VCF instances passwords.

   c. From the navigation pane, click **Fleet Management > Tags** to create and import the tags.

   d. From the navigation pane, click **Fleet Management > Fleet Settings** to set the Password Policy, DNS Settings and NTP Settings.

   e. From the navigation pane, click **Capacity > Assess** to view the capacity available in both the Management and Workload Domain data centers.

   f. From the navigation pane, click **Licensing > Licenses & Registration** to view the licenses for version 9+ and earlier versions.

   g. From the navigation pane, click **Licensing > Usage Analytics** to view the licenses usage by product.

4. From the VMware Cloud Foundation Operations home page, click **Protect** tab.

   a. Click **Security Operations** to view the user security and infrastructure security information.

   b. From the navigation pane, click **Compliance** to view and enable the **VMware SDDC Benchmarks**, **VCF Benchmarks**, **Custom benchmark** details and to Activate **Regulatory Benchmarks** from Repository**.**

c. From the navigation pane, click **Protection and Recovery** to monitor Protection and Recovery.

5. From the VMware Cloud Foundation Operations, click **Build** tab.

   a. From the navigation pane, click Software Depot to view the Software depot connection status and to view the Storage Space used for the data storage.

   b. From the navigation pane, click **Lifecycle** to view the VCF Management and VCF Instances Component details and the version.

   c. From the navigation pane, click Tasks to view the Task, Subtask, Task Status and Last Occurrence details.

   d. From the navigation pane, click **Developer Center > APIs & SDKs** to view the APIs and VCF PowerCLI, and management packs information.

   e. From the navigation pane, click **Developer Center > Management Pack Builder** to design a management pack.

6. Navigate to the top-right corner of the VMware Cloud Foundation Operations UI, click the **admin** user icon, and select **Log Out**.

7. Close **X** to close the **ops-a.site-a.vcf.lab** tab from Firefox.

Task 4: Manage VCF Networking with NSX

You use the NSX UI to familiarize yourself to view VCF Networking in the VMware Cloud Foundation private cloud environment. Making modifications directly in NSX might impact some VCF Operations workflows.

1. Log in to NSX Manager.

   a. Open a new tab in Firefox.

   b. Click **Region A > nsx-mgmt-01a** from the bookmarks bar.

   c. If the `Your connection is not private` message appears, click **Advanced** and click **Proceed to nsx-mgmt-a.site-a.vcf.lab (unsafe)**.

   d. Log in to NSX Manager.

      - User name: admin
      - Password: VMware123!VMware123!

IMPORTANT

You will find the following banner displayed on the login page.
`This NSX Manager is managed by VCF Operations. Making modifications directly in NSX might impact some VCF Operations workflows.`

i. Click **DO NOT SHOW THIS AGAIN** button to close.

2. On the **Home** page, click **Alarms** to view the triggered alarms and the alarm details.

   You can ignore the Minimum Capacity Threshold and Remote Logging Not Configured alarms shown in Open Alarm State.

3. On the **Home** Page, click the **System Health** tab to view the health of Management cluster, hosts, and Edge details.

   The single-node Management cluster is deployed with an IP address 10.1.1.21. assigned to the node with a Virtual IP 10.1.1.20.

4. On the **Home** page, click **Monitoring Dashboards > Networking& Security** to view the dashboard details.

5. On the **Home** page, click **Monitoring - Networking & Security > Compliance Report** to view the list of noncompliant resources.

6. Click the **Networking** tab to view the network overview and manage connectivity, network services, IP management, and settings.

7. Click the **VPCs** tab for the VPC overview, topology, virtual private clouds, and profiles used in VPC configuration.

8. Click the **Security** tab to view the security overview and manage threat event monitoring, policy management, service chain management, and settings.

   You use the vertical scroll bar in Firefox to view these sections.

9. Click **Inventory** to view the Inventory view and manage services, groups, profiles, virtual machines, containers, bare-metal servers, tags, and resource sharing details.

10. Click the **Plan & Troubleshoot** tab to view the Discover & Plan and Troubleshooting tools.

11. Click the **System** tab to view the system overview, configuration, life cycle management, and settings.

Lab 5 Verifying the VCF Operation Integration with the Management Domain

Objective and Tasks

Verify the VCF Operations integration with the Management domain:

1. Prepare for the Lab
2. Verify the VCF Operation Integration with vCenter, vSAN, and NSX

Task 1: Prepare for the Lab

You log in to the VCF Operations UI.

1. Log in to VCF Operations.
    a. Open a new tab in Firefox.
    b. On the bookmarks bar, click **Region A > VMware Cloud Foundation Operations.**
    c. Log in to VCF Operations.
        - User name: admin
        - Password: VMware123!VMware123!

Task 2: Verify the VCF Operation Integration with vCenter, vSAN, and NSX

1. From the navigation pane, click **Administration > Integrations.**
2. On the **Accounts** tab, expand **VMware Cloud Foundation**.
3. Expand **instance-a > mgmt-a** and verify that the status is Collecting.

The status must be collecting for vCenter, vSAN and NSX.

4. Expand **OS and Application Monitoring** and verify that the **Application Monitoring Adapter (opscollector-01a.site-a.vcf.lab)** status is Collecting.
5. Expand **VCF Automation for All Apps Organization** and verify that the **VCF Automation** status is Collecting.

   You can ignore the warning that appears on VCF Automation for All Apps Organization because it will automatically clear the warning.
6. On the **Integrations > Accounts** page, click **COLLECTION STATUS** to view the VCF Operations collecting status.
7. From the navigation page, click **Administration > Integrations** to return to the integrations view.
8. Click **admin** user drop-down menu from the top-right corner and click **Log Out.**
9. Click **X** to close the VMware Cloud Foundations page.
10. Click **X** to close the Firefox application.

Lab 6 (Simulation) Deploying Log Management

Objective and Tasks

Deploy and log in to the Log Management component:

1. Prepare for the Lab
2. Deploying Log Management
3. Using Log Management UI
4. Shutting down Log Management

In the web browser in your local desktop, go to https://core-vmware.bravais.com/s/21939/Deploying_Log_Management to open the simulation.

Perform the following steps in the simulated environment. Refer to these steps as you work through the simulation.

IMPORTANT

Do not refresh, navigate away from, or minimize the browser tab hosting the simulation. Such actions might pause the simulation and prevent it from progressing.

Task 1: Prepare for the Lab

You log in to VCF Operations.

1. Log in to the VCF Operations with the local admin account.
    a. Open **Chrome**.

b. On the Firefox bookmarks bar, click **Region A > VMware Cloud Foundation Operations.**

c. Select **Local Account** as the login method and click **Log In**.

- User name: admin
- Password: VMware123!VMware123!

Task 2: Deploying Log Management

You deploy Log Management component using VCF Operations:

1. Click **Build** tab.
2. In the navigation pane on the left, click **Lifecycle** > **VCF Management**
3. Click **Components** tab.
4. Click **ADD COMPONENT** > **Log Management**
5. On the **Deployment** wizard, select the Deploy Parameters.

Option	Value
Version	9.1.0.0
Import configuration from an earlier instance	Verify that the option is not selected.
Log Management FQDN	opslogs-01a.site-a.vcf.lab
Size	Small
Number of Replicas	1

6. Select **I am aware of possible service disruption and want to continue with deployment** checkbox.
7. Click **NEXT**.
8. Click **FINISH.**

   The deployment takes 2 to 3 hours. Log Management is deployed in two different stages, and you can monitor the deployment status under Tasks.

   You wait for the deployment to complete before proceeding to the next task.

9. Click **View Details** link next Log Management is being deployed to monitor the Log Management deployment status.

Task 3: Using Log Management UI

You log in to the VCF Operations and use Log Management capabilities:

1. In Chrome, Open a **new** tab.
2. Enter the following URL and press **Enter** key.

   `https://opslogs-01a.site-a.vcf.lab`
3. Log in to VCF Log Management.
   - User name: admin
   - Password: VMware123!VMware123!
4. In the top-right corner, click the **Logout** option to log out from the UI.
5. Close the **VCF Operations for Logs** tab in Chrome.

Task 4: Shutting down VCF Log Management

1. Click on **VMware Cloud Foundation Operations** tab in Chrome.
2. Click **Build** > **Lifecycle**.
3. Select **VCF Management.**
4. Click **Components** tab.
5. Click **Log Management**.
6. Click **ACTIONS** drop-down menu next to Log Management.
7. Click **STOP SERVICE**.

   This will gracefully shutdown the Log Management service.

Lab 7 Configuring VCF Single Sign-On for the Management Domain

Objective and Tasks

Configure VCF SSO for VCF instances and VCF Management components:

1. Prepare for the Lab
2. Configuring Authentik
3. Enable Single Sign-on
4. Configure SCIM in Authentik
5. Enable Single Sign-on on Management Domain vCenter
6. Verify the Integration of VCF Instances with SSO
7. Verify VCF Automation and VCF Operation Integration with VCF SSO

Task 1: Prepare for the Lab

You log in to the vSphere Client and SDDC Manager.

1. Log in to the vSphere Client.
    a. Open Firefox.
    b. On the bookmarks bar, click **Region A > vc-mgmt-a Client**.
    c. Log in to the vSphere Client.
        - User name: administrator@vsphere.local
        - Password: VMware123!VMware123!
2. Log in to VCF Operations.
    a. Open a new tab in Firefox.

b. On the bookmarks bar, click **Region A > VMware Cloud Foundation Operations**.

c. Log in to VCF Operations.

- User name: admin
- Password: VMware123!VMware123!

3. Launch SDDC Manager in a new tab.

a. On the Firefox bookmarks bar, right-click **Region A > SDDC Manager** and select **Open in new tab**.

b. Click the **SDDC Manager** browser tab.

- User name: administrator@vsphere.local
- Password: VMware123!VMware123!

4. Launch NSX Manager.

a. Open a new tab.

b. On the Firefox bookmarks bar, click **Region A > nsx-mgmt-01a**.

c. Log in to NSX Manager.

- User name: admin
- Password: VMware123!VMware123!

Task 2: Configuring Authentik

You log in to the vSphere Client and SDDC Manager.

1. Fix CoreDNS Forwarder in the Console.

a. Open **Terminal** from Console.

b. Run the following command:

```
sshpass -p 'VMware123!VMware123!' ssh root@router "sed -i
's/10.1.1.1/10.244.0.1/g' /holodeck-
runtime/k8s/coredns_configmap.yaml && kubectl delete -f
/holodeck-runtime/k8s/coredns_configmap.yaml && kubectl
apply -f /holodeck-runtime/k8s/coredns_configmap.yaml"
```

2. Log in to Authentik.

a. Open a **new tab** in Firefox.

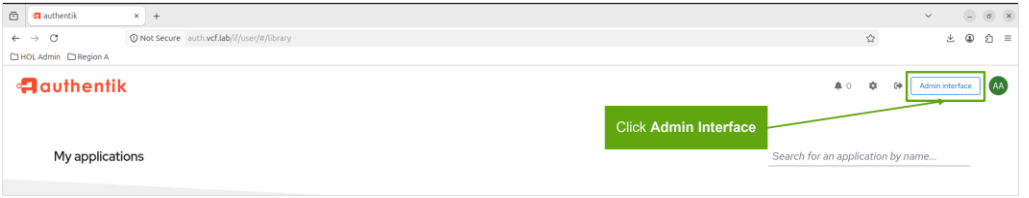
b. On the bookmarks bar, click **HOL Admin > Authentik**.

c. If you receive "Be careful. Something doesn't look right", click **Advanced > proceed to auth.vcf.lab (Risky)**

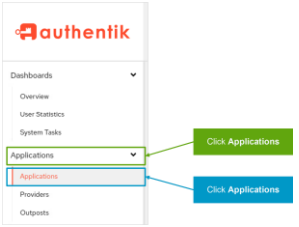
d. Log in to authentik.

- User name: akadmin
- Password: VMware123!VMware123!

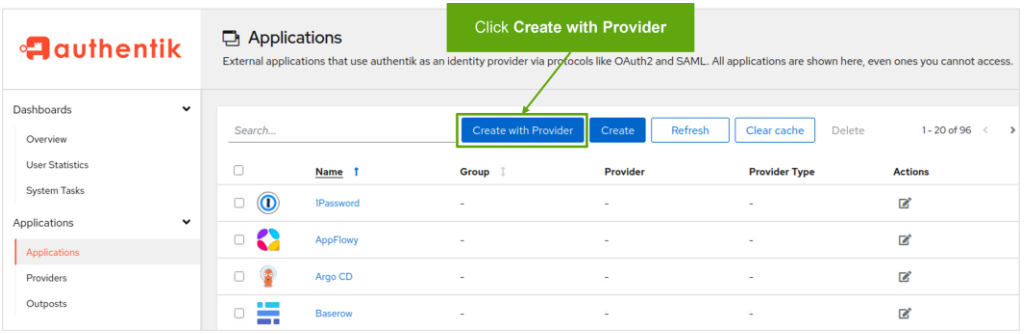
3. Click **Admin Interface** from the top right corner.



4. Click **Applications > Applications** from the navigation pane on the left-hand side.



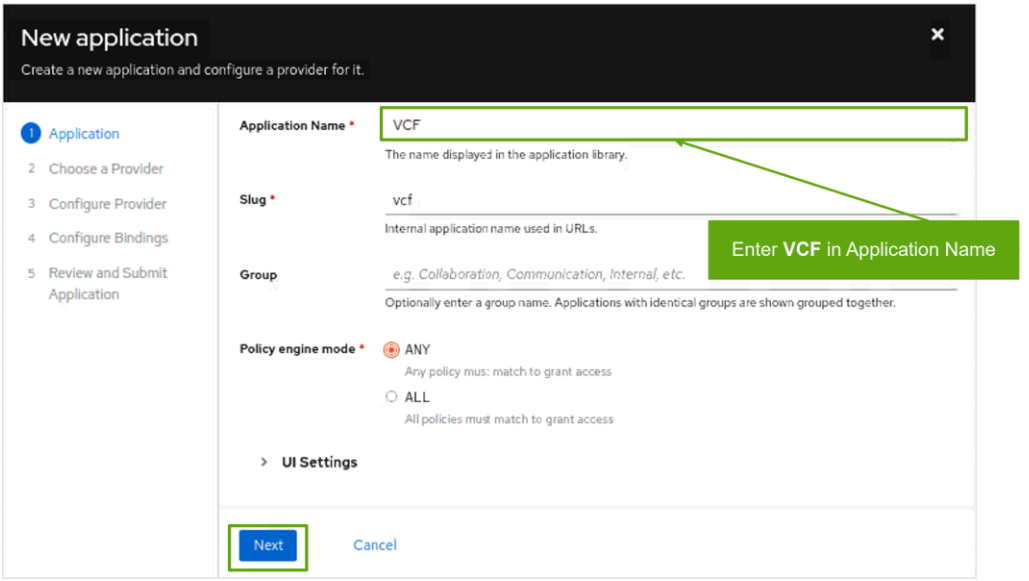
5. Click **Create with Provider**.



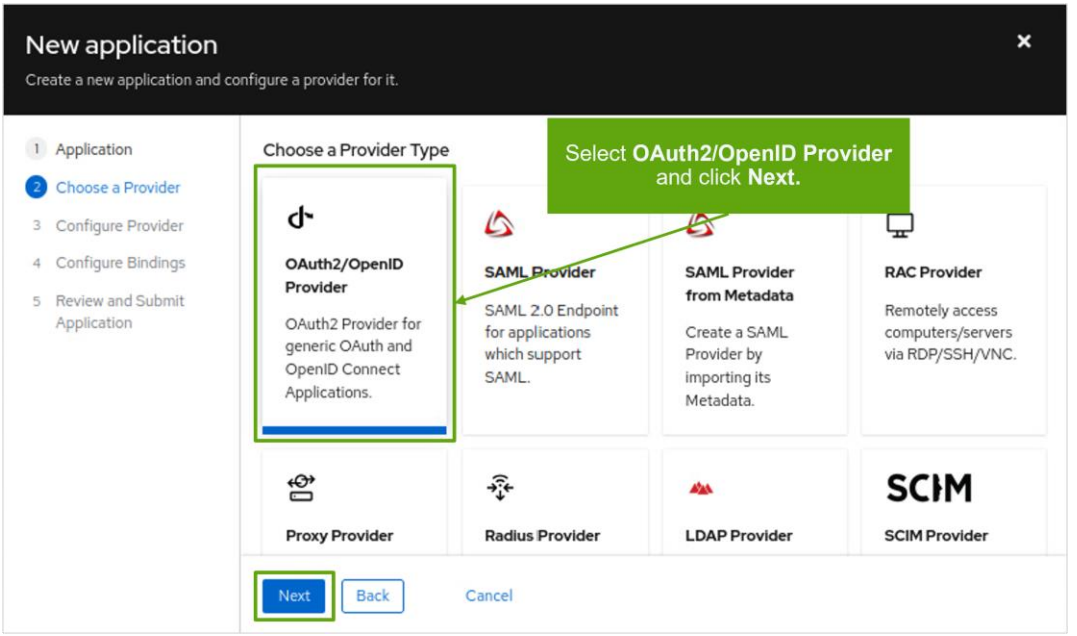
6. On the New application wizard, configure the Application with the following details and click **Next**.

- **Application Name**: VCF
- **Slug:** vcf
- **Group:** Leave it blank

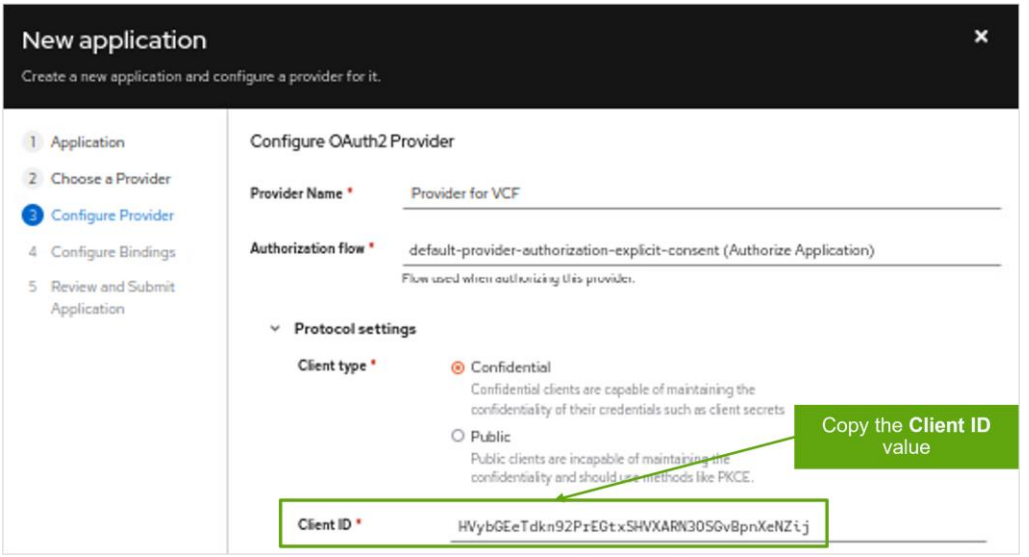
- **Policy engine mode:** Any



7. Select **OAuth2/OpenID Provider** and click **Next**.



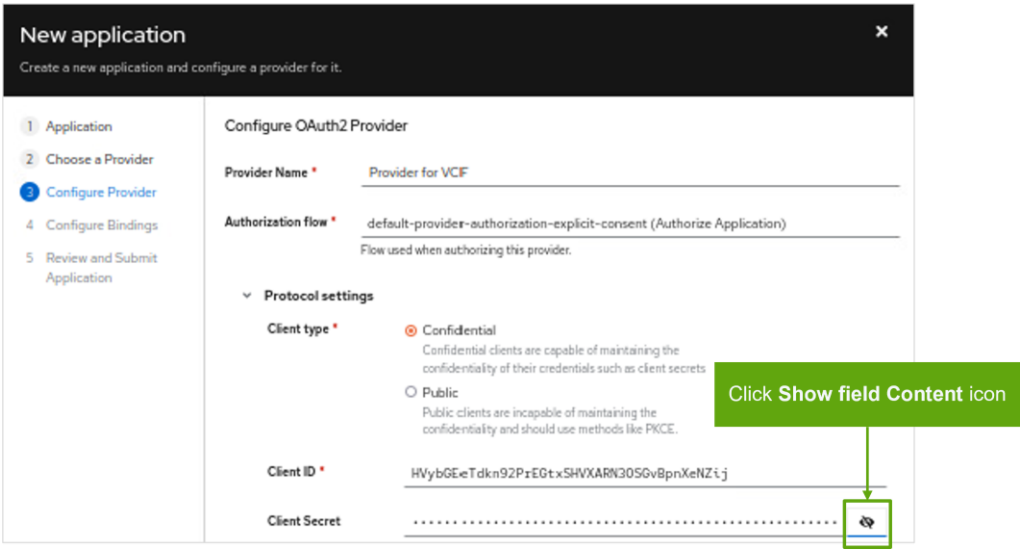
8. On the New application wizard, configure the OAuth2 Provider with the following details and click **Next**.
   - **Provider Name**: Provider for VCF
   - **Authorization flow**: default-provider-authorization-explicit-consent (Authorize Application)
   - **Client type:** Confidential
   - **Client ID**: Press Ctrl+A to select all text and press Ctrl+C to copy the client ID



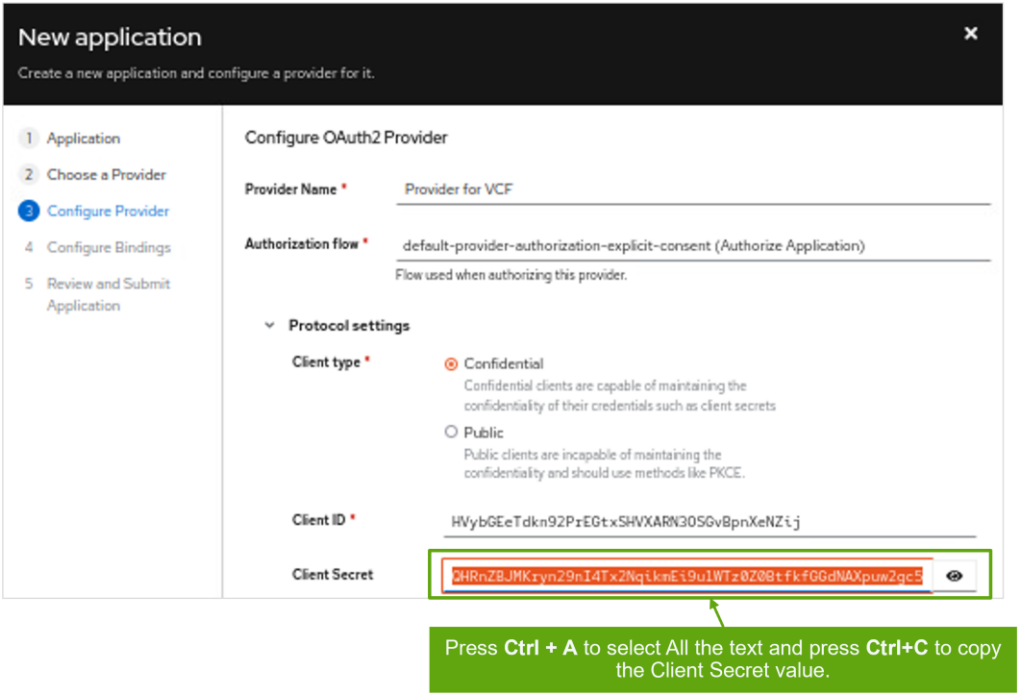
Open **authentik.txt** file from desktop and paste the client ID value under Client ID.



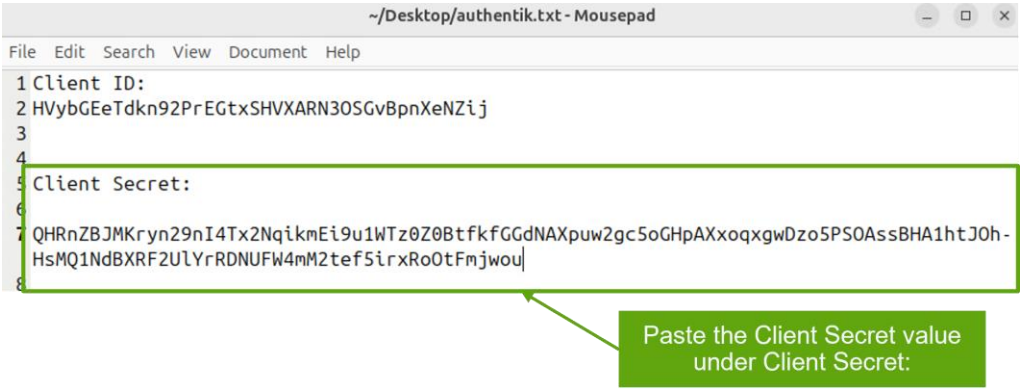
9. Click **Show Field content** icon from Client Screen text box.



10. Click **Ctrl + A** to select the Client Secret value and click **Ctrl+C** to copy the client secret.

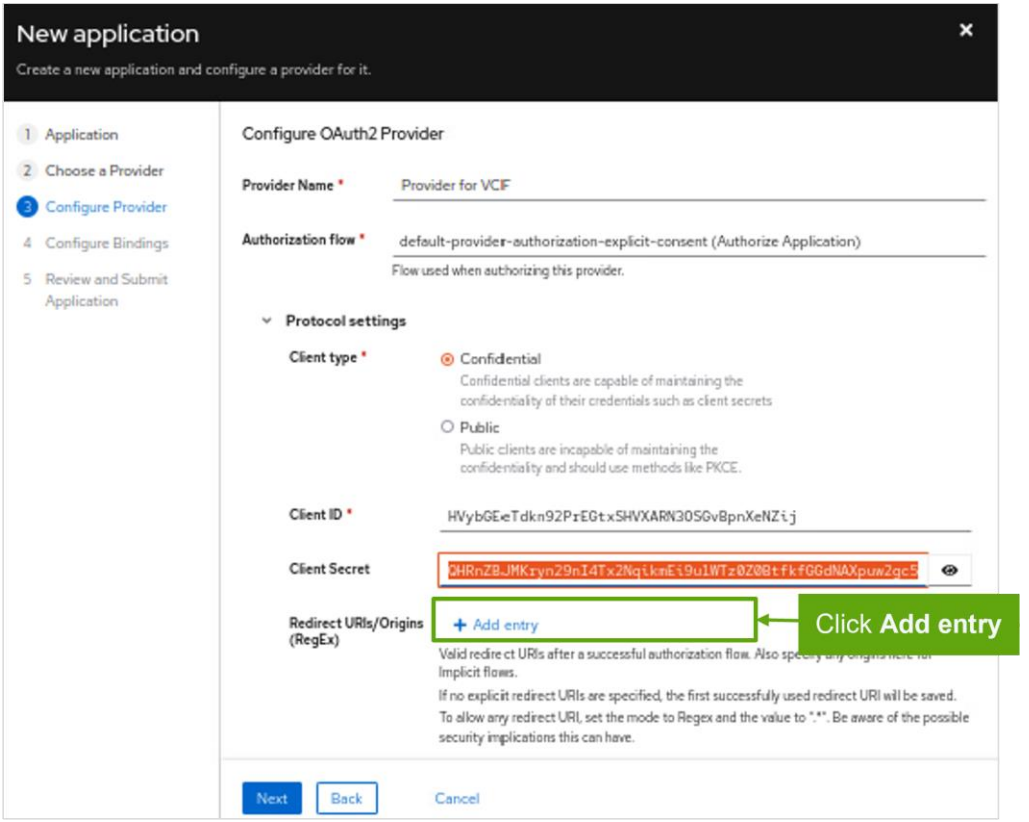


11. Open **authentik.txt** and paste the Client Secret value under Client Secret.



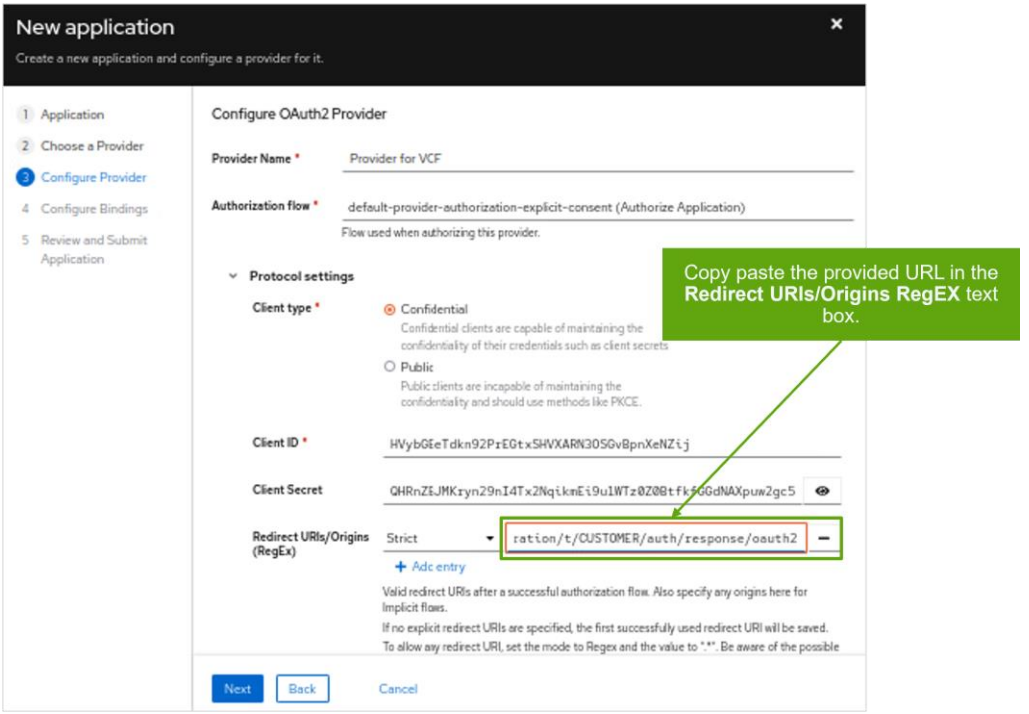
a. Press **Ctrl + S** to save the authentik.txt file.

12. Click **+ Add entry** next to Redirect URIs/Origins (RegEx)

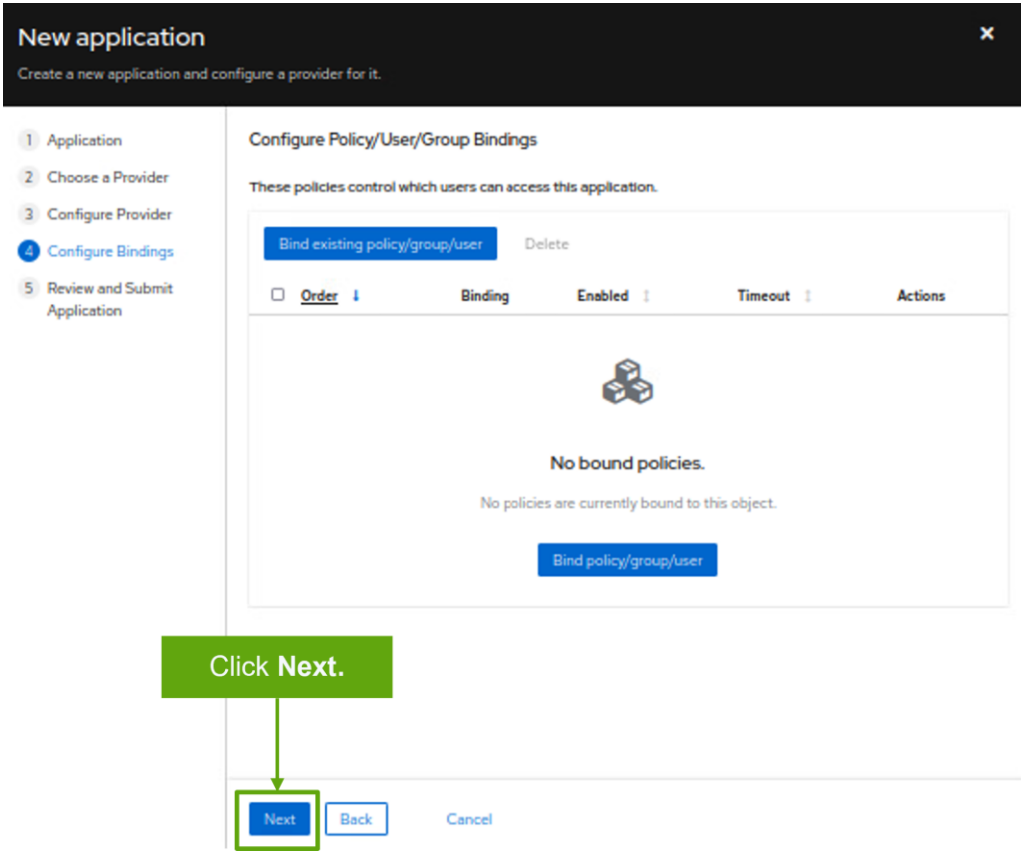


13. Copy the following link and paste it in the URL field and click **Next**.

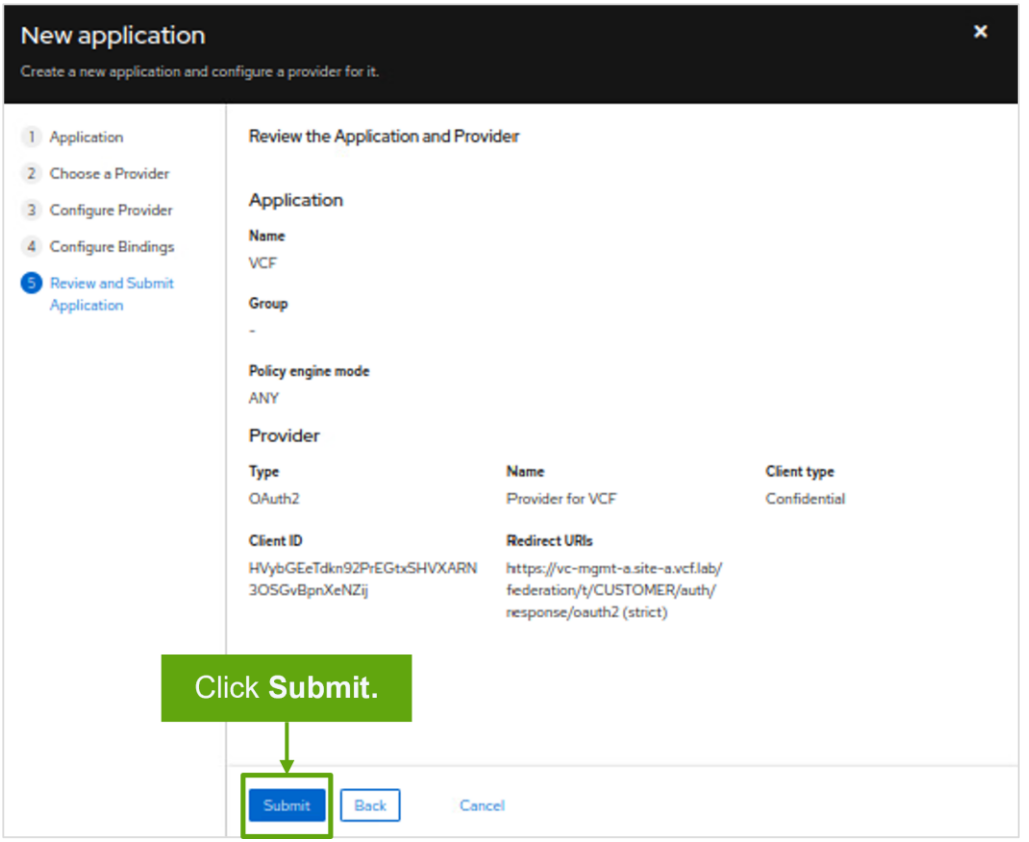
```
https://vc-mgmt-a.site-
a.vcf.lab/federation/t/CUSTOMER/auth/response/oauth2
```



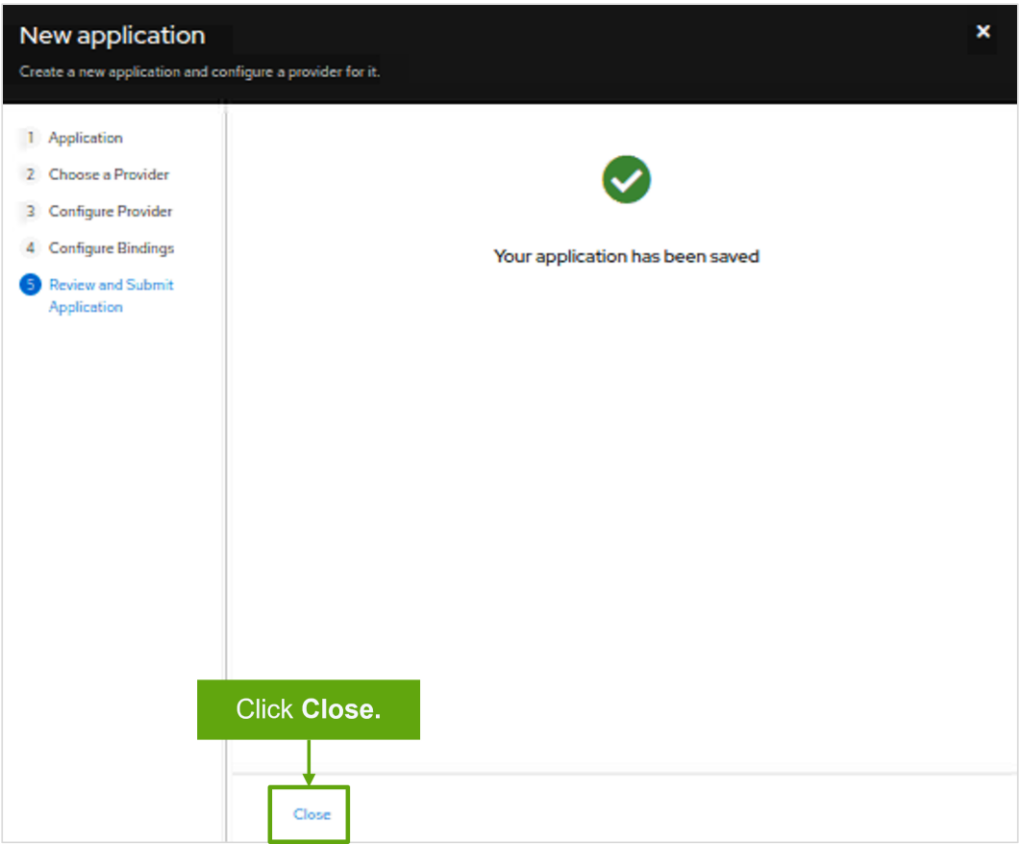
14. Click **Next** on **Configure Policy/User/Group Bindings.**



15. Click **Submit** on Review the Application and Provider.



16. Click **Close** on Your application has been saved.

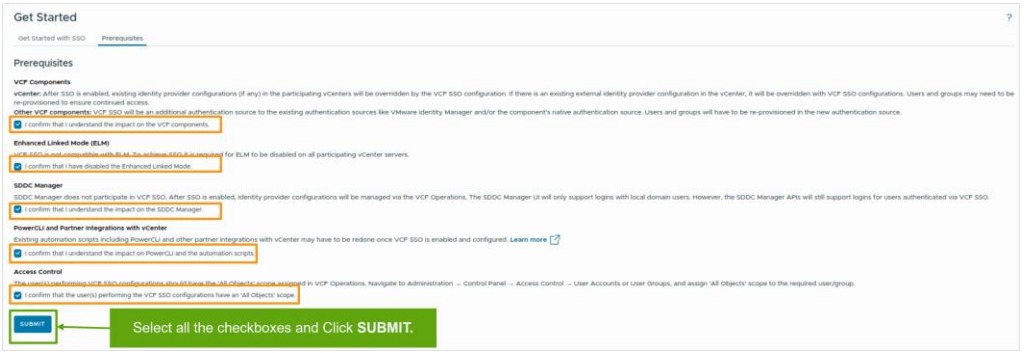


Task 3: Enable Single Sign-On

You use the Guided Setup wizard to connect SDDC Manager to an Active Directory (AD) identity provider.

1. In VMware Cloud Foundation Operations, click the **Manage** tab.
2. Click **Fleet Management > Identity & Access.**
3. Under **Get Started with SSO** tab, click **CONFIGURE SSO.**
4. Under Prerequisites, select the VCF components to configure for SSO.

   You must read all the prerequisites shown on this screen before continuing with the lab.



a. Under Other VCF Components, select the **I confirm that I understand the impact on the VCF components** checkbox.

b. Under Enhanced Linked Mode (ELM), select the **I confirm that I have disabled the Enhanced Linked Mode** checkbox.

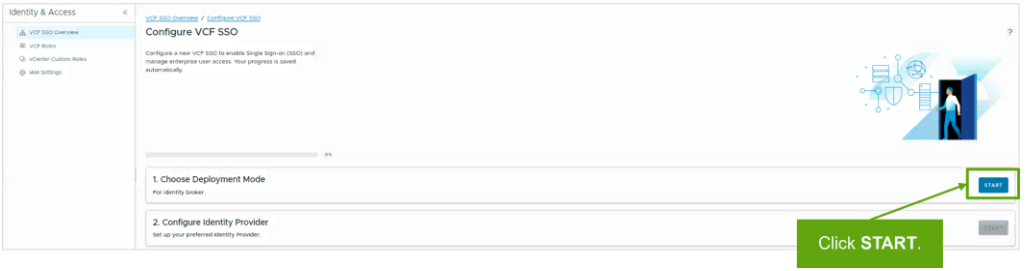
c. Under SDDC Manager, select the **I confirm that I understand the impact on the SDDC Manager** checkbox.

d. Under PowerCLI and Partner Integrations with vCenter, select the **I confirm that I understand the impact on PowerCLI and the automation scripts** checkbox.

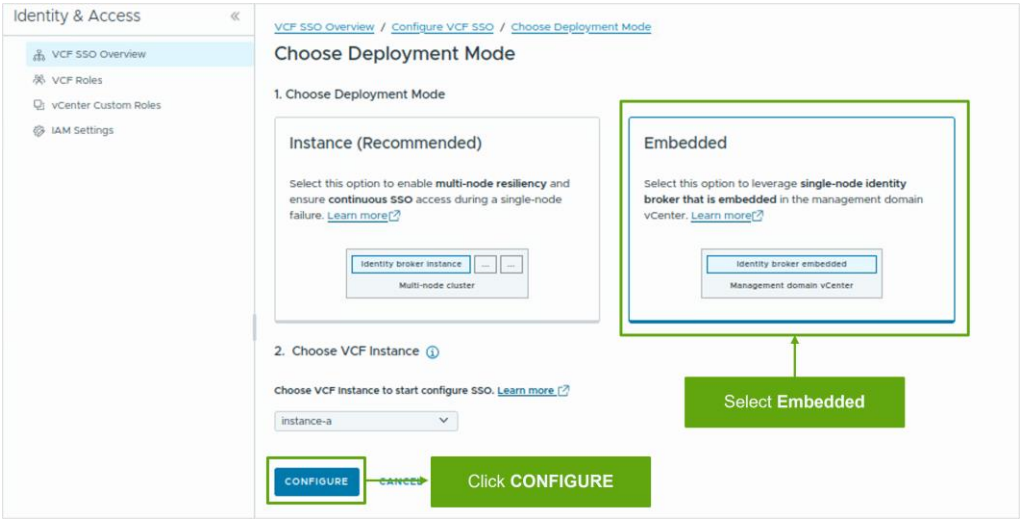
e. Under Access Control, select the **I confirm that the user(s) performing the VCF SSO configurations have an 'All Objects' scope** checkbox.

f. Click **SUBMIT**.

5. On the Configure VCF SSO page, click START to go to the 1. Choose deployment mode page.



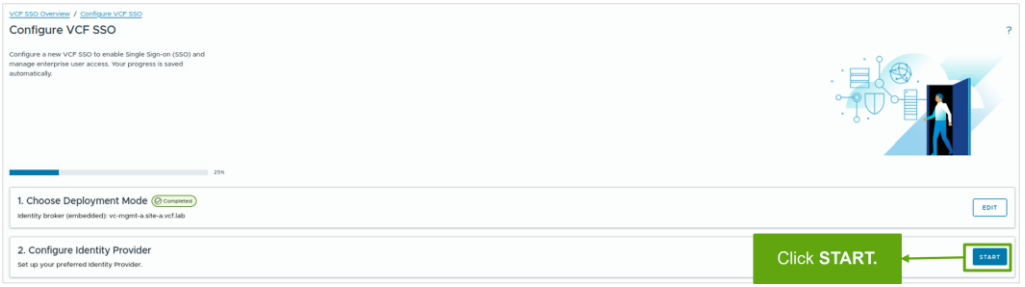
6. Under Choose the deployment mode, select the following:



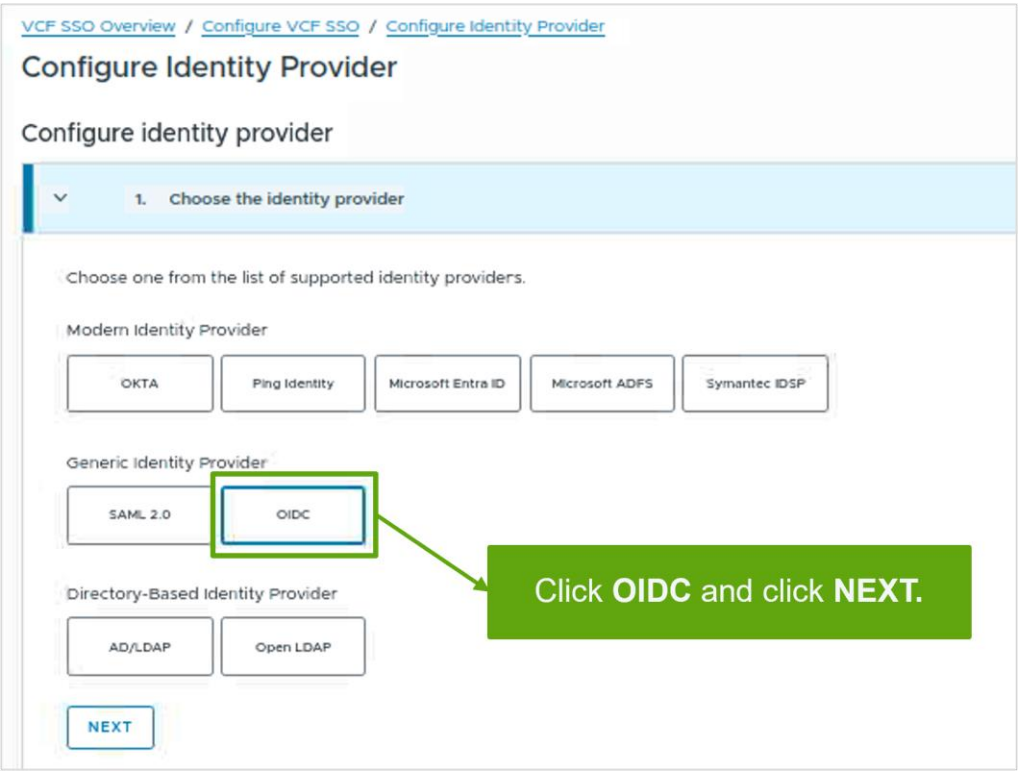
a. Under 1. Choose the deployment mode, select the **Embedded** option to leverage single-node identity broker that is embedded in the management domain vCenter.

b. Under 2. Choose VCF Instance, verify that **instance-a** is selected and click **CONFIGURE**.

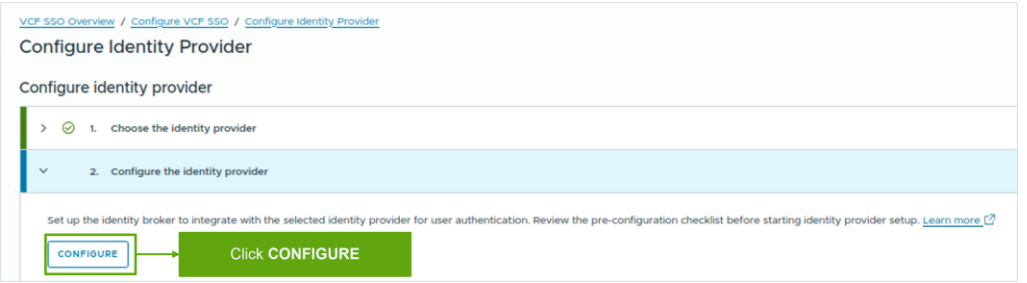
7. Under 2. Configure the Identity Provider, click **START**.



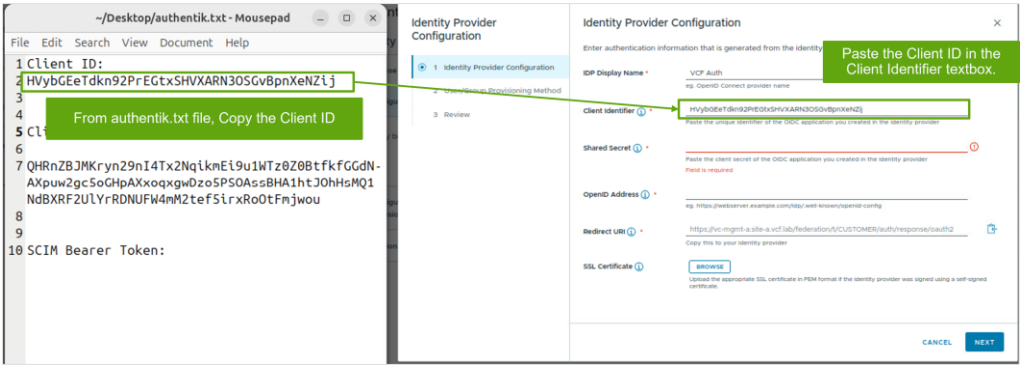
8. Under 1. Choose the Identity Provider, select **OIDC** and click **NEXT**.



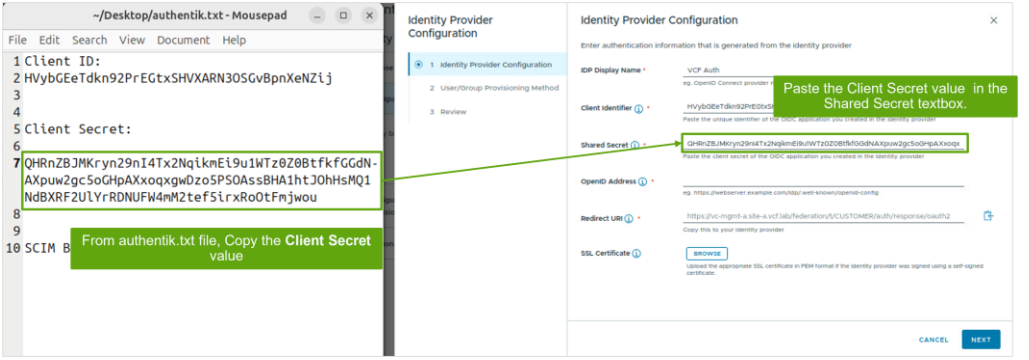
9. Under 2. Configure the Identity provider, click **CONFIGURE**.



10. On Identity Provider Configuration, configure the identity provider.
    - **IDP Display Name**: VCF Auth
11. Copy the Client ID from the authentik.txt file and paste it in the **Client Identifier** text box.

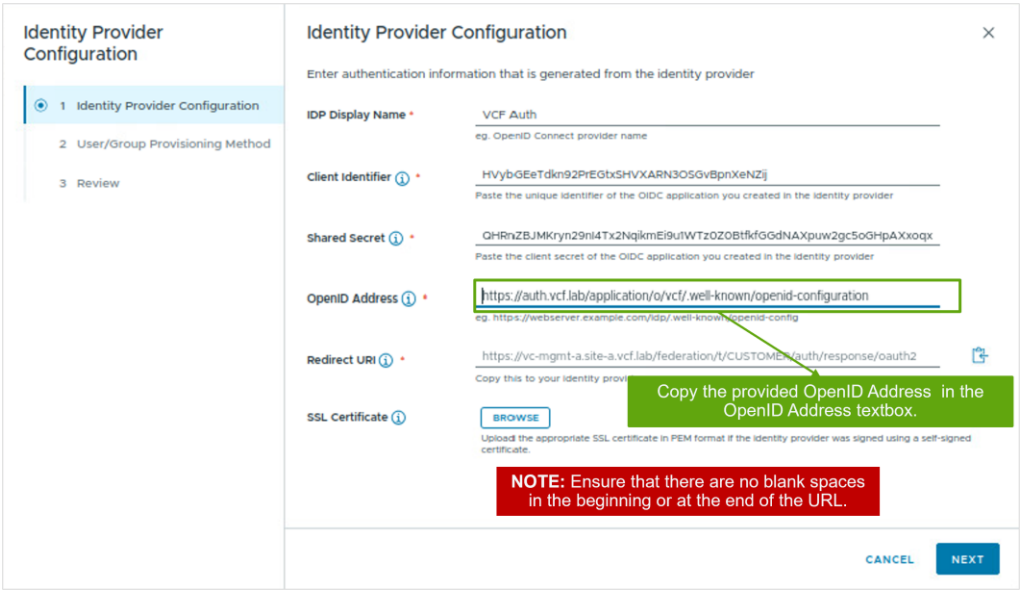


12. Copy the client secret from the authentik.txt file and paste it in the **Shared Secret** text box.

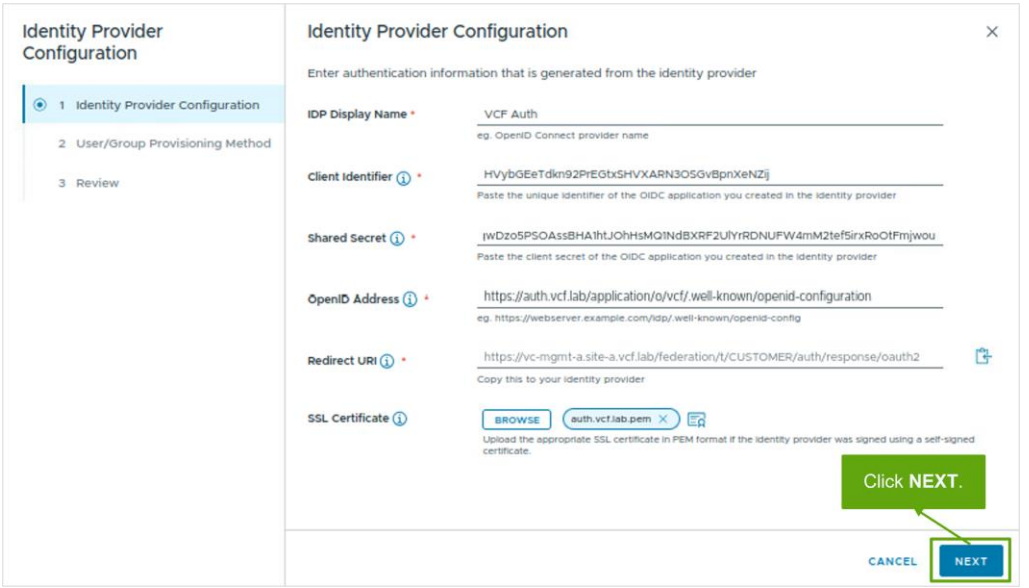


13. Copy the following URL and paste in **OpenID Address** text box.

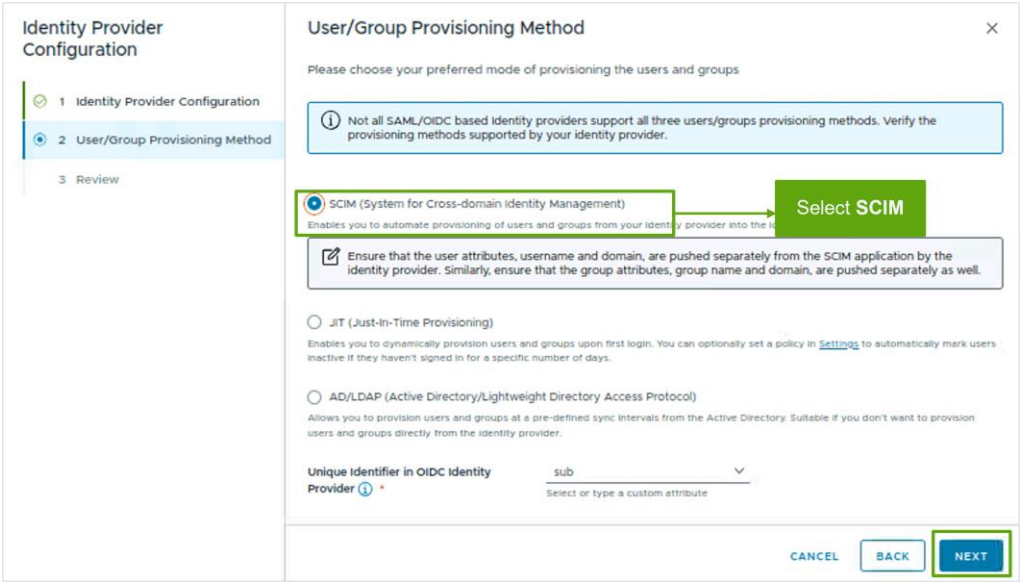
    `https://auth.vcf.lab/application/o/vcf/.well-known/openid-configuration`



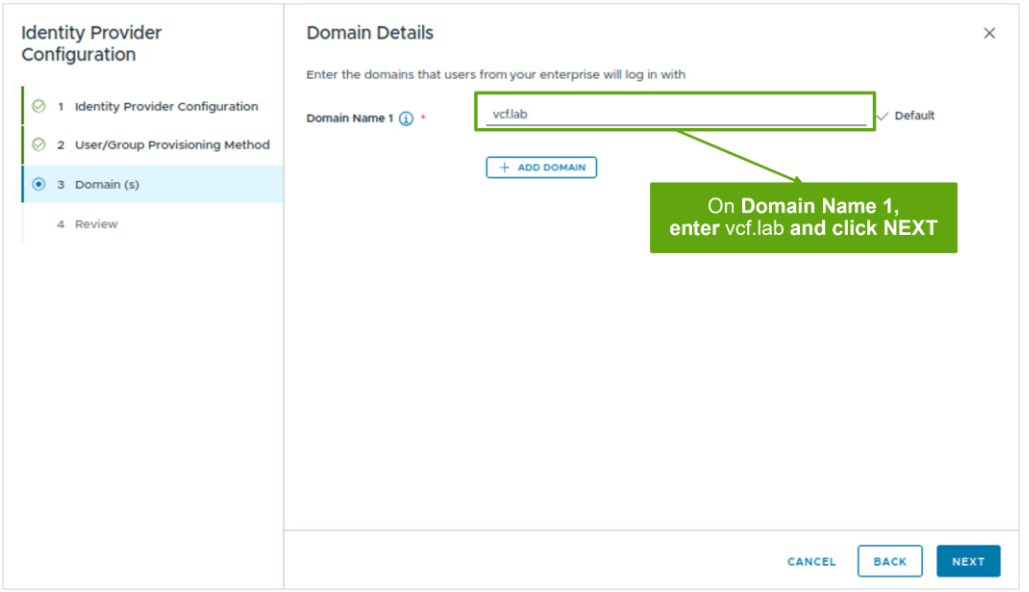
14. Download PEM Certificate.
    a. Open a **new tab** in Firefox.
    b. Enter **`ca.vcf.lab`** in the address bar and press enter.
    c. Username: Administrator
    d. Password: VMware123!VMware123!
    e. Click **Download a CA Certification, certificate chain or CRL**
    f. Click **auth.vcf.lab** link.
    g. Click **Download PEM** from the certificate details.
    h. Click on **VCF Operations tab** in Firefox.
    i. Click **BROWSE** next to **SSL Certificate**, click **Downloads** on the left and click `auth.vcf.lab.pem` file and click **Select**.
15. Click **NEXT** on the Identity Provider Configuration.



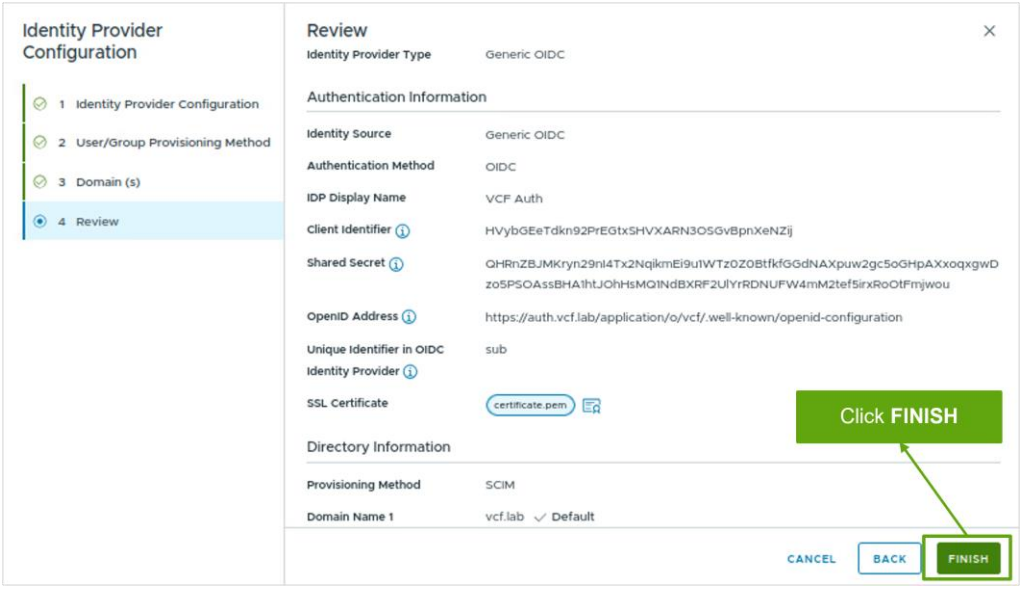
16. Select **SCIM (System for Cross-domain Identity Management)** and Click **NEXT**.



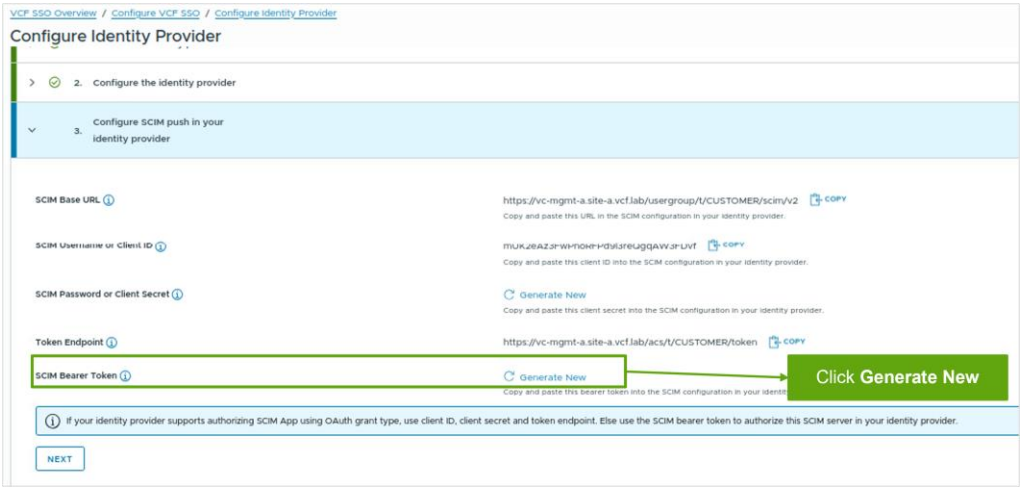
17. On **Domain Name 1, enter `vcf.lab` and click NEXT.**



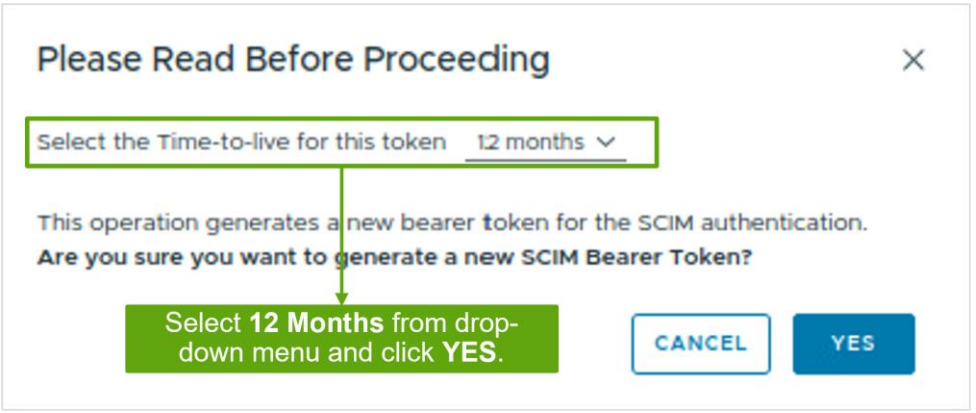
18. Click **FINISH.**



19. On the Configure Identity Provider page, click **Generate New** next to SCIM Bearer Token.



a. On the **Please Read before Proceeding** pop up window, Click Select the Time-to-live for this token drop-down menu and select **12 months** and Click **YES**.



b. On the Newly Generated SCIM Bearer Token, click **COPY**.

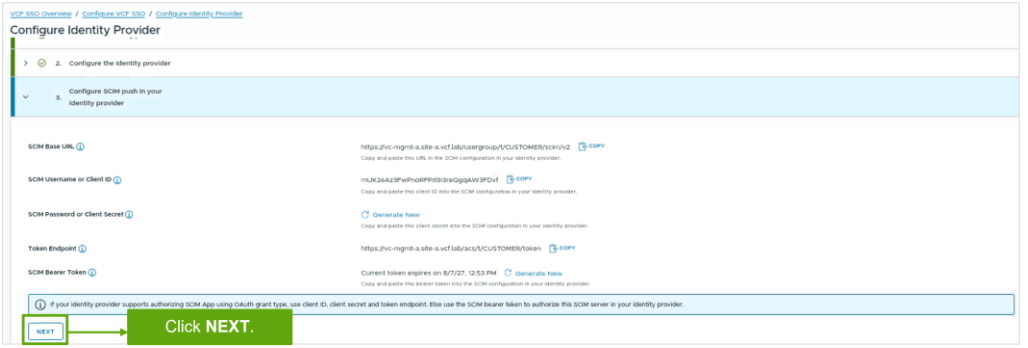


c. Open **authentik.txt** and paste it under **SCIM Bearer Token** and **Save** the file.

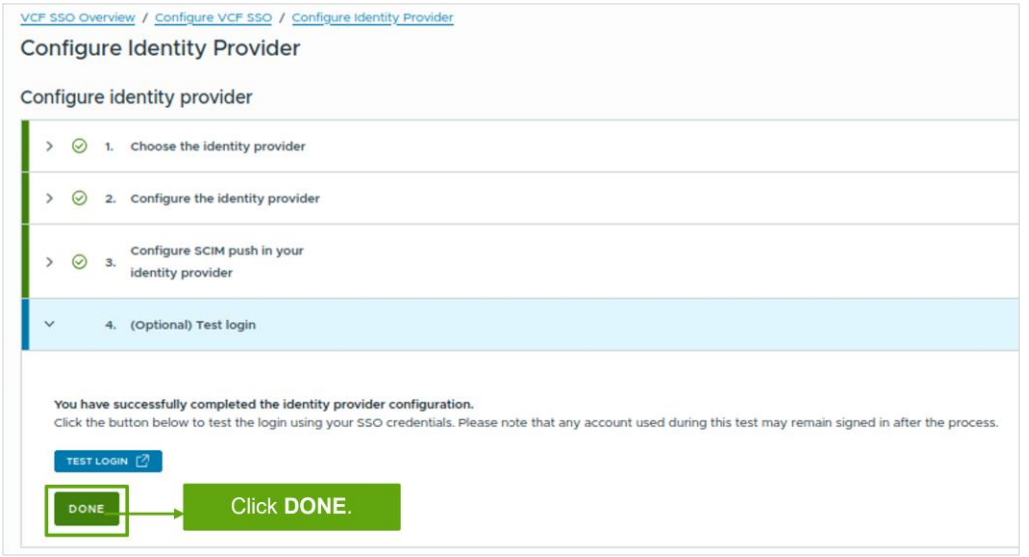


d. Click **CONTINUE** on Newly Generated SCIM Bearer Token.

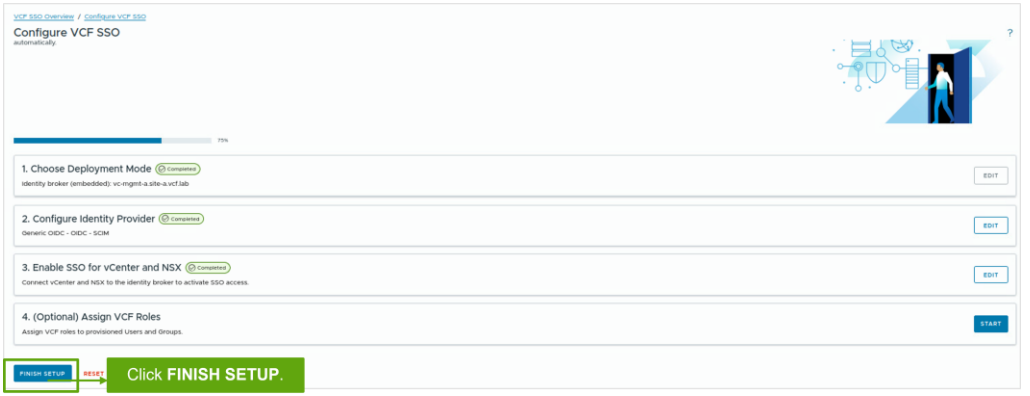
20. Click **NEXT** on Configure Identity Provider wizard.



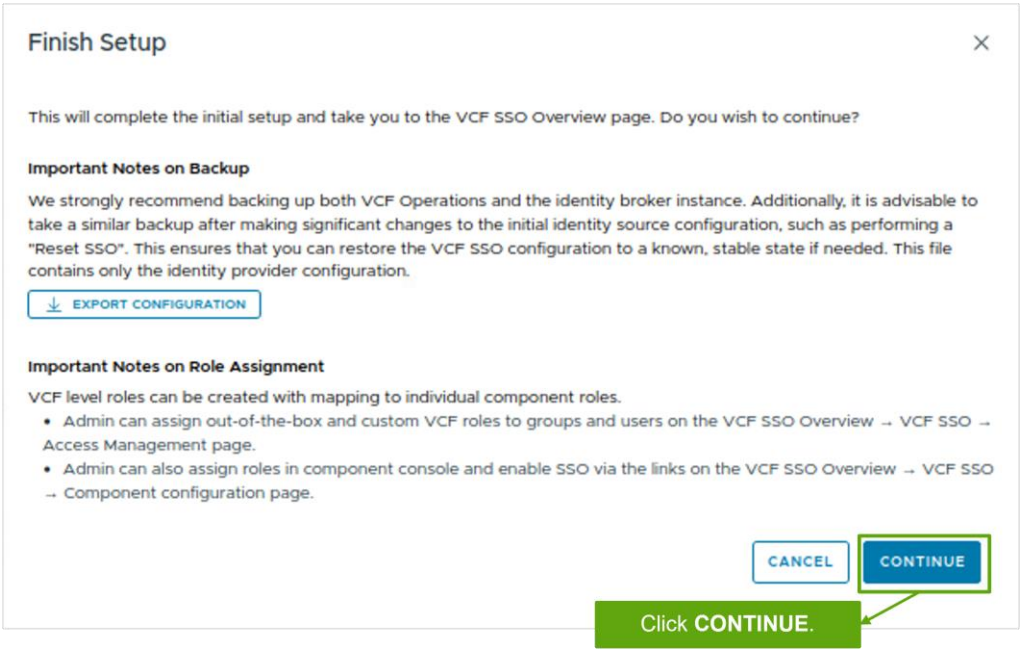
21. Click **DONE**.



22. Click **FINISH SETUP**.



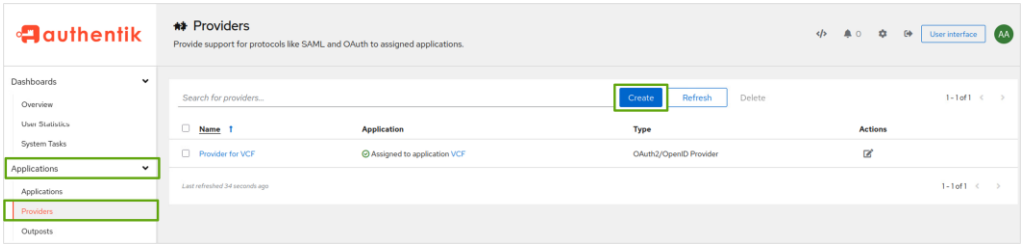
23. Click **CONTINUE**.



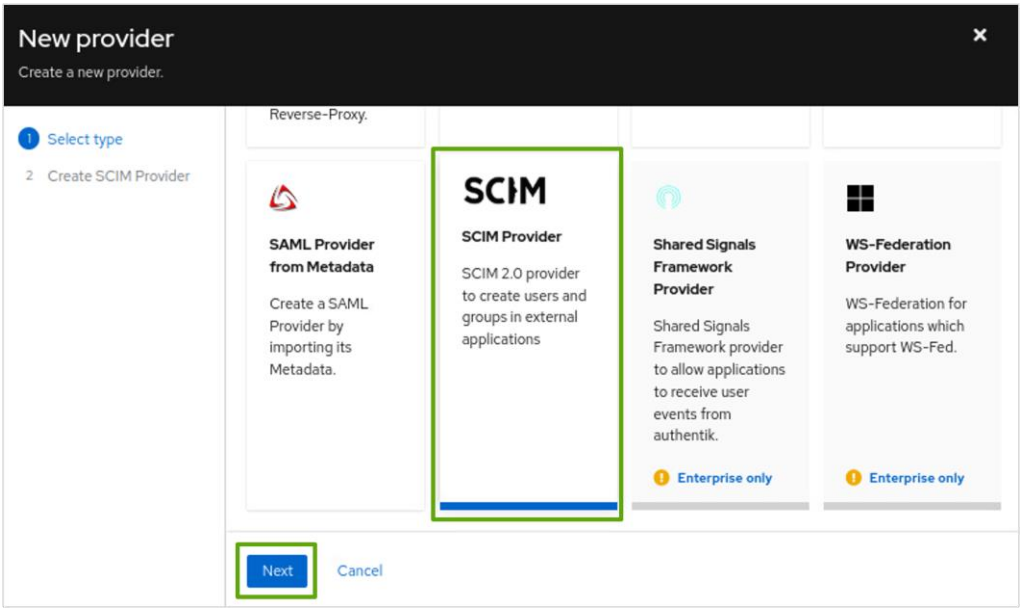
Task 4: Configure SCIM in Authentik

You configure SCIM Provider in authentik.

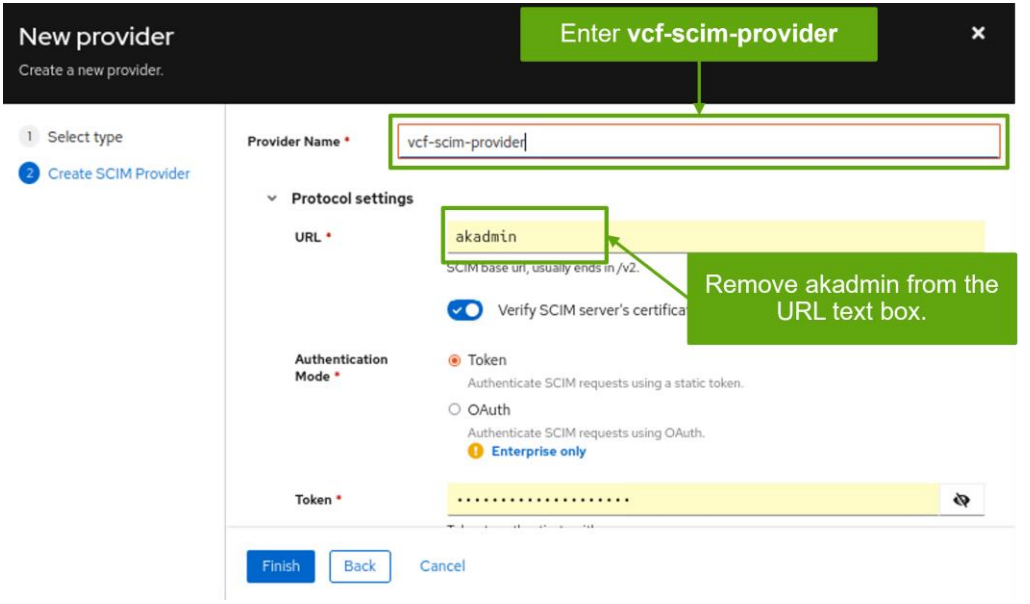
1. In Firefox, click **authentik** tab**.**
2. Click **Applications > Providers** from the navigation pane on the left-hand side and click **Create**.



3. Select **SCIM Provider** on the New Provider wizard and click **Next**.

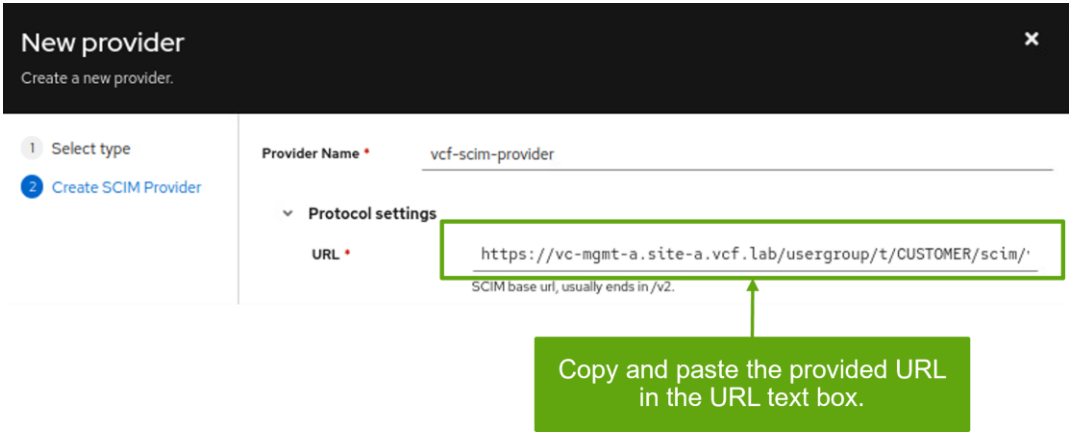


4. On the Create SCIM Provider wizard, enter the configuration details:
   - **Provider Name**: vcf-scim-provider
   - **URL**: Remove `akadmin` from the URL textbox.

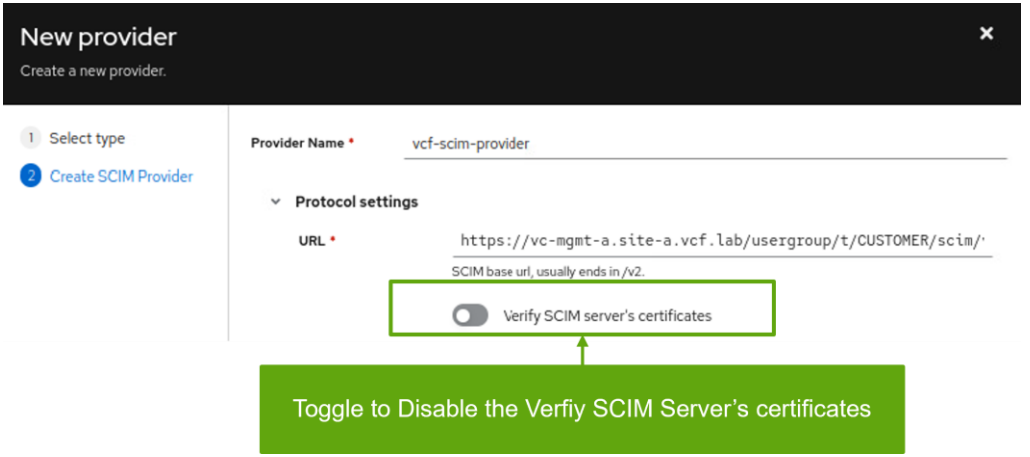


5. Copy and paste the following URL in the URL text box.

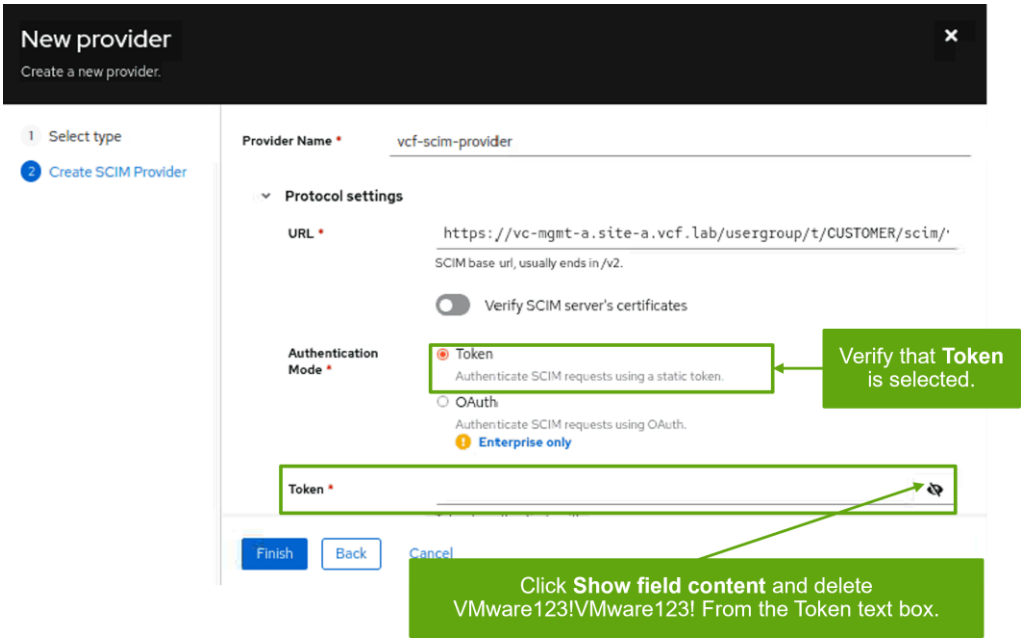
```
https://vc-mgmt-a.site-
a.vcf.lab/usergroup/t/CUSTOMER/scim/v2
```



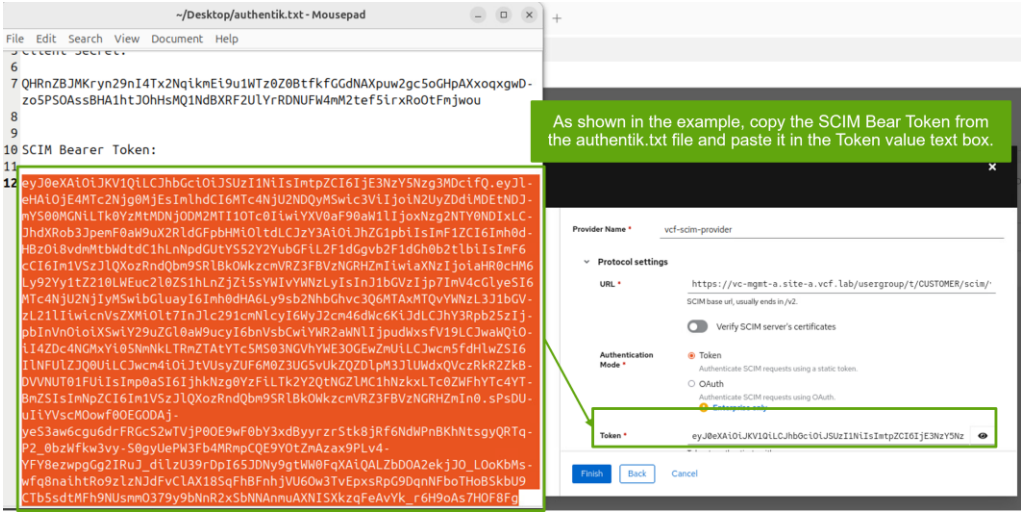
6. Toggle to **Disable** Verify SCIM Server's certificates.



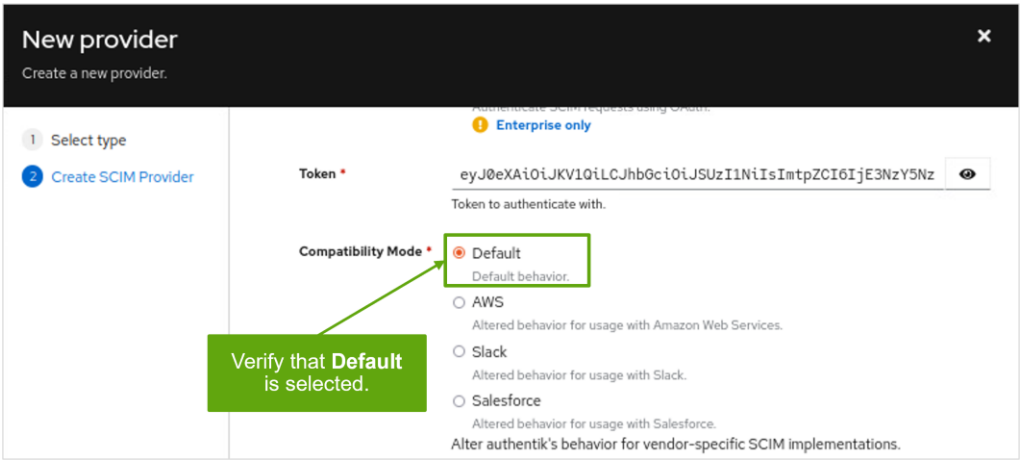
7. Verify that **Token** is selected for **Authentication Mode**, click **Show Field Content** and delete the `VMware123!VMware123!` from the **Token** field.



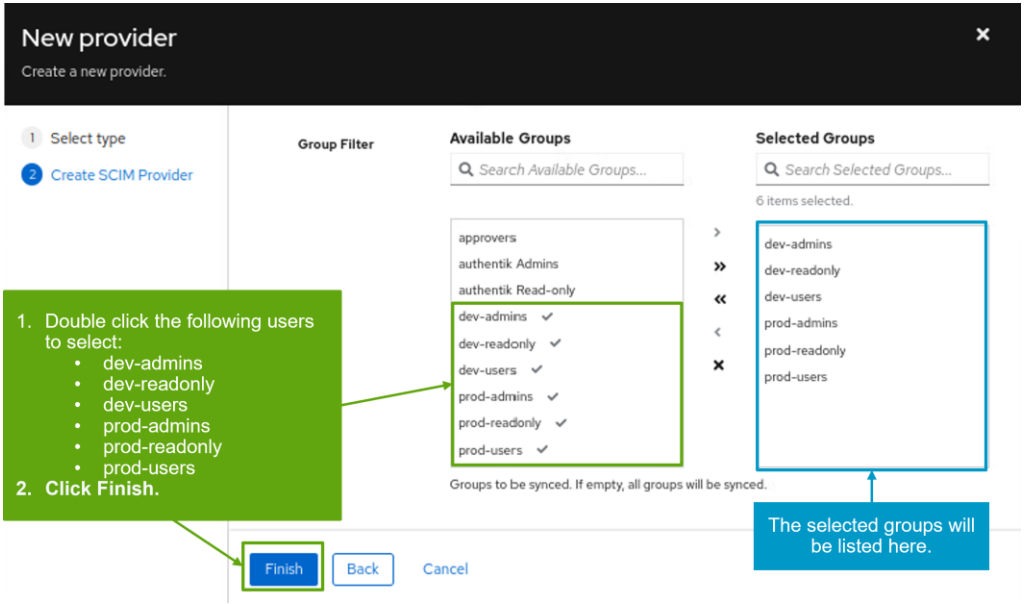
8. Copy SCIM Bearer token from the authentik.txt file and paste the value in Token text box.



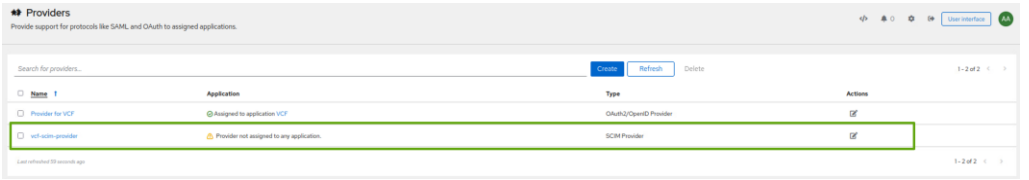
9. Verify that Default is selected for the Compatibility mode.



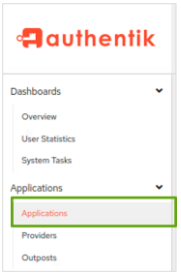
10. Double click to select the following groups and click **Finish**.
    dev-admins
    dev-readonly
    dev-users
    prod-admins
    prod-readonly
    prod-users



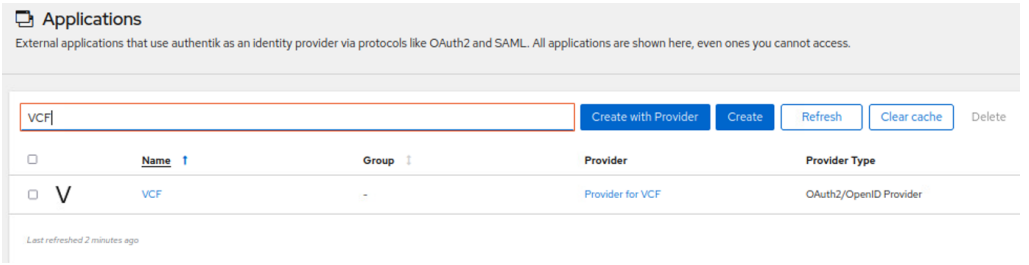
11. You are back to Provider page in authentik, verify that the vcf-scim-provider is listed.



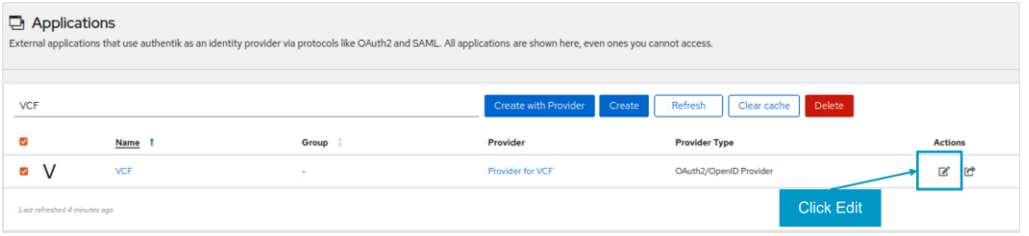
12. Click **Applications > Applications** from the navigation pane on the left-hand side.



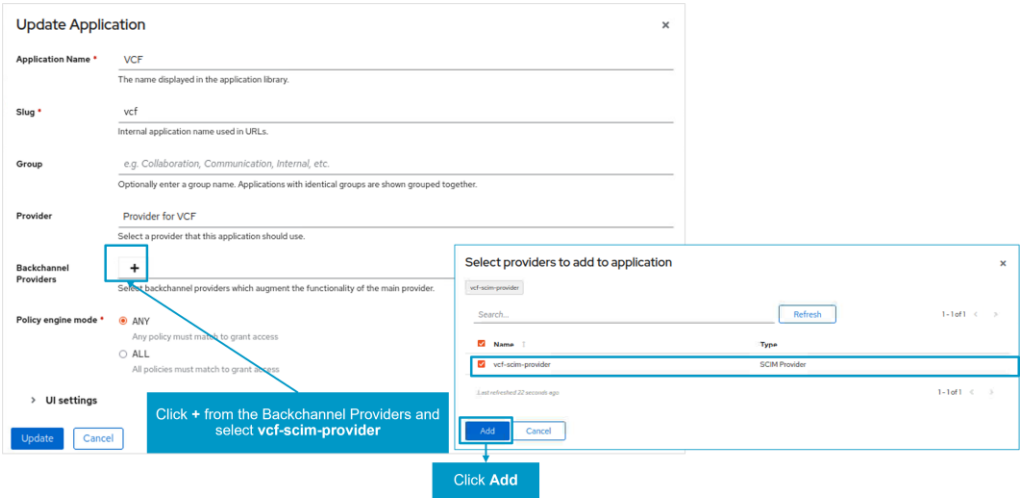
13. On the **Applications** page, type **VCF** in the **Search** text box and press **Enter**.



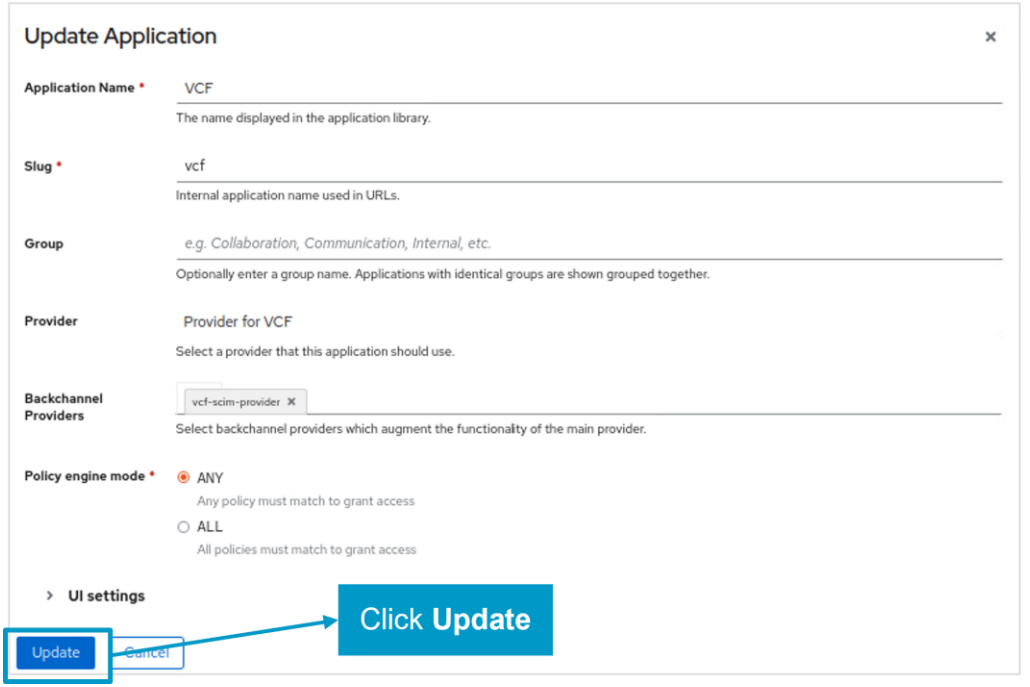
14. Click **Edit** from actions column to edit VCF application.



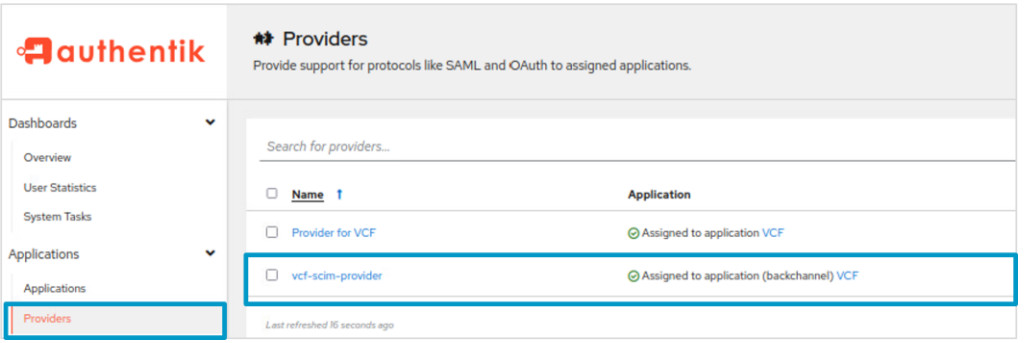
15. Click + from the Backchannel Providers text box and select **vcf-scim-provider** checkbox and click **Add**.



16. Click **Update**.



17. Click **Applications > Providers** from the navigation pane on the left-hand side and click **vcf-scim-provider.**



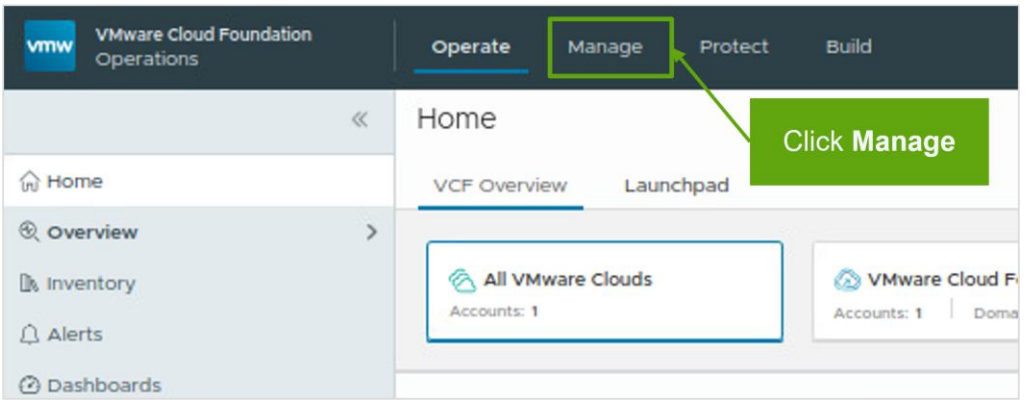
18. Scroll down to Schedules section and click the **play** button under the Actions column to run a Full sync for SCIM provider.



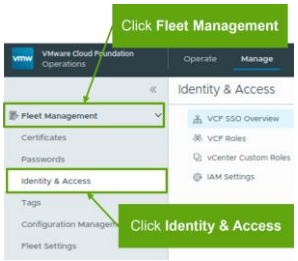
Task 5: Enable Single Sign-On on Management Domain vCenter.

You enable and configure the SSO on Management Domain vCenter.

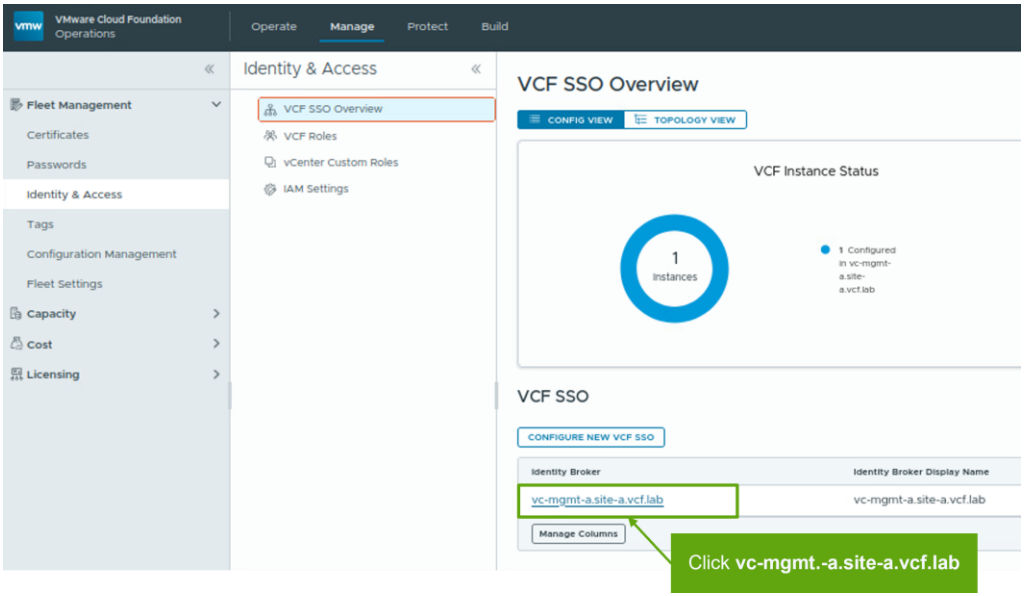
1. In VMware Cloud Foundation Operations, click **Manage** tab.



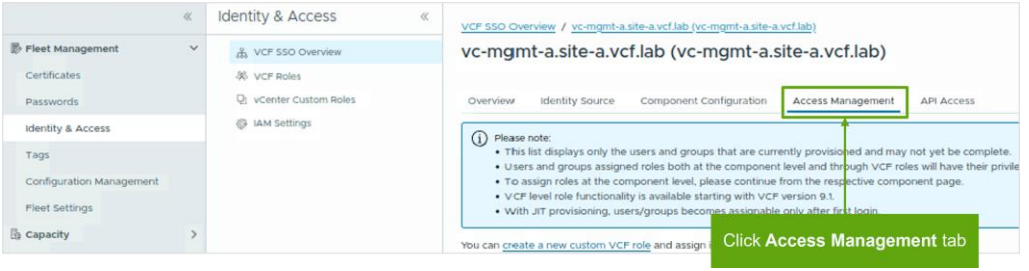
2. Click **Fleet Management > Identity & Access.**



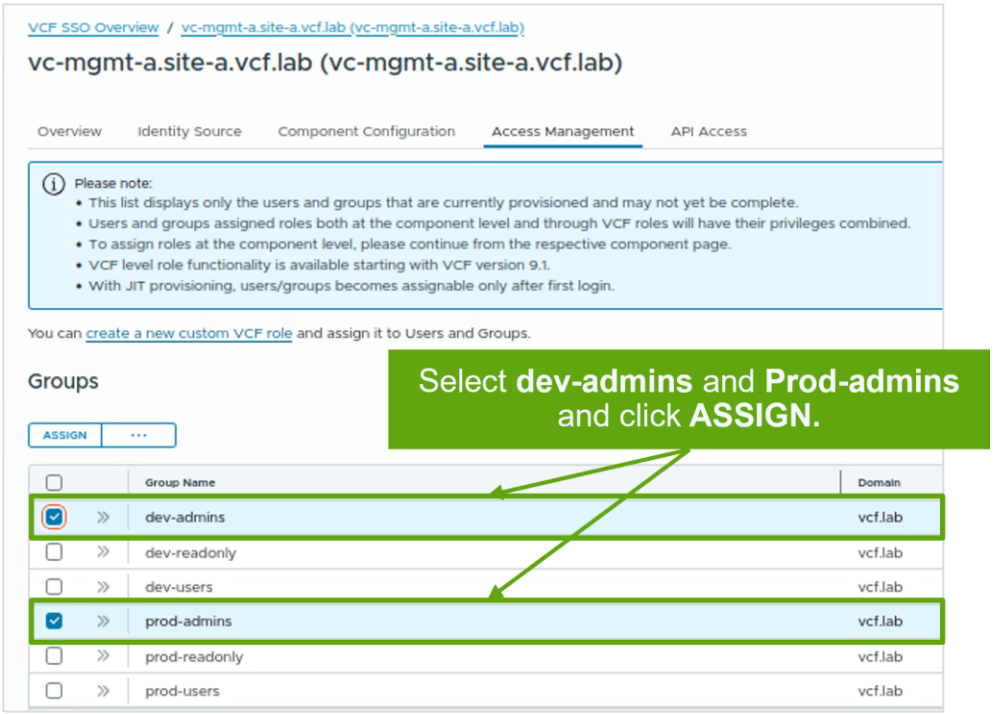
3. Under **VCF SSO Overview,** click **vc-mgmt-a.site-a.vcf.lab** link.



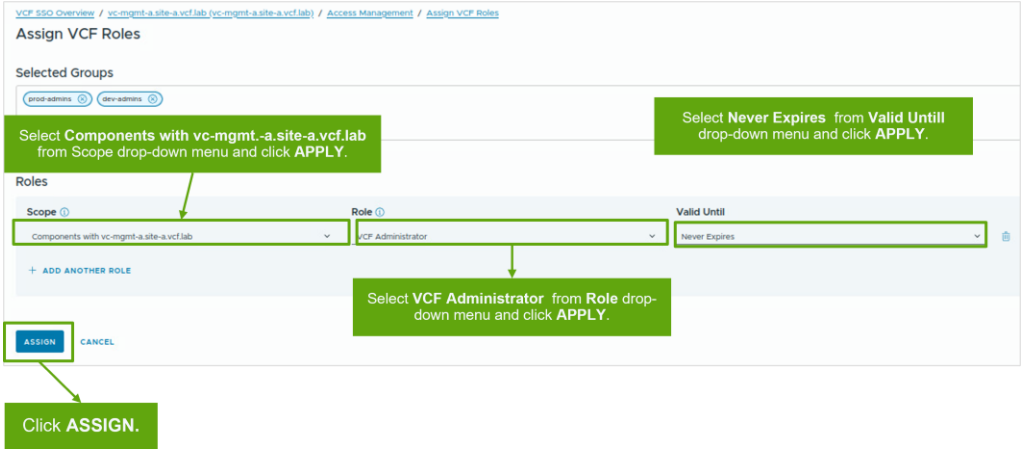
4. Click **Access Management** tab.



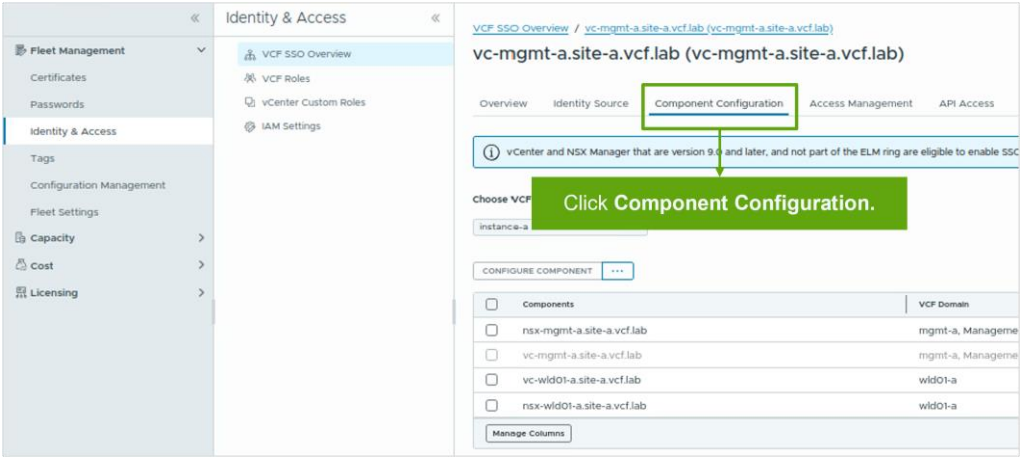
5. Select **Prod-admins** and **dev-admins** checkbox and click **ASSIGN**.



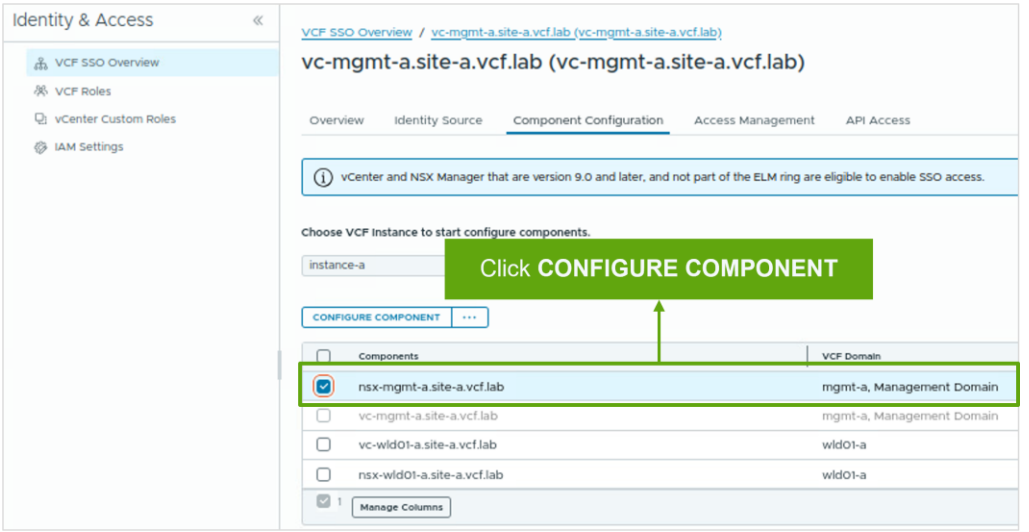
6. Select the following options under the Roles and click **ASSIGN**:
   - **Scope**: Select **Components with vc-mgmt-a.site-a.vcf.lab** and click **APPLY**.
   - **Role**: Select **VCF Administrator** and click **APPLY**.
   - **Valid Until**: Select **Never Expires**



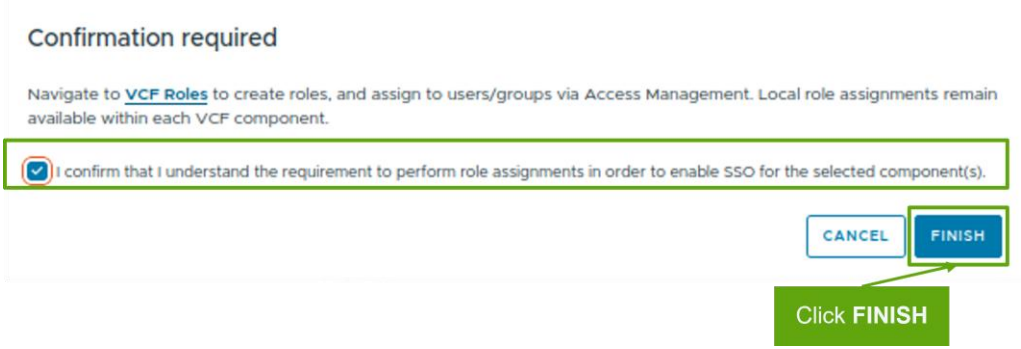
7. Click **Component Configuration** tab.



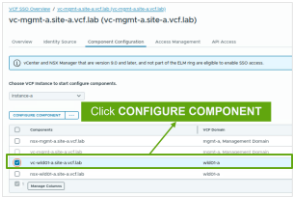
8. Select **nsx-mgmt-a.site-a.vcf.lab** checkbox and click **CONFIGURE COMPONENT**.



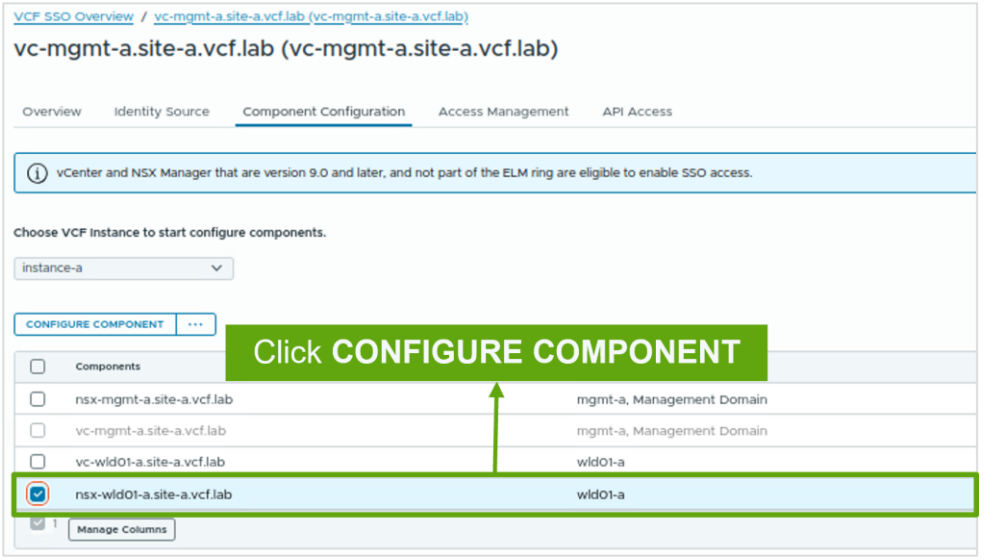
9. Select the **I confirm that I understand the requirement to perform role assignments in order to enable SSO for the selected component(s)** checkbox, and click **FINISH.**



10. Verify that the Status is **Configured MM/DD/YYYY** for the nsx-mgmt-a.site-a.vcf.lab.
11. Select **vc-wld01-a.site-a.vcf.lab** checkbox and click **CONFIGURE COMPONENT**.

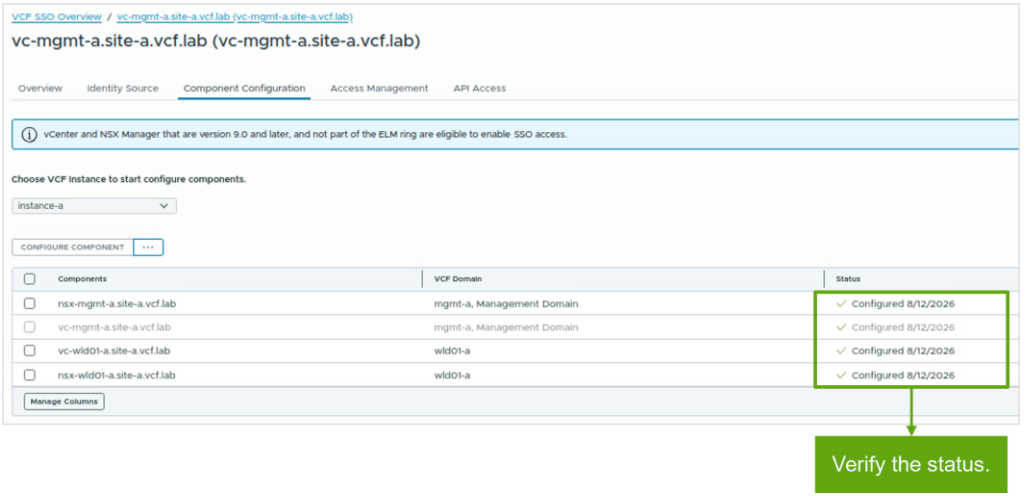


12. Select the **I confirm that the requirement to perform role assignments in order to enable SSO for the selected components** checkbox and click **FINISH**.
13. Verify that the Status is **Configured MM/DD/YYYY** for the nsx-wld01-a.site-a.vcf.lab.
14. Select **nsx-wld01-a.site-a.vcf.lab** checkbox and click **CONFIGURE COMPONENT**.

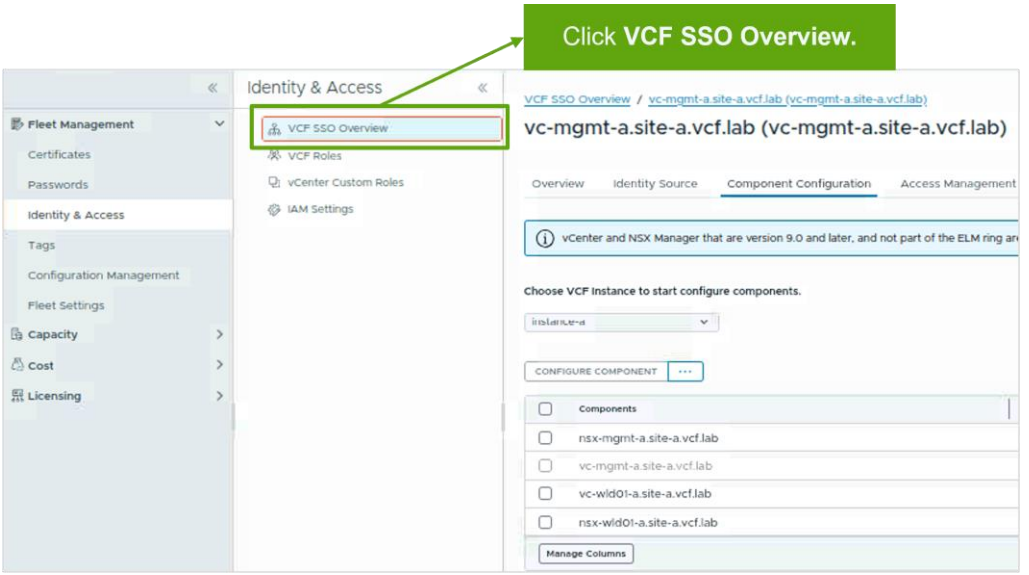


15. Select the **I confirm that the requirement to perform role assignments in order to enable SSO for the selected components** checkbox and click **FINISH**.

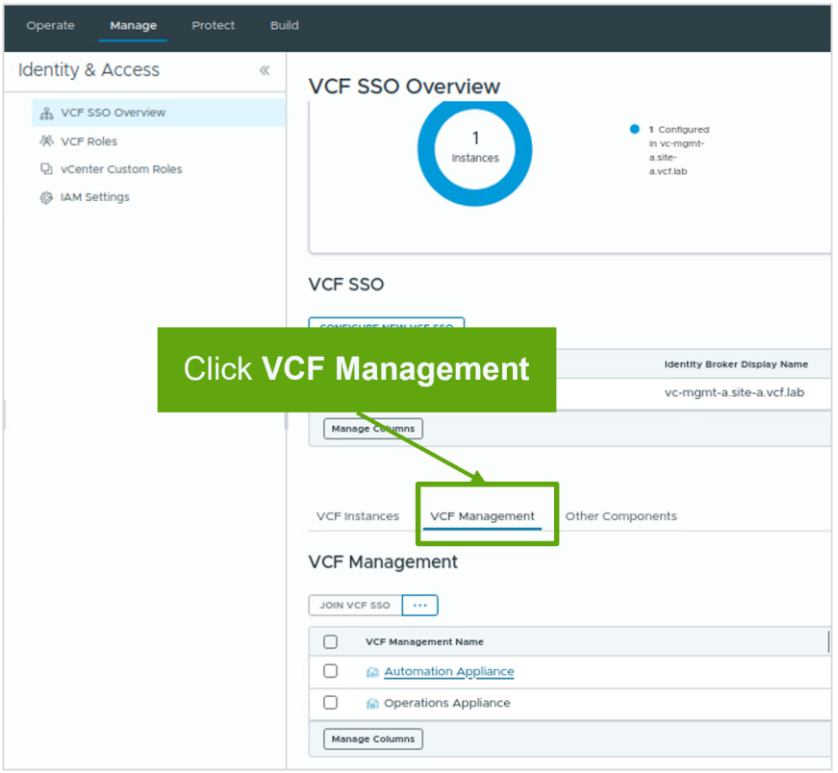
16. Verify that the Status is **Configured MM/DD/YYYY** for the nsx-wld01-a.site-a.vcf.lab.



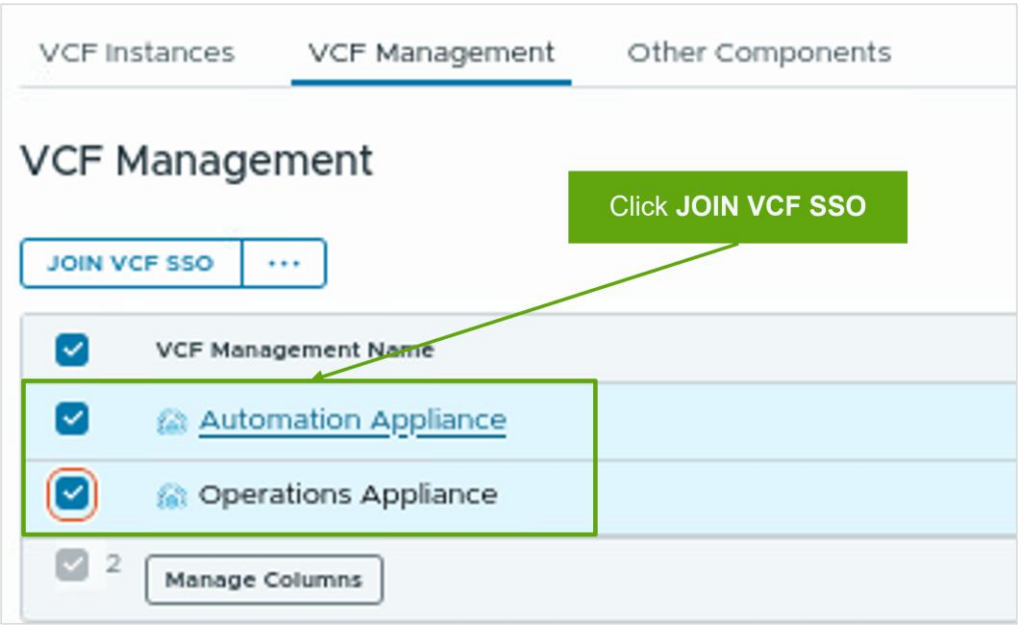
17. Click **VCF SSO Overview** tab.



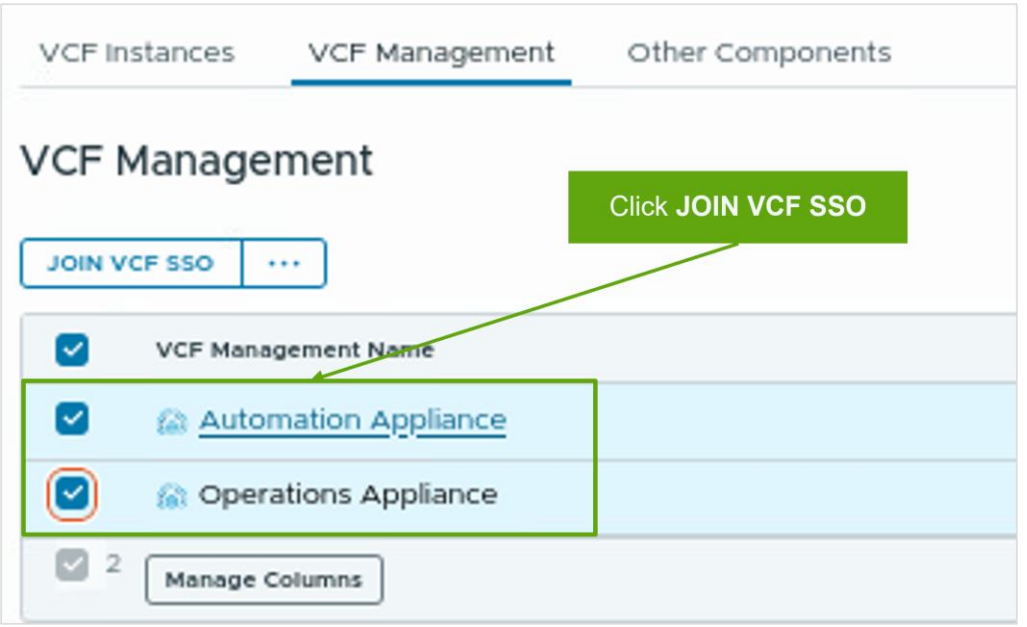
18. Click **VCF Management** tab.



19. Select **Automation Appliance** and **Operation Appliance** checkbox and click **JOIN VCF SSO**.

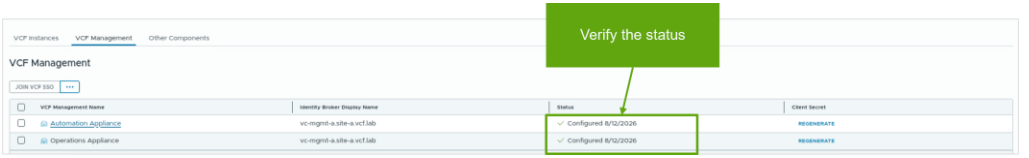


20. Under Choose SSO, select **vc-mgmt-a.site-a.vcf.lab** and click **JOIN**.



21. Select the **I confirm that the requirement to perform role assignments in order to enable SSO for the selected components** checkbox and click **FINISH**.

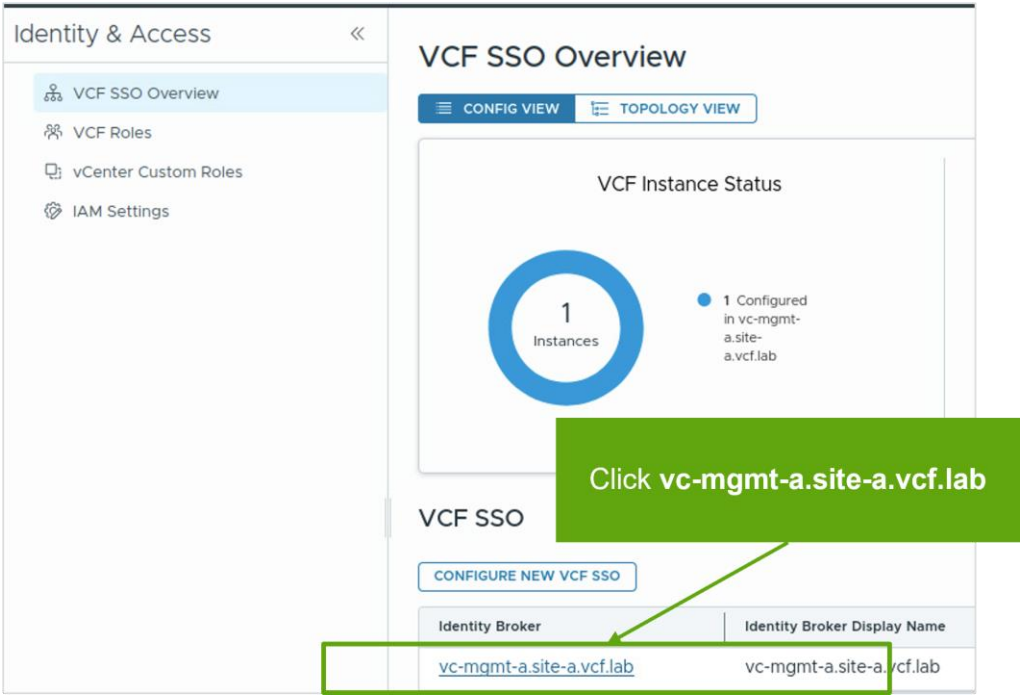
22. Verify that the Status is **Configured MM/DD/YYYY** for the auto-a-a.site-a.vcf.lab.



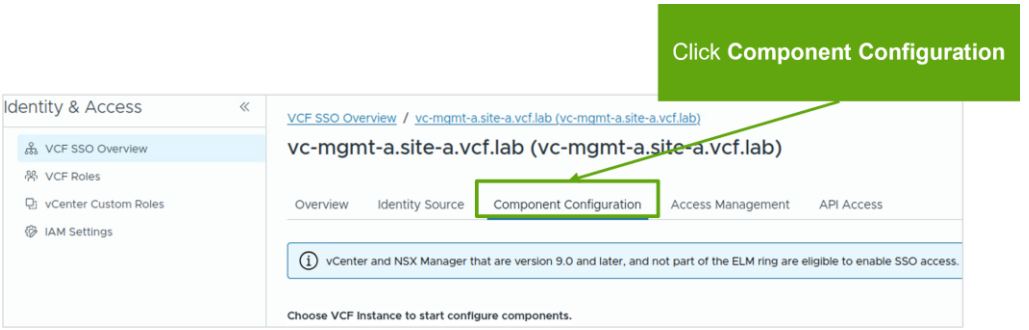
Task 6: Verify the Integration of VCF Instances with SSO

You log in to vCenter and NSX Manager to verify the SSO integration.

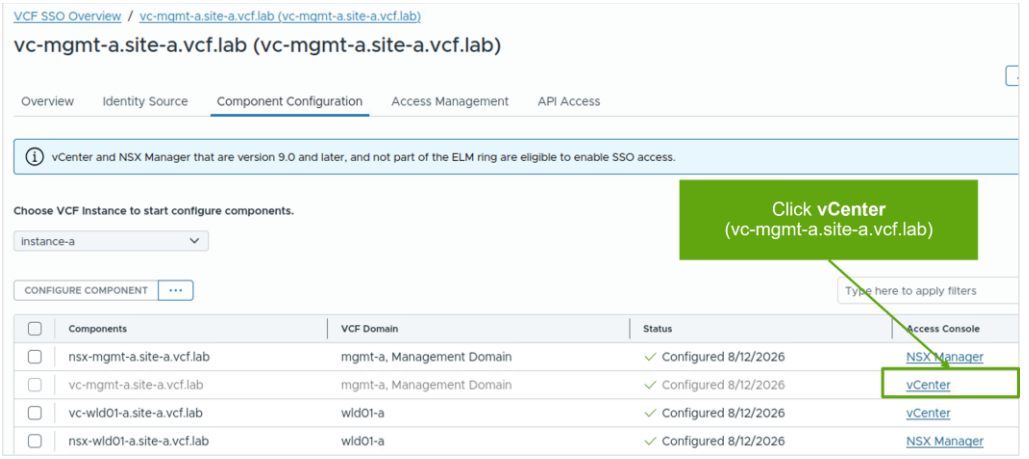
1. From the **VCF SSO Overview** menu, click **vc-mgmt-a.site-a.vcf.lab** under VCF SSO**.**



2. Click C**omponent Configuration** tab.



3. Click the **vCenter** link next to vc-mgmt-a.site-a.vcf.lab to open the vSphere Client.

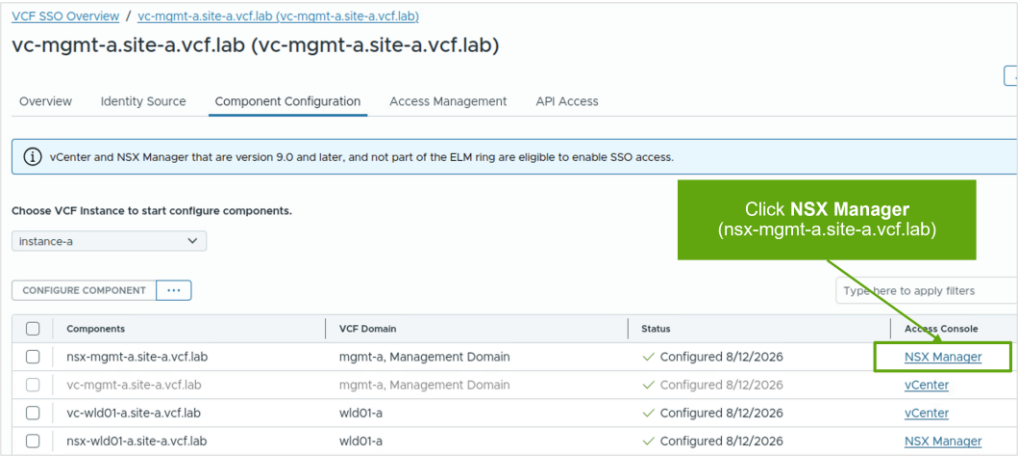


A new tab opens in Firefox and connects you to the vSphere Client and logs you in as administrator@vsphere.local user.

4. On the vSphere Client login page, select **Local Account** as the login method.
   - User name: administrator@vsphere.local
   - Password: VMware123!VMware123!
5. From the **vSphere Client** menu, select **Administration.**
6. Under Single Sign On, click **Configuration**.
7. On the **Identity Provider** tab, verify that a message appears to confirm that vCenter is integrated with VCF SSO.

   ```
   VCF SSO is configured through the VCF Identity Broker. To
   manage Identity provider configuration, log in to the VCF
   Operations Console.
   ```

8. Click the **administrator@vsphere.local** user name at the top right-corner of the vSphere Client and select **Logout**.
    a. On the **vSphere Client login** page, verify that **VCF SSO** appears in the **Login Method** drop-down menu.
    b. Close the **vSphere Client** tab.
9. Click on the **VCF Operations** tab in Firefox.
10. On the **Component Configuration** tab, click the **NSX Manager** link next to nsx-mgmt-a.site-a.vcf.lab to log in to NSX Manager.



11. On the **NSX Manager** login page, verify that **VCF SSO** is listed in the **Login Method** drop-down menu.
12. From the **Login Method** drop-down menu, select **Local Account.**
    a. Log in to NSX Manager.
        - User name: admin
        - Password: VMware123!VMware123!
13. In the NSX UI home page, click **System > Settings > User Management**.
14. Under User Management, click **Authentication Providers.**
    a. Under VCF SSO, verify that the following message appears.

    ```
    VCF SSO is configured through VCF Identity Broker. To
    manage the identity source, go to the VCF Operations
    console.
    ```

    b. In the VCF SSO section, verify the VCF SSO configuration and connection status.

Option	Value
Identity Provider Name	VCF SSO
Identity Broker Issuer	https://vc-mgmt-a.site-a.vcf.lab:443/acs/t/CUSTOMER/.well-known/openid-configuration
Connection to Identity Broker	Successful

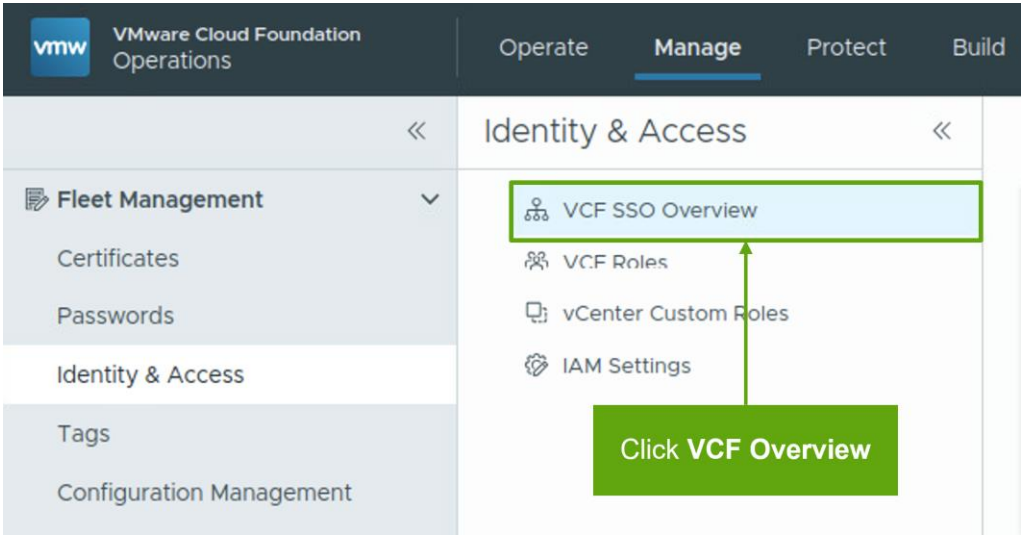
c. From the top-right corner of NSX Manager, click the **admin** user and click **Logout**.

d. Close the **NSX Manager** tab.

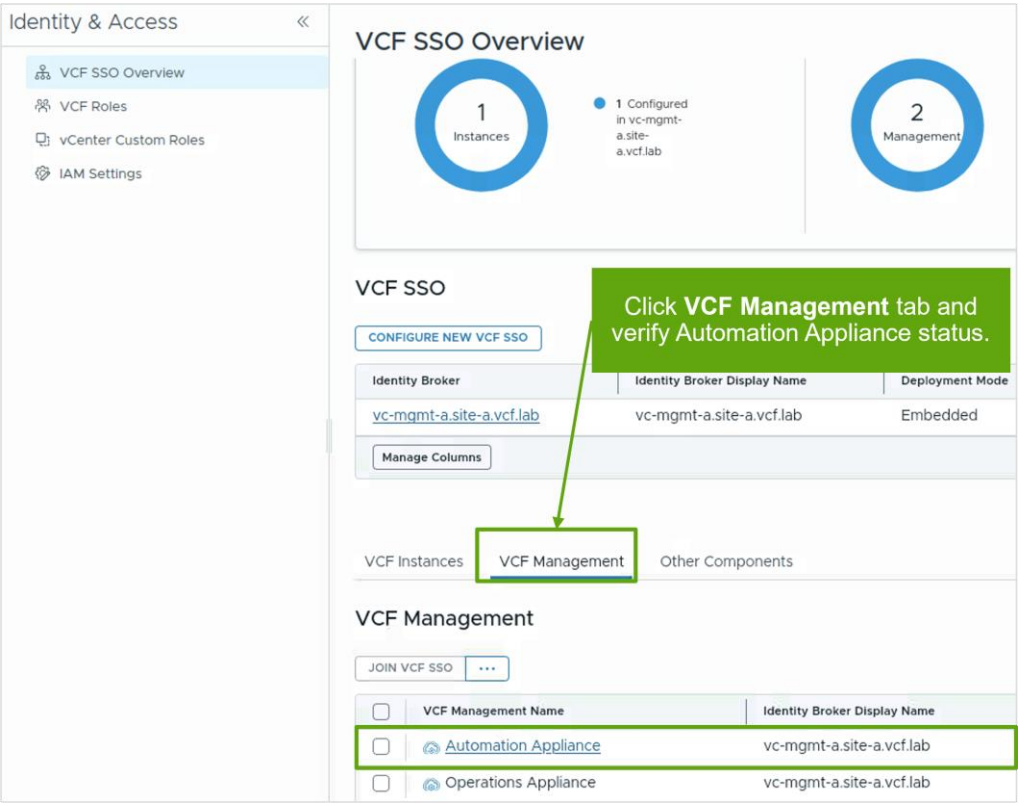
Task 7: Verify VCF Automation and VCF Operation Integration with VCF SSO

You log in to the VCF Automation as prod-admin@vcf.lab user.

1. In the VCF Operations UI, expand **Fleet Management > Identity & Access**.
2. Click **VCF SSO Overview**.



3. Click **VCF Management** tab and select **Automation Appliance**.



Verify that the Status is Configured for Automation Appliance and Operations Appliance

4. Click **Automation Appliance** hyperlink.

   It opens an Automation Appliance Login UI and redirects automatically to auth.vcf.lab page for single sign-on login.

5. Click **Continue** on the authentik Redirecting to VCF page.

6. On the welcome to **authentik page , login to continue to vcf.**

   - Email or Username: prod-admin@vcf.lab
   - Password: VMware123!VMware123!

   Click **Save** on the Save Password for vcf.lab

   a. Click **Continue** on the Redirecting to vcf page on the authentik.

   b. You are now logged into VCF automation as a **prod-admin@vcf.lab** SSO user.

7. Click **prod-admin@vcf.lab** from the top-right corner and click **Log Out**.

8. Click **X** to close the VCF Automation tab.

9. In the VMware Cloud Foundation Operations page, click **admin** user drop-down menu and select **Logout**.
10. On the VMware Cloud Foundation Operations login page, verify that **VCF SSO** is selected and click **LOG IN**.

    The SSO will automatically log you in to VCF Operations.
11. Click **prod-admin@vcf.lab** from the top-right corner and click **Log Out**.
12. **Close** the Firefox browser.

Lab 8 Assigning Permissions to VCF SSO Users

Objective and Tasks

Configure VCF SSO as an identity source and configure appropriate roles for users in vSphere and SDDC Manager:

1. Prepare for the Lab
2. Import an SSO User and Assign Permissions to VCF Operations
3. Add VCF SSO Users to vCenter
4. Add VCF SSO Users to NSX Manager
5. Verify VCF SSO Access to VCF Fleet

Task 1: Prepare for the Lab

You log in to the vSphere Client and SDDC Manager.

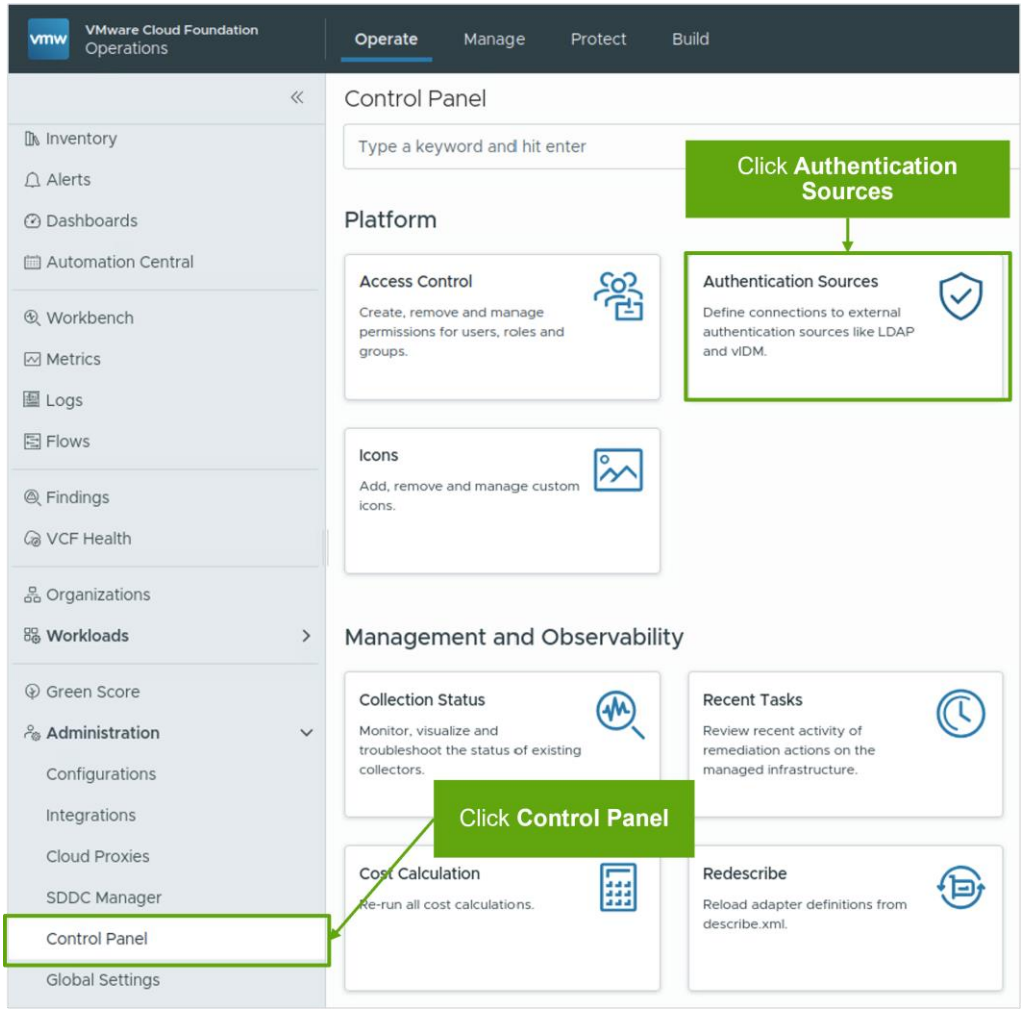
1. Log in to the vSphere Client.
    a. Open **Firefox**.
    b. On the bookmarks bar, click **Region A > vc-mgmt-a Client**.
    c. Log in to the vSphere Client with **Local Account** as the login method.
        - User name: administrator@vsphere.local
        - Password: VMware123!VMware123!
2. Log in to VCF Operations.
    a. Open a **new tab** in Firefox.
    b. On the bookmarks bar, click **Region A > VMware Cloud Foundation Operations**.

c. Log in to VCF Operations with **Local Account** as the login method.
    - User name: admin
    - Password: VMware123!VMware123!

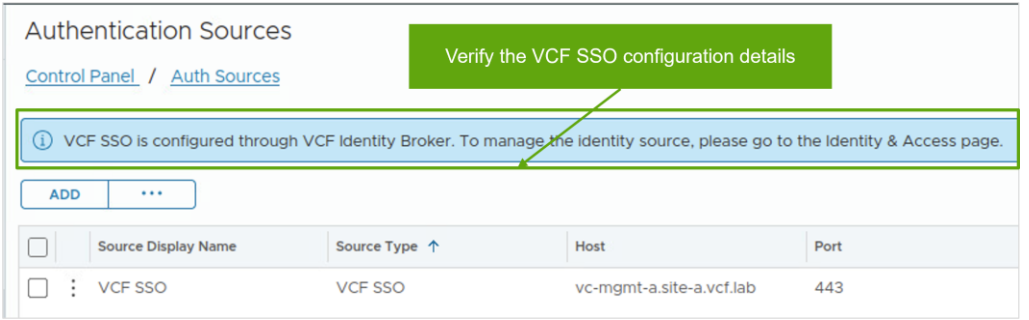
Task 2: Import an SSO User and Assign Permissions to VCF Operations

You import an SSO user and assign permissions to VCF Operations.

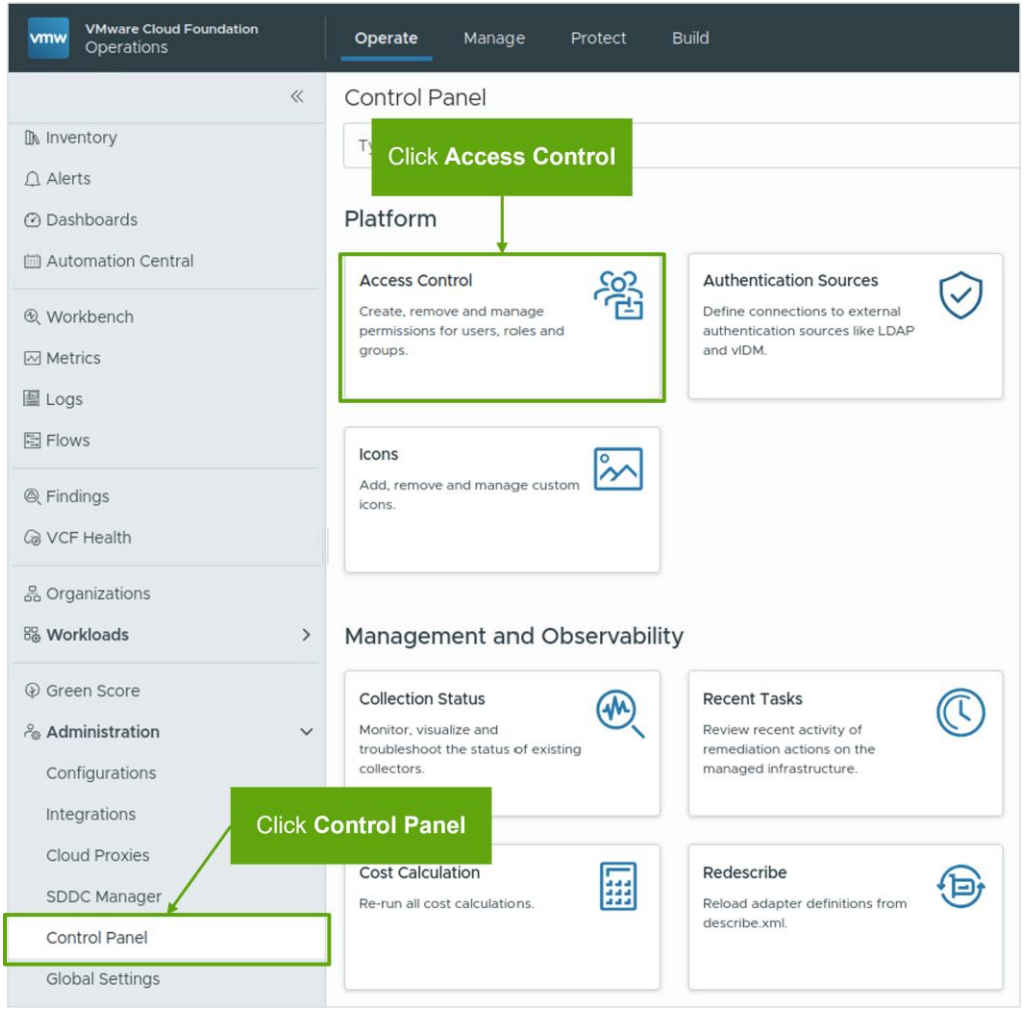
1. In the VCF Operations UI, click **Operate** tab.
2. Click **Administration > Control Panel > Authentication Sources**.



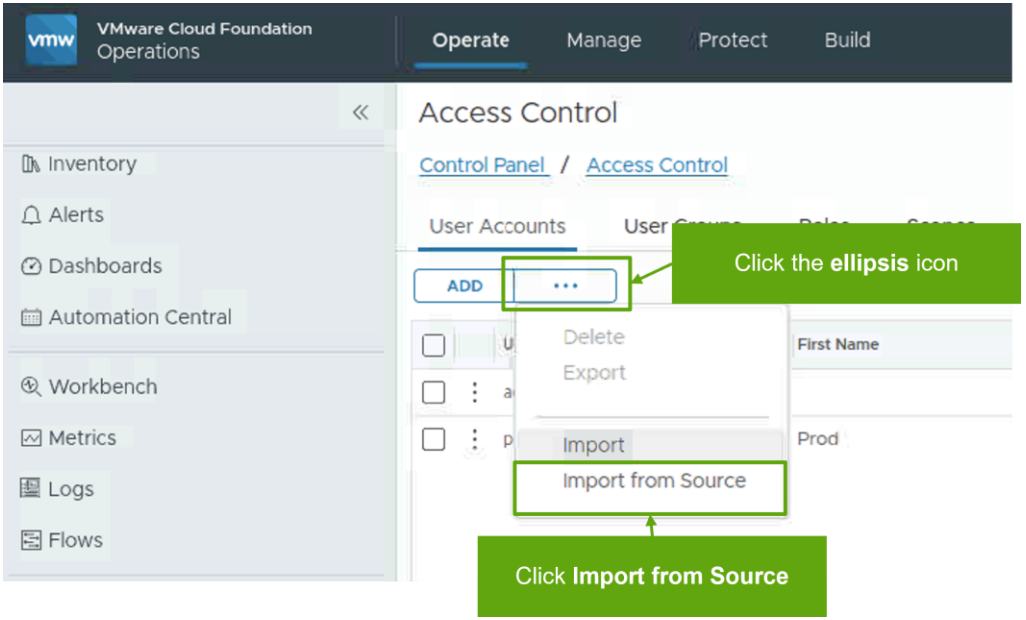
3. Verify that the VCF SSO is configured through VCF Identity Broker.



4. Click **Control Panel > Access Control**.

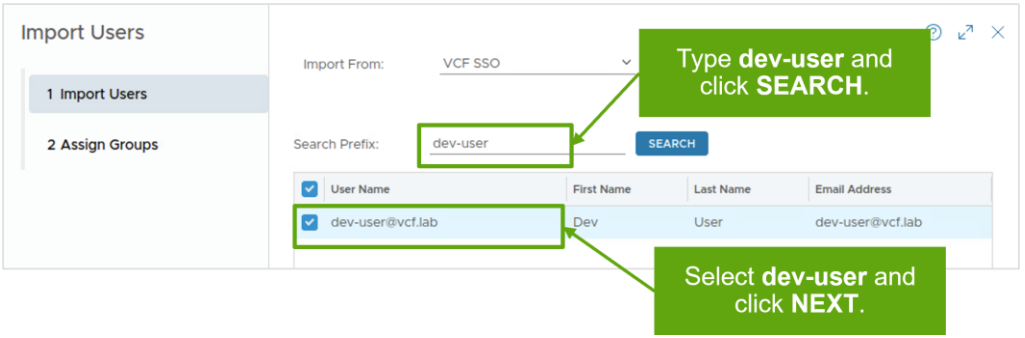


5. On the **User Accounts** tab, click the **ellipsis** icon and click **Import from Source.**

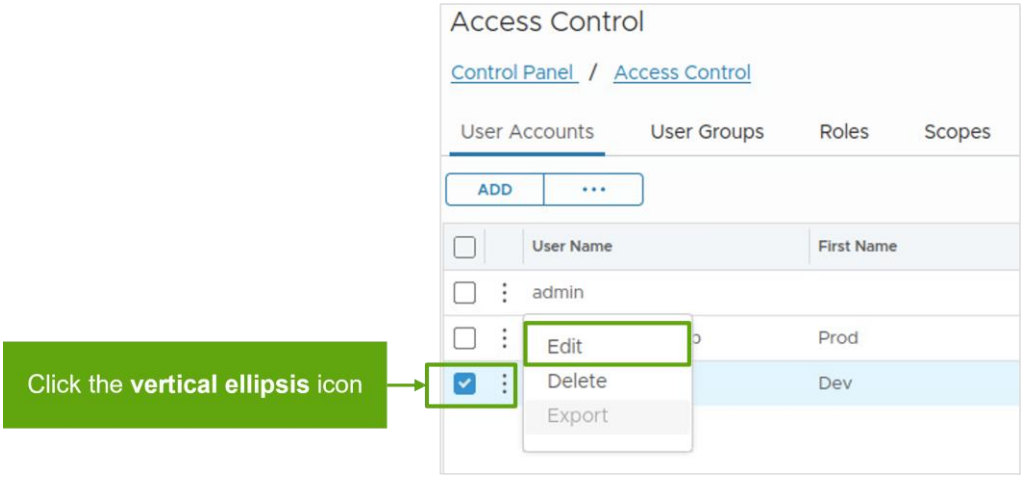


6. In the **search prefix** text box, enter **`dev-user`** and click **SEARCH**.

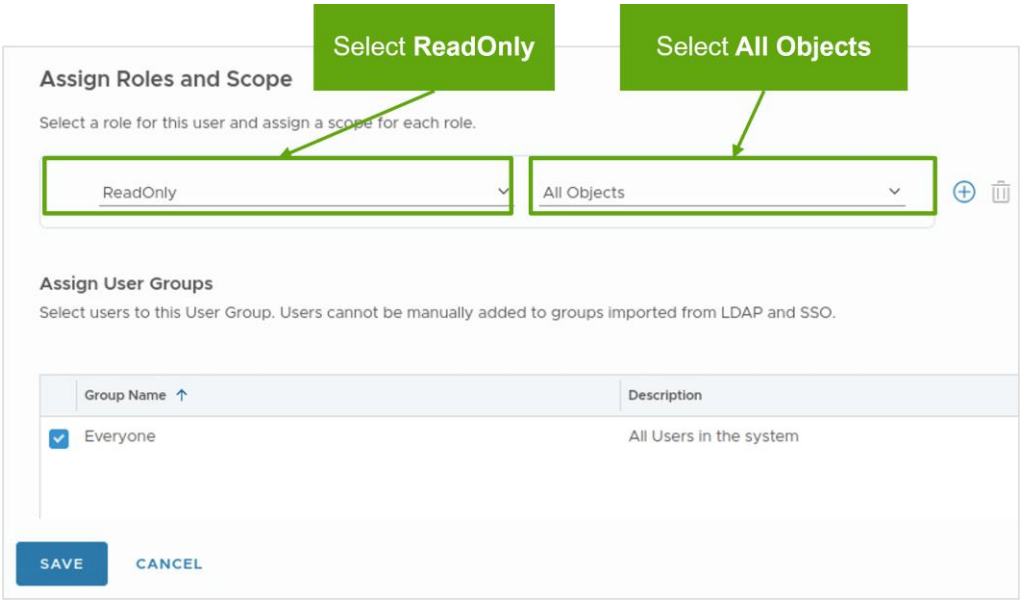
   Select **dev-user@vcf.lab** from the search result and click **NEXT**.



7. Click **FINISH**.
8. From User Accounts, verify that the source type is **VCF SSO** for the **dev-user@vcf.lab** user.
9. From User Accounts, select **dev-user@vcf.lab**, click the vertical ellipsis icon, and click **EDIT.**



10. Under Assign Roles and Scope, select **ReadOnly** from the **Select Role** drop-down menu, and select **All Objects** from the **Select Scope** drop-down menu, and click **SAVE**.



11. Open a New Private window in Firefox.

    a. Click the vertical ellipsis icon at the top-right corner of Firefox and select **New private window.**

    b. Click the **Region A > VMware Cloud Foundation Operations** bookmark from the bookmark bar.

c. On the **VCF Operations Login** page, select **VCF SSO** as a login method and click **LOG IN**.

d. On the authentik log in page, log in to the VCF SSO.

- User name: dev-user@vcf.lab
- Password: VMware123!VMware123!

e. On the Redirecting to VCF page, click **Continue**.

12. Verify that you can log in to VCF Operations.
13. Click **Manage** tab.
14. Click **Fleet Management > Configuration Management**.
15. Click **Configuration Templates** tab.
16. Click **Create vCenter Configuration Template** link.

    You will see the **Access Denied** error because the dev-user@vcf.lab has Read-Only access and cannot create or modify new templates.
17. Close the **Private browsing** window.

Task 3: Add VCF SSO Users to vCenter

You configure the permissions for the dev-user@vcf.lab account in vSphere.

1. In Firefox, click the **vSphere Client** tab.
2. In the vSphere Client, select **Administration** from the main menu in the top-left corner.
3. In the navigation pane on the left, select **Access Control > Global Permissions**.
4. Add the dev-user user as an read-only user..
   a. On the **Global Permissions** page in the right pane, click **ADD**.
   b. In the **Add Permission** dialog box, select **vcf.lab** from the **Domain** drop-down menu.
   c. In the **User/Group** search box, enter **dev** and select **dev-user** from the results.
   d. In the **Role** drop-down menu, leave **Read-only** selected.
   e. Select the **Propagate to children** checkbox and click **OK**.
   f. Click the refresh icon at the top of the vSphere Client and verify that `VCF.LAB\dev-user` appears in the Global Permissions list with the role of Read-only.
5. Click the **Administrator@VSPHERE.LOCAL** user and select **Logout**.
6. Log in to the vSphere Client as dev-user.

a. On the **vSphere Client** login page, verify that the **VCF SSO** is selected as the login method and click **LOG IN**.

- Email or Username: dev-user@vcf.lab
- Password: VMware123!VMware123!

Click **Save** on the S**ave Password for vcf.lab?** pop up window.

b. In the **Hosts and Clusters** inventory list in the left navigation pane, expand **vc-mgmt-a.site-a.vcf.lab** to see the hosts and resource pools.

c. Expand **dc-a**.

d. Expand **cluster-mgmt-01a**.

7. Open a new tab in Firefox.
8. Click **Region A > SDDC Manager** on the bookmarks bar to try to log in to SDDC Manager as dev-user.
   - User name: dev-user@vcf.lab
   - Password: VMware123!VMware123!

   Q1. Can you access SDDC Manager with the dev-user account?
9. Click **X** to close the vSphere login page that you opened to log in as dev-user.

Task 4: Add VCF SSO Users to NSX Manager

You add VCF SSO user to NSX Manager.

1. Launch NSX Manager in a new tab.
   a. Click + to open a new tab.
   b. On the Firefox bookmarks bar, click **Region A > nsx-mgmt-01a**.
   c. Log in to NSX Manager with **Local Account** as the login method.
      - User name: admin
      - Password: VMware123!VMware123!
2. From the NSX Manager home page, click **System > Settings > User Management.**
3. Under User Role Assignment, click **ADD ROLE FOR VCF SSO USER/GROUP**.
4. In the **VCF SSO** text box, enter **`dev-user`** and select the **dev-user@vcf.lab** user.
   a. Under Roles, click **Set**.
   b. Click **ADD ROLE** and select **Enterprise Admin.**

c. Click **ADD**.

d. Click **APPLY**.

e. Click **SAVE**.

f. Under User Role Assignment, verify that **dev-user@vcf.lab** has the **Enterprise Admin** role and is of the **VCF SSO** user type.

5. Open a New private window and click **Region A > nsx-mgmt-01a** from the bookmarks.

   a. From the Login Method, select **VCF SSO** and click **LOG IN**.

      The authentik login screen appears.

   b. Log in to NSX Manager.

      - User name: dev-user@vcf.lab
      - Password: VMware123!VMware123!

   c. On the Notifications dialog box, click **Clear All** to clear any alarm notifications.

   d. From the top-right side of NSX Manager, click the **dev-user@vcf.lab** user and click **Log Out**.

   e. **Close** the private browsing window.

Task 5: Verify VCF SSO Access to VCF Fleet

You verify VCF SSO user access to VCF Fleet.

1. Open a **New private window** in Firefox.
2. Click **Region A > VMware Cloud Foundation Operations** from the bookmarks bar.

   a. Select **VCF SSO** from Login Method and click **LOGIN**.

      The authentik log in screen appears.

   b. Log in to VCF Operations:

      - User name: dev-user@vcf.lab
      - Password: VMware123!VMware123!

3. Open a new tab and click **Region A > vc-mgmt-a Client.**

   a. Select **VCF SSO** as Login Method and click **LOG IN**.

   b. Verify that you can log in to vCenter without providing any user credentials.

4. Open a new tab and click **Region A> nsx-mgmt-01a** from the bookmarks.

   a. Select **VCF SSO** as Login Method and click **LOG IN.**

b. Verify that you can log in to NSX without providing any user credentials.

   Reach out to the instructor for help if you are prompted to enter user credentials.

5. Close the private browsing window.

Lab 9 Managing Passwords

Objective and Tasks

Rotate management domain ESX host passwords and create a file to document new passwords for components:

1. Prepare for the Lab
2. Rotate the ESX Host Passwords
3. Schedule Password Rotation for the vCenter Root Account
4. Re-Create a Primary Password File
5. Use the API to Create a Primary Password List
6. Managing Passwords from VCF Operations

Task 1: Prepare for the Lab

You log in to SDDC Manager.

1. Log in to the vSphere Client.
    a. Open Firefox.
    b. On the Firefox bookmarks bar, click **Region A > vc-mgmt-a Client**.
    c. Select **Local Account** as the login method and click **Login**.
        - User name: administrator@vsphere.local
        - Password: VMware123!VMware123!
2. Log in to VCF Operations.
    a. Open a new tab in Firefox.
    b. On the bookmarks bar, click **Region A** > **VMware Cloud Foundations Operations**.
    c. Select **Local Account** as the login method and click **Login**.
        - User name: admin

- Password: VMware123!VMware123!

3. Launch SDDC Manager in a new tab.
   a. On the bookmarks bar, right-click **SDDC Manager** and select **Open in new tab**.
   b. Click the **SDDC Manager** browser tab.
   c. (Optional) Log in to SDDC Manager.
      - User name: administrator@vsphere.local
      - Password: VMware123!VMware123!

Task 2: Rotate the ESX Host Passwords

You use SDDC Manager to rotate the passwords for all ESX hosts.

1. In Firefox, click the **SDDC Manager** tab.
2. In the navigation pane on the left, select **Security > Password Management**.
3. In the **Quick Filter** text box, enter **`root`** and select all the ESX hosts from the list.
4. Click **ROTATE PASSWORD**.
5. In the **Rotate Password** dialog box, click **ROTATE**.
6. Expand the **Tasks** pane to verify that the Credentials rotate operation completes successfully.
7. Close the **Tasks** pane.

Task 3: Schedule Password Rotation for the vCenter Root Account

You use SDDC Manager to schedule password rotation for the vCenter root account.

1. Enter **`root`** in the **Quick Filter** text box.

   The Quick Filter will list 10 components per page, so you might need to navigate to the next page to view the vCenter.
2. Select the checkbox where the resource type is vCenter.
3. Select the checkbox for the root account.
4. From the **SCHEDULE ROTATION** drop-down menu, select **90 DAYS**.
5. In the **Confirm changes** dialog box, click **YES**.
6. Expand the **Tasks** pane to verify that the Credentials update auto rotate policy operation completes successfully.
7. Close the **Tasks** pane.

Task 4: Re-Create a Primary Password File

You use the `lookup_passwords` command to create a password file for the lab environment.

1. Click the **Remmina Remote Desktop Client** icon from the Linux taskbar.



2. In the Remmina Remote Desktop Client window, double-click **SDDC Manager** to open an SSH session configured with an automatic login.
3. Perform the password lookup operation using the administrator@vsphere.local account.
   a. Run the `lookup_passwords` command.
   b. Enter **`ESXI`** at the `Enter an entity type from above list` prompt.
   c. At the `Enter page number (optional)` prompt, leave the value blank and press **Enter**.
   d. At the `Enter page size (optional, default=50)` prompt, leave the value blank and press **Enter**.
   e. Enter **`administrator@vsphere.local`** at the `Enter Username` prompt.
   f. Enter **`VMware123!VMware123!`** at the `Enter password` prompt.

      The command output lists the ESX hosts, user name, and password details. Review the passwords from the command output.
4. Run the `exit` command to close the SSH session to SDDC Manager.
5. Close the **Remmina** window.

Task 5: Use the API to Create a Primary Password List

You create a primary password file containing all passwords in JSON format.

1. Click the **Firefox** icon on the taskbar.
2. Click the **VCF Operations** tab in Firefox.
3. Click the **Build** tab in the VCF Operations.
4. Click **Developer Center** from the navigation pane on the left.
5. Click **APIs & SDKs.**
6. Click **API Explorer** from the **SDDC Manager API tile.**
7. Under API Categories, expand **Credentials**.
8. Expand **GET /v1/credentials**.

9. In the **Try it out** section, enter **`ESXI`** in the **resourceType** text box.
10. Enter **`USER`** in the **accountType** text box.
11. Leave the other options blank and click **EXECUTE**.
12. Under Response, expand **PageOfCredential** to expand the link if it is not already expanded.
13. Click the down arrow icon next to PageOfCredential to download the user credentials JSON file.



The `PageofCredential.json` file is automatically downloaded to the `Downloads` folder.

14. Open the `PageOfCredential.json` file in Text Editor and verify that it contains all ESX passwords.
15. Close the `PageOfCredential.json` file.

Task 6: Managing Passwords from VCF Operations

1. In VCF Operations, click the **Manage** tab.
2. Click **Fleet Management > Passwords**.
3. Under VCF Management, select the **VCF Automation** checkbox in the **Filters** box and click **APPLY**.
4. Under VCF Component, select **VCF Automation** for the **admin** user name and click **UPDATE PASSWORD**.
5. Enter the passwords in the **Update Password** dialog box and click **UPDATE PASSWORD**.
   - Current Password: VMware123!VMware123!
   - New Password: VMware1234!VMware1234!
   - Confirm Password: VMware1234!VMware1234!

   After the password is updated, the following message appears:
   `Password update for appliance auto-a.site-a.vcf.lab is successful. Changes may take some time to appear.`

Lab 10 (Simulation) Commissioning Hosts

Objective and Tasks

Use vCenter to commission hosts for a workload domain:

1. Prepare for the Lab
2. Configure a New Network Pool
3. Commission a Host
4. Bulk-Commission Additional Hosts

In the web browser in your local desktop, go to https://core-vmware.bravais.com/s/21796/Commissioning_Hosts to open the simulation.

Perform the following steps in the simulated environment. Refer to these steps as you work through the simulation.

IMPORTANT

Do not refresh, navigate away from, or minimize the browser tab hosting the simulation. Such actions might pause the simulation and prevent it from progressing.

NOTE

If you're stuck anywhere in the simulation, press the **TAB** key to reveal the next clickable area.

Task 1: Prepare for the Lab

You log in to the vSphere Client and VCF Operations.

1. Log in to the vSphere Client with the administrator account.
    a. Open Chrome.
    b. From the Chrome bookmarks bar, click **Region A > vc-mgmt-a Client**.
    c. Login to vSphere Client:
        - User name: administrator@vsphere.local
        - Password: VMware123!VMware123!
2. Log in to VCF Operations.
    a. Open a new tab in Chrome.
    b. On the bookmarks bar, click **Region A** > **VMware Cloud Foundation Operations**.
    c. Log in to VCF Operations:
        - User name: admin
        - Password: VMware123!VMware123!

Task 2: Configure a New Network Pool

You configure a new network pool for a virtual infrastructure (VI) workload domain.

IMPORTANT

In most environments, the MTU is set to 9000. In this lab environment, an MTU of 8000 is used to accommodate the additional packet headers because of nested networking.

---

1. In Chrome, click the **vSphere Client** tab.
2. From the vSphere Client menu, select **Global Inventory Lists**.
3. Click **Resources > Hosts.**
4. Click the **Network Pools** tab.
5. Click + **CREATE NETWORK POOL**.
6. Create a network pool.
    a. On the **Create Network Pool** page, enter **`wld-a-np01`** in the **Network Pool Name** text box.

b. Select the **vSAN** and **vMotion** checkboxes next to Network Type,

c. Under vSAN Network Information, configure the vSAN network pool to use the 10.1.5.0/24 network.

Option	Value
IP scheme	Select **IPv4**
VLAN ID	15
MTU	8000
Gateway (CIDR Notation)	10.1.4.1/25
Included IP Address Ranges	Enter **`10.1.4.2 to 10.1.4.20`** and click **ADD.**

d. Under vMotion Network Information, configure the vMotion network pool to use the 10.1.5.1/25 network.

Option	Value
IP Scheme	Select **IPV4**
VLAN ID	17
MTU	8000
Gateway (CIDR Notation)	10.1.5.1/25
Included IP Address Ranges	Enter **`10.1.5.2 to 10.1.5.20`** and click **ADD**.

7. Click **SAVE**.
8. On the **View Network Pool details** page, expand **wld-a-np01** to verify that the vSAN and vMotion network pools are configured correctly.

Task 3: Commission a Host

You use the management domain vCenter to commission the esx-05a.site-a.vcf.lab host.

1. Click the **Unassigned Hosts** tab.
2. On the **Hosts** page, click **COMMISSION HOSTS**.
3. In the **Checklist** dialog box, review the checklist criteria, select the **Select All** checkbox, and click **PROCEED**.
4. In the **Commission Hosts** wizard, configure a new host called esx-05a.site-a.vcf.lab.

a. Under Add Hosts on the **Host Addition and Validation** page, verify that **Add new** is selected.

b. Configure the options for the new esx-05a.site-a.vcf.lab host.

Option	Action
Host FQDN	Enter **`esx-05a.site-a.vcf.lab`** in the text box.
Storage Type	Click **vSAN**.
vSAN Type	Click **vSAN HCI**.
vSAN ESA	**Enabled**
Network Pool Name	**wld-a-np01** from the drop-down menu.
Username	Enter **`root`** in the text box and select root from drop-down menu.
Password	Enter `VMware123!VMware123!` in the text box.

c. Click **ADD**.

The host is added to the list of hosts under Hosts Added at the bottom of the wizard.

d. Under Hosts Added, turn on the **Confirm all Fingerprints** toggle.

The icon under Confirm Fingerprint becomes a solid green circle with a white check mark.

e. Click **VALIDATE ALL**.

The `Host Validated Successfully` message appears if the host is configured correctly.

f. After the validation completes, click **NEXT**.

5. On the **Review** page, click **COMMISSION**.
6. Expand the **Tasks** pane and monitor the Commissioning Host(s) task until it completes successfully.

   This task takes several minutes to complete. You must click **REFRESH** periodically to see the accurate progress status.
7. Close the **Tasks** pane.
8. On the **Hosts** page, click the **UNASSIGNED HOSTS** tab and verify that esx-05a.site-a.vcf.lab is listed.

Task 4: Bulk-Commission Additional Hosts

You use a custom JSON file to bulk-commission two additional hosts in the vSphere Client Commission Hosts wizard.

1. On the **Hosts** page, click **COMMISSION HOSTS**.
2. In the **Checklist** dialog box, select the **Select All** checkbox and click **PROCEED**.
3. Customize a JSON file to bulk-commission two new hosts called `esx-06a.site-a.vcf.lab` and `esx-07a.site-a.vcf.lab`.
   a. Under Add Hosts, click **Import** and click the **JSON** link to download the JSON template necessary for bulk commissioning.
   b. From the **Download** panel of the Chrome browser, click **sample_bulk_commission.json** to open the file in the Text Editor application.
   c. In the first host section under the `hosts` line, delete the default text between the quotation marks on the right for each parameter and replace it with the correct value for the esx-06a.site-a.vcf.lab host.

Parameter	Value
`fqdn`	`esx-06a.site-a.vcf.lab`
`username`	`root`
`storageType`	`VSAN_ESA`
`password`	`VMware123!VMware123!`
`networkPoolName`	**`wld-a-np01`**
`vVolStorageProtocolType`	`""`

   d. In the second host section under the `hosts` line, delete the default text between the empty quotation marks on the right for each parameter and replace it with the correct value for the esx-07a.site-a.vcf.lab host.

Parameter	Value
`fqdn`	`esx-07a.site-a.vcf.lab`
`username`	`root`
`storageType`	`VSAN_ESA`

`password`	`VMware123!VMware123!`
`networkPoolName`	**`wld-a-np01`**
`vVolStorageProtocolType`	`""`

NOTE

In a production environment, you can add as many hosts as you require to the JSON file by copying and modifying additional sections separated by curly braces under the `hostsSpec` line. In this lab, you add the three hosts specified.

e. Click **Save** and close Text Editor.

4. Return to the **Commission Hosts** wizard in Chrome.
5. Bulk-commission the esx-06a.site-a.vcf.lab and esx-07a.site-a.vcf.lab .
   a. Under Add Hosts, click **Import** and click **BROWSE**.
   b. In the **Open File** window, navigate to the `Downloads` folder and double-click **sample_bulk_commission-esx-06a-esx-07a.json** file.

      The **sample_bulk_commission-esx-06a-esx-07a.json** file is preconfigured with the two hosts.
   c. On the **Host Addition and Validation** page, click **UPLOAD**.
   d. Under Hosts Added, verify that the esx-06a.site-a.vcf.lab , and esx-07a.site-a.vcf.lab hosts were added successfully.
   e. Turn on the **Confirm all Fingerprints** toggle to confirm the fingerprints for all hosts.
   f. Click **VALIDATE ALL**.
   g. When the hosts are validated successfully, click **NEXT**.
   h. On the **Review** page, click **COMMISSION**.
6. Expand the **Tasks** pane and monitor the Commissioning Host(s) task until completion.

   This commissioning task takes 3 to 5 minutes to complete.

IMPORTANT

As a result of the nested environment, you might observe some errors. If necessary, alert your instructor and restart any tasks that require it. The tasks should complete even with these errors.

7. When the bulk-commissioning task is complete, close the **Tasks** pane.
8. On the **Hosts** page, click **UNASSIGNED HOSTS** and verify that the esx-05a.site-a.vcf.lab, esx-06a.site-a.vcf.lab and esx-07a.site-a.vcf.lab hosts are listed.

   You must alert your instructor if any of the three hosts are missing from the **Unassigned Hosts** tab.
9. If any of the hosts are not listed, refresh the browser every 2 minutes until they appear.

Lab 11 (Simulation) Creating Workload Domain

Objective and Tasks

Create a workload domain using three unassigned hosts:

1. Prepare for the Lab
2. Create a Workload Domain
3. Retrieve Passwords and Log in to vCenter and NSX Manager

In the web browser in your local desktop, go to https://core-vmware.bravais.com/s/21797/Creating_Workload_Domain to open the simulation.

Perform the following steps in the simulated environment. Refer to these steps as you work through the simulation.

IMPORTANT

Do not refresh, navigate away from, or minimize the browser tab hosting the simulation. Such actions might pause the simulation and prevent it from progressing.

NOTE

If you're stuck anywhere in the simulation, press the **TAB** key to reveal the next clickable area.

Task 1: Prepare for the Lab

You log in to the vSphere Client and the VCF Operations UI.

1. Log in to the vSphere Client.
    a. Open Chrome.
    b. From the Chrome bookmarks bar, click **Region A > vc-mgmt-a Client**.
    c. Login to the vSphere Client.
        - User name: administrator@vsphere.local
        - Password: VMware123!VMware123!
2. Log in to the VCF Operations.
    a. Open a new tab in Chrome.
    b. On the bookmarks bar, click **Region A > VMware Cloud Foundations Operations.**
    c. Login to the VCF Operations.
        - User name: admin
        - Password: VMware123!VMware123!

Task 2: Create a Workload Domain

You create a workload domain in VCF Operations using the three unassigned hosts.

1. In Chrome, click the **VCF Operations** tab.
2. Click **Operate** tab.
3. Create a Workload domain.
    a. Click **Inventory** from the navigation pane.
    b. Click **instance-a > mgmt-a > vc-mgmt-a.site-a.vcf.lab** to expand.
    c. Select **instance-a** under VCF Instances and select **Create New** from the **ADD WORKLOAD DOMAIN** drop-down menu.
    d. On Workload Domain Prerequisites, select the **Select All** checkbox and click **PROCEED**.
4. Configure the options on the **General Information** page and click **NEXT**.

Option	Action
Workload Domain Name	Enter **`wld01-a`** in the text box.
Select a Deployment Type	Full deployment with cluster

Enable vSphere Supervisor	Toggle to disable.

5. Configure the options on the **vCenter** page and click **NEXT**.

Option	Action
vCenter FQDN	Enter **vc-wld01-a.site-a.vcf.lab** in the text box and click **eye icon** to show/hide the IP configuration.
SSO Domain Name	Enter **wld.sso** in the text box.

6. Configure the options on the **Cluster** page and click **NEXT**.

Option	Action
Cluster Name	Enter **cluster-wld01-01a** in the text box.

7. On the **Image** page, select the **Management-Domain-ESXi-Personality** image and click **NEXT**.
8. Configure the options on the **NSX Manager** page and click **NEXT**.

NOTE

When the deployment is complete, the NSX cluster IP address, the NSX FQDN, the first NSX Manager IP address, and the first NSX Manager FQDN are accessible, providing the same experience as a three-node NSX Manager cluster.

Option	Action
NSX Manager instance options	Create new NSX Manager instance
Deployment Size	Simple
Appliance Size	Medium
Appliance 1 FQDN	Enter **nsx-wld01-01a.site-a.vcf.lab** in the text box.

Appliance 1 IP Address	This value is populated automatically through a lookup of the FQDN.
Appliance Cluster FQDN	Enter **`nsx-wld01-a.site-a.vcf.lab`** in the text box.
Appliance Cluster Virtual IP Address	This value is populated automatically through a lookup of the FQDN.
Configure Network Connectivity and VPC External IP Blocks	**Centralized Connectivity**

9. On the **Storage** page, select **vSAN** and verify that the **Enable vSAN ESA** checkbox is selected.
   a. Click **NEXT**.
10. On the **vSAN Storage** page, verify that the vSAN Cluster Type is **vSAN HCI** and click **NEXT.**
11. On the **Hosts** page, select all three available hosts and click **NEXT**.
12. Configure the options on the **Distributed Switch Configuration** page.
    a. Under Storage Traffic Separation, click **SELECT**.
    b. Click **Edit** next to the Transport `VLAN is mandatory and has to be updated before proceeding` message.
    c. Under Details of vSphere Distributed Switch, scroll down to the **NSX** section and enter **14** in the **VLAN ID** text box.
    d. Click **SAVE CHANGES.**
    e. Click **Acknowledge** next to the `Transport VLAN ID. It is mandatory and has to be updated before proceeding message` and click **NEXT**.
13. On the **Review** page, review the configuration thoroughly for any errors.

NOTE

The deployment of the workload domain takes several hours to complete.

Comparing the details on the **Review** page is critical to ensure that the deployment does not fail as a result of user input error.

14. After verifying that all inputs are correct, click **FINISH**.

15. In the **Workload Domain creation has started** pop-up window, click **CLOSE**.
16. Click on the **Build** tab.
17. Click **Tasks** from the navigation pane.
18. On the **Workload Domain creation** has started dialog box, click **VIEW SDDC MANAGER TASKS**.
19. Click the **Creating Workload Domain: wld01-a** task to monitor the subtasks and the status.

    IMPORTANT

    While the **Creating Workload domain: wld01-a** task is running, the SDDC Manager runs few health check operations for SDDC and syncs vSAN HCL to vCenter instances**.** You can safely ignore If the **Health-Check operation for SDDC** and **Syncing vSAN HCL to vCenters** tasks fail.

    When you reach this step, notify your instructor.

    Because this process takes around 2 hours to complete, you might take a break before VMware Cloud Foundation finishes deploying the workload domain.

    ---

20. Retry any failed subtasks.
    a. If a subtask fails, return to the main task view and click **RESTART TASK**.
    b. Return to the subtask view and continue to monitor the subtasks.

       If the same subtask fails a second time, notify your instructor to help you resolve the problem.

       When the **Tasks** pane displays the `Resource Release Locks with successful` status, the workload domain deployment is complete.
21. Click **Operate** tab to return to the Inventory.
22. Click **Inventory** from the navigation pane and expand **wld01-a > vc-wld01-a.site-a.vcf.lab.**
    a. Verify that the workload domain vCenter, sa-wld01-vc01.vcf.sddc.local, is listed.

Task 3: Retrieve Passwords and Log in to vCenter and NSX Manager

1. Click **Build** tab.
2. Click **Developer Center > APIs & SDKs**

3. Click **API Explorer** under SDDC Manager API tile.
4. Under API Categories, expand **Credentials**.
5. Click **GET /v1/credentials.**
6. In resourceType text box, type **PSC** and click **EXECUTE**.

NOTE

You can also use the following resource Types to query and retrieve the passwords: `ESXI, VCENTER, PSC, NSXT_MANAGER, NSXT_EDGE, NSX_ALB, BACKUP, HCX_MANAGER, VSP`

---

7. From the Response, click on the first link to expand **Credential (UUID 1.....)**

   Note down the Password without the quotes for the administrator@vsphere.local user.
8. From the Response, click on the Second link to expand **Credential (UUID 2**.....)

   Note down the Password without the quotes for the administrator@wld.sso user.
9. In Chrome, open a new tab and click the **Region A > vc-wld01-a Client** from bookmark bar.
   a. Log in to Workload Domain vCenter.
      - User name: administrator@wld.sso
      - Password: <administrator@wld.sso password>
10. Click on **VCF Operations** tab in Chrome.
11. In resourceType text box, type **NSXT_MANAGER** and click **EXECUTE**.
12. From the Response, click to expand **Credential (UUID 1.....) until you find the Credentialtype:" is API and username is admin.**

    Note down the Password without the quotes for the admin user.
13. In Chrome, open a new tab and click the **Region A > nsx-wld01-a** from bookmark bar.
    a. Click Login Method drop down menu and select Local Account and Log in to Workload Domain NSX.
       - User name: admin
       - Password: <admin-user-password>
    b. Click **SAVE** on the Join the VMware Customer Experience Improvement Program.

Lab 12 Creating Workload Domain without vSphere Cluster

Objective and Tasks

Create a workload domain using three unassigned hosts:

1. Prepare for the Lab
2. Create a Workload Domain
3. Log In to vCenter and NSX Manager

Task 1: Prepare for the Lab

You log in to the vSphere Client and the VCF Operations UI.

1. Log in to the vSphere Client.
    a. Open Firefox.
    b. From the Firefox bookmarks bar, click **Region A > vc-mgmt-a Client**.
    c. Select **VCF SSO** as the login Method and click **LOG IN** on the VMware vSphere login page.
    d. On the welcome to authentik, enter the following credentials.
        - Email or Username: prod-admin@vcf.lab
        - Password: VMware123!VMware123!
    e. From the navigation pane on the left hand side, click **>** to expand **vc-mgmt-a.site-a.vcf.lab > dc-a > cluster-mgmt-01a**
2. Log in to VCF Operations.
    a. Open a new tab in Firefox.

b. On the bookmarks bar, click **Region A > VMware Cloud Foundations Operations.**

c. Select **VCF SSO** as the login Method and click **LOG IN** on the VMware Cloud Foundation Operations login page..

The VCF SSO should allow you to automatically logs you in to VCF Operations UI without any password.

Task 2: Create a Workload Domain (Deploy Infrastructure only)

You create a workload domain using the "Deploy Infrastructure Only" deployment type in VCF Operations.

1. In Firefox, click the **VCF Operations** tab.
2. Click **Operate** tab.
3. Create a Workload domain.
   a. Click **Inventory** from the navigation pane.
      i. Click **vc-mgmt-a.site-a.vcf.lab** to switch to the Detailed view.
   b. Select **instance-a** under VCF Instances and select **Create New** from the **ADD WORKLOAD DOMAIN** drop-down menu.
   c. On Workload Domain Prerequisites, select the **Select All** checkbox and click **PROCEED**.
4. Configure the options on the **General Information** page and click **NEXT**.

Option	Action
Workload Domain Name	Enter **`wld02-a`** in the text box.
Select a Deployment Type	Deploy Infrastructure only

5. Configure the options on the **vCenter** page and click **NEXT**.

Option	Action
vCenter FQDN	Enter **`vc-wld02-a.site-a.vcf.lab`** in the text box and click **eye icon** to show/hide the IP configuration.
SSO Domain Name	Enter **`wld02.sso`** in the text box.

6. Configure the options on the **NSX Manager** page and click **NEXT**.

NOTE

When the deployment is complete, the NSX cluster IP address, the NSX FQDN, the first NSX Manager IP address, and the first NSX Manager FQDN are accessible, providing the same experience as a three-node NSX Manager cluster.

Option	Action
NSX Manager instance options	Create new NSX Manager instance
Deployment Size	Simple
Appliance Size	Medium
Appliance 1 FQDN	Enter **`nsx-wld02-01a.site-a.vcf.lab`** in the text box.
Appliance 1 IP Address	This value is populated automatically through a lookup of the FQDN.
Appliance Cluster FQDN	Enter **`nsx-wld02-a.site-a.vcf.lab`** in the text box.
Appliance Cluster Virtual IP Address	This value is populated automatically through a lookup of the FQDN.
Configure Network Connectivity and VPC External IP Blocks	**Centralized Connectivity**

7. On the **Review** page, review the configuration thoroughly for any errors.

NOTE

The deployment of the workload domain takes several hours to complete.

Comparing the details on the **Review** page is critical to ensure that the deployment does not fail as a result of user input error.

8. After verifying that all inputs are correct, click **NEXT**.
9. After all the validation checks are Succeeded, click **FINISH**.

10. On the **Workload Domain creation has started** has started dialog box, click **VIEW SDDC MANAGER TASKS**.
11. Click the **Creating Workload Domain: wld02-a** task to monitor the subtasks and the status.

    IMPORTANT

    While the **Creating Workload domain: wld02-a** task is running, the SDDC Manager runs several precheck tasks to deploy and configure the vCenter server virtual appliance and NSX Manager. You will see that the **Validate Platform infrastructure state for Domain Creation workflow** task is skipped, which is expected for the **Deployment Infrastructure Only** deployment type.

    When you reach this step, notify your instructor.

    Because this process takes around 2 hours to complete, you might take a break before VMware Cloud Foundation finishes deploying the workload domain.

    ---

12. Retry any failed subtasks.
    a. If a subtask fails, return to the main task view and click **RESTART TASK**.
    b. Return to the subtask view and continue to monitor the subtasks.

       If the same subtask fails a second time, notify your instructor to help you resolve the problem.

       When the **Tasks** pane displays the `Refresh Inventory Cache` status is Successful, the workload domain deployment is complete.
13. Click **Operate** tab to return to the Inventory.
14. Click **Inventory** from the navigation pane and expand **wld02-a.**
    a. Verify that the workload domain vCenter, vc-wld02-a.site-a.vcf.lab, is listed.

Task 3: Retrieve Passwords and Log in to vCenter and NSX Manager

1. Click **Build** tab.
2. Click **Developer Center > APIs & SDKs**
3. Click **API Explorer** under SDDC Manager API tile.
4. Under API Categories, expand **Credentials**.
5. Click **GET /v1/credentials.**

6. In resourceType text box, type **PSC** and click **EXECUTE**.

NOTE

You can also use the following resource Types to query and retrieve the passwords: `ESXI, VCENTER, PSC, NSXT_MANAGER, NSXT_EDGE, NSX_ALB, BACKUP, HCX_MANAGER, VSP`

7. From the Response, click on the first link to expand **Credential (UUID 1.....)**

   Note down the Password without the quotes for the administrator@vsphere.local user.

8. From the Response, click on the Second link to expand **Credential (UUID 2**.....)

   Note down the Password without the quotes for the administrator@wld.sso user.

9. In Firefox, open a new tab and click the **Region A > vc-wld02-a Client** from bookmark bar.
   a. Log in to Workload Domain vCenter.
      - User name: administrator@wld02.sso
      - Password: <administrator@wld02.sso password>
10. Click on VCF Operations tab in Firefox.
11. In resourceType text box, type **NSXT_MANAGER** and click **EXECUTE**.

    The response provides links to user account **Credentials** across different workload domains. Click each link to view the credentials for a specific domain and user.

12. From the Response, click to expand **Credential (UUID 1.....)** until you find the **Credentialtype:" is API** and **username is admin.**

    You click **AuthenticatedResource** link to expand for domainName, resourceName and resourceType details.

    Note down the Password without the quotes for the admin user.

You may need to expand the section to find the admin user password for the wld02-a domain.

13. In Firefox, open a new tab and click the **Region A > nsx-wld02-a** from bookmark bar.
    a. Log in to Workload Domain NSX.
       - User name: admin

- Password: <admin-user-password>

b. Click **SAVE** on the Join the VMware Customer Experience Improvement Program.

c. Select the **Don't Show this again** checkbox and click **SKIP** on the **Welcome to NSX** dialog box.

Lab 13 (Simulation) Creating a Workload Domain with the Supervisor Cluster

Objective and Tasks

Create a workload domain with the Supervisor cluster:

1. Prepare for the Lab
2. Create a Workload Domain with the Supervisor Cluster
3. Retrieve Passwords
4. Silence vSAN Alerts

In the web browser in your local desktop, go to https://core-vmware.bravais.com/s/21890/Creating_a_Workload_Domain_with_the_Supervisor_Cluster to open the simulation.

Perform the following steps in the simulated environment. Refer to these steps as you work through the simulation.

IMPORTANT

Do not refresh, navigate away from, or minimize the browser tab hosting the simulation. Such actions might pause the simulation and prevent it from progressing.

NOTE

If you're stuck anywhere in the simulation, press the **TAB** key to reveal the next clickable area.

Task 1: Prepare for the Lab

You log in to the vSphere Client and VCF Operations.

1. Log in to the vSphere Client with the administrator account.
   a. Open Chrome.
   b. From the Chrome bookmarks bar, click **vSphere Client**.
   c. Select Local Account from the Login Metho to login the vSphere Client.
      - User name: administrator@vsphere.local
      - Password: VMware123!VMware123!
2. Log in to VCF Operations.
   a. Open a new tab in Chrome.
   b. On the bookmarks bar, click Region A > **VMware Cloud Operations**.
   c. Log in to VCF Operations.
      - User name: admin
      - Password: VMware123!VMware123!

Task 2: Create a Workload Domain with Supervisor Cluster

You create a workload domain in VCF Operations using the three unassigned hosts.

1. In Chrome, click the **VCF Operations** tab.
2. Click **Operate** tab.
3. Create a Workload domain.
   a. Click **Inventory** from the navigation pane.
      i. Click **vc-mgmt-a.site-a.vcf.lab** to switch to the Detailed view.
   b. Select **instance-a** under VCF Instances and select **Create New** from the **ADD WORKLOAD DOMAIN** drop-down menu.
   c. On Workload Domain Prerequisites, select the **Select All** checkbox and click **PROCEED**.
4. Configure the options on the **General Information** page and click **NEXT**.

Option	Action
Workload Domain Name	Enter **`wld01-a`** in the text box.

Select a Deployment Type	Full deployment with cluster
Enable vSphere Supervisor	Verify that the Toggle is enabled.

5. Configure the options on the **vCenter** page and click **NEXT**.

Option	Action
vCenter FQDN	Enter **`vc-wld01-a.site-a.vcf.lab`** in the text box and click **eye icon** to show/hide the IP configuration.
SSO Domain Name	Enter **`wld.sso`** in the text box.

6. Configure the options on the **Cluster** page and click **NEXT**.

Option	Action
Cluster Name	Enter **`cluster-wld01-01a`** in the text box.
vSphere Zone Name	z-wld-a

7. On the **Image** page, select the **Management-Domain-ESXi-Personality** image and click **NEXT**.
8. Configure the options on the **NSX Manager** page and click **NEXT**.

NOTE

When the deployment is complete, the NSX cluster IP address, the NSX FQDN, the first NSX Manager IP address, and the first NSX Manager FQDN are accessible, providing the same experience as a three-node NSX Manager cluster.

Option	Action
NSX Manager instance options	Create new NSX Manager instance
Deployment Size	Simple
Appliance Size	Medium

Appliance 1 FQDN	Enter **`nsx-wld01-01a.site-a.vcf.lab`** in the text box.
Appliance 1 IP Address	This value is populated automatically through a lookup of the FQDN.
Appliance Cluster FQDN	Enter **`nsx-wld01-a.site-a.vcf.lab`** in the text box.
Appliance Cluster Virtual IP Address	This value is populated automatically through a lookup of the FQDN.
Configure Network Connectivity and VPC External IP Blocks	**Centralized Connectivity**

9. On the **Storage** page, select **vSAN** and verify that the **Enable vSAN ESA** checkbox is selected.
    a. Click **NEXT**.
10. On the **vSAN Storage** page, verify that the vSAN Cluster Type is **vSAN HCI** and click **NEXT.**
11. On the **Hosts** page, select all the Four available hosts and click **NEXT**.
12. Configure the options on the **Distributed Switch Configuration** page.
    a. Under Storage Traffic Separation, click **SELECT**.
    b. Click **Edit** next to the Transport `VLAN is mandatory and has to be updated before proceeding` message.
    c. Under Details of vSphere Distributed Switch, scroll down to the **NSX** section and enter **14** in the **VLAN ID** text box.
    d. Click **SAVE CHANGES.**
    e. Click **Acknowledge** next to the `Transport VLAN ID. It is mandatory and has to be updated before proceeding message` and click **NEXT**.
13. On the **vSphere Supervisor** page, enter the configuration details and click **NEXT**.

Option	Value
Supervisor Name	sup-wld-01a
Service CIDR	172.29.0.0/16

Use ESXi Management VMK Settings	Verify that the Use ESXi Management VMK Settings is selected.
Control Plane IP Range	10.1.1.165-10.1.1.169
NSX Project	Default Project
VPC Connectivity Profile	Default Profile
Private CIDR	172.16.201.0/24
Workload DNS	10.1.1.1
Workload NTP	10.1.1.1

14. On the **Review** page, review the configuration thoroughly for any errors.

    NOTE

    The deployment of the workload domain takes several hours to complete.

    Comparing the details on the **Review** page is critical to ensure that the deployment does not fail as a result of user input error.

15. After verifying that all inputs are correct, click **FINISH**.
16. In the **Workload Domain creation has started** pop-up window, click **CLOSE**.
17. Click on the **Build** tab.
18. Click **Tasks** from the navigation pane.
19. On the **Workload Domain creation** has started dialog box, click **VIEW SDDC MANAGER TASKS**.
20. Click the **Creating Workload Domain: wld01-a** task to monitor the subtasks and the status.

IMPORTANT

While the **Creating Workload domain: wld01-a** task is running, the SDDC Manager runs few health check operations for SDDC and syncs vSAN HCL to vCenter instances**.** You can safely ignore If the **Health-Check operation for SDDC** and **Syncing vSAN HCL to vCenters** tasks fail.

When you reach this step, notify your instructor.

Because this process takes around 2 hours to complete, you might take a break before VMware Cloud Foundation finishes deploying the workload domain.

21. Retry any failed subtasks.
    a. If a subtask fails, return to the main task view and click **RESTART TASK**.
    b. Return to the subtask view and continue to monitor the subtasks.

       If the same subtask fails a second time, notify your instructor to help you resolve the problem.

       When the **Tasks** pane displays the `Resource Release Locks with successful` status, the workload domain deployment is complete.
22. Click **Operate** tab to return to the Inventory.
23. Click **Inventory** from the navigation pane and expand **wld01-a.**
    a. Verify that the workload domain vCenter, vc-wld01-a.site-a.vcf.lab, is listed.
    b. You will set up Edge Cluster and configure the Tier-0 gateway in the Networking labs section.

Task 3: Retrieve Passwords

1. Click **Build** tab.
2. Click **Developer Center > APIs & SDKs**
3. Click **API Explorer** under SDDC Manager API tile.
4. Under API Categories, expand **Credentials**.
5. Click **GET /v1/credentials.**
6. In resourceType text box, type **PSC** and click **EXECUTE**.

NOTE

You can also use the following resource Types to query and retrieve the passwords: `ESXI, VCENTER, PSC, NSXT_MANAGER, NSXT_EDGE, NSX_ALB, BACKUP, HCX_MANAGER, VSP`

7. From the Response, click on the first link to expand **Credential (UUID 1.....)**

   Note down the Password without the quotes for the administrator@vsphere.local user.

8. From the Response, click on the Second link to expand **Credential (UUID 2**.....)

   Note down the Password without the quotes for the administrator@wld.sso user.

9. In Chrome, open a new tab and click **Region A > vc-wld01-a Client** from bookmark bar.
   a. Log in to Workload Domain vCenter.
      - User name: administrator@wld.sso
      - Password: <administrator@wld.sso password>

Task 4: Silence vSAN Alerts

In a production environment, alerts and warnings must remain active and unsilenced.

1. In the vSphere Client, click the **Hosts and Cluster** view.
2. Click to expand **vc-wld01-a.site.vcf.lab > wld01-a DC.**
3. Clear Issues and Alarms.
   a. Click to select the **cluster-wld01-01a** cluster.
   b. For vSAN hardware alarm, click **Actions > Reset to Green**.
   c. For vSAN cluster alarm 'vSAN Managed disk claim', click **Actions > Reset to Green**.
   d. For vSAN cluster alarm 'vSAN cluster compliance', click **Actions > Reset to Green**.
4. Click the **Monitor** tab.
5. Click **vSAN > vSAN Health.**
6. Scroll down to the **Health finding** section.
7. On the **UNHEALTHY** tab, silence the alerts.
   a. Click **Silence Alert** under the **VSAN Managed disk Claim** and click **YES** on the **Silence Alert** pop-up window.

b. Click **Silence Alert** under **vSAN cluster compliance** and click **YES** on the **Silence Alert** pop-up window.

c. Click **Silence Alert** under **NVMe device is VMware Certified** and click **YES** on the **Silence Alert** pop-up window.

8. On the **INFO** tab, silence the alerts.

   a. Click **Silence Alert** under Advisor and click **YES** on the **Silence Alert** pop-up window.

   b. Click **Silence Alert** under vSAN Support Insight and click **YES** on the **Silence Alert** pop-up window.

Lab 14 Verifying the Workload Domain Components Integration with VCF Operations

Objective and Tasks

Verify the workload domain integration with VCF Operations:

1. Prepare for the Lab
2. Verify the Workload Domain Components Integration with VCF Operations

Task 1: Prepare for the Lab

You log in to the vSphere Client and SDDC Manager.

1. Log in to VCF Operations.
    a. Open a new tab in Firefox.
    b. On the bookmarks bar, click **Region A > VMware Cloud Foundation Operations**.
    c. Select **VCF SSO** as the login method and click **Login**.
        - User name: prod-admin@vcf.lab
        - Password: VMware123!VMware123!

Task 2: Verify the Workload Domain Components Integration with VCF Operations

1. Click **Operate** tab**.**
2. Click **Administration > Integrations.**
3. Expand **VMware Cloud Foundation** on the **Accounts** tab**.**

4. Expand **instance-a** > **wld01-a** and verify that the status is Collecting for the Workload domain's vCenter, vSAN, and NSX components.
    a. Expand **mgmt-a** to verify the Management domain's vCenter, vSAN, and NSX component status.

Lab 15 (Simulation) Assigning a License to the Workload Domain vCenter

Objective and Tasks

You assign the VCF license to the Workload domain:

1. Prepare for the Lab
2. Assign a License to the Workload Domain vCenter

In the web browser in your local desktop, go to https://core-vmware.bravais.com/s/21798/Assigning_a_License_to_the_Workload_Domain_vCenter to open the simulation.

Perform the following steps in the simulated environment. Refer to these steps as you work through the simulation.

NOTE

If you're stuck anywhere in the simulation, press the **TAB** key to reveal the next clickable area.

IMPORTANT

Do not refresh, navigate away from, or minimize the browser tab hosting the simulation. Such actions might pause the simulation and prevent it from progressing.

Task 1: Prepare for the Lab

You log in to the vSphere Client and VCF Operations.

1. Log in to the vSphere Client.
    a. Open Chrome.
    b. On the bookmarks bar, click **Region A > vc-wld01-a Client**.
    c. Login to the vCenter.
        - User name: administrator@wld.sso
        - Password: VMware123!VMware123!
2. Log in to VCF Operations.
    a. Open a new tab in Chrome.
    b. On the bookmarks bar, click **Region A > VMware Cloud Foundation Operations**.
    c. Log in to VCF Operations.
        - User name: **admin**
        - Password: VMware123!VMware123!

Task 2: Assign a License to the Workload Domain vCenter

You assign a license to the workload domain vCenter.

1. In Chrome, click the VCF Operations tab.
2. Click the Manage tab in VCF Operations.
3. In the navigation pane on the left, select **Licensing > Licenses & Registration**.
4. Under vCenter Systems, verify that the `Fully licensed` status for the **vc-wld01-a.site-a.vcf.lab** is **No**.
5. Select **vc-wld01-a.site-a.vcf.lab** and click **ASSIGN PRIMARY LICENSE.**
6. Verify that the Fully Assigned status is **Yes** for vc-wld01-a.site-a.vcf.lab.

Lab 16 (Simulation) Adding an ESX Host to the Workload Domain

Objective and Tasks

Expand the workload domain with a fourth host:

1. Prepare for the Lab
2. Commission Hosts
3. Add a Host to the Workload Domain Using vCenter

In the web browser in your local desktop, go to https://core-vmware.bravais.com/s/21810/Adding_an_ESX_Host_to_the_Workload_Domain to open the simulation.

Perform the following steps in the simulated environment. Refer to these steps as you work through the simulation.

IMPORTANT

Do not refresh, navigate away from, or minimize the browser tab hosting the simulation. Such actions might pause the simulation and prevent it from progressing.

NOTE

If you're stuck anywhere in the simulation, press the **TAB** key to reveal the next clickable area.

Task 1: Prepare for the Lab

You log in to the vSphere Client.

1. Log in to the vSphere Client with the vcfadmin account.
    a. Open Chrome.
    b. On the Chrome bookmarks bar, click **Region A > vc-wld01-a**.
    c. Log In to vcenter.
        - User name: administrator@wld.sso
        - Password: VMware123!VMware123!
2. Log in to vc-mgmt-01a Client.
    a. Open a new tab in Chrome.
    b. On the Chrome bookmarks bar, click **Region A > vc-mgmt-01a Client**.
    c. Log in to vCenter.
        - User name: administrator@vsphere.local
        - Password: VMware123!VMware123!

Task 2: Commission Hosts

You add a fourth host to the newly deployed workload domain to expand its resource capabilities.

NOTE

Before you proceed with adding a new host, you must ensure that the vSAN HCL DB Auto Update alert is silenced.

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1. In vSphere Client, click the **vSphere Client menu** and select **Global Inventory Lists.**
2. Click **Resources** and select **Hosts.**
3. Under Hosts, click the **Unassigned Hosts** tab.
4. Click **COMMISSION HOSTS**.
5. In the **Checklist** dialog box, select the **Select All** checkbox and click **PROCEED**.
6. In the **Commission Hosts** wizard, configure a new host called esx-08a.site-a.vcf.lab.
    a. Under Add Hosts on the **Host Addition and Validation** page, verify that **Add new** is selected.

b. Configure the options for the new esx-08a.site-a.vcf.lab host and click **ADD**.

Option	Action
Host FQDN	Enter **`esx-08a.site-a.vcf.lab`** in the text box.
Storage Type	Click **vSAN**.
vSAN Type	Click **vSAN HCI**.
vSAN ESA	**Enabled**.
Network Pool Name	Select **wld-a-np01** from the drop-down menu.
Username	Enter **`root`** in the text box.
Password	Enter **`VMware123!VMware123!`** in the text box.

c. Under Hosts Added, turn on the **Confirm all Fingerprints** toggle.

d. Click **VALIDATE ALL**.

e. When the validation is complete, click **NEXT**.

f. Click **COMMISSION**.

g. Expand the **Tasks** pane and monitor the `Commissioning host(s) esx-08a.site-a.vcf.lab to VMware Cloud Foundation` task.

h. When the commissioning task completes successfully, close the **Tasks** pane.

Task 3: Add a Host to the Workload Domain

You add a fourth host to the newly deployed workload domain to expand its resource capabilities.

1. Add the newly commissioned esx-08a.site-a.vcf.lab host

   a. Open a new tab in Chrome.

   b. Click **Region A > SDDC Manager.**

   c. Click **Workload Domains.**

   d. **Scroll down** to the workload domains

   e. Click **wld01-a.**

   f. Click **Clusters** tab.

   g. Click **cluster-wld01-01a.**

h. Click **ACTIONS** drop-down menu.

i. Click **Add Host**.

j. On Host Selection, select **esx-08a.site-a.vcf.lab** and click **NEXT**.

k. On vSphere Distributed Switch, click **NEXT**.

l. On the **Review** page, click **NEXT**.

m. On the **Validation** page, verify that the status for all tasks is succeeded and click **FINISH**.

   Wait for 2-3 minutes for the `Adding new host(s) to cluster` task to appear in the **Tasks** pane.

2. Expand the **Tasks** pane and monitor the subtasks of the `Adding new host(s) to cluster` task.

3. When the `Adding new host(s) to cluster` task is complete, close the **Tasks** pane.

4. Click vsphere tab to switch to workload domain vcenter and verify that the esx-08a host is added to the cluster-wld01-01a cluster.

Lab 17 Creating vCenter Group

Objective and Tasks

You create a vCenter group with the vCenter instances in the Management domain and the Workload domain:

1. Prepare for the Lab
2. Create a vCenter Group with vCenter Instances
3. Verify vCenter Linking

Task 1: Prepare for the Lab

You log in to the vSphere Client and VCF Operations.

1. Log in to the vSphere Client.
    a. Open **Firefox**.
    b. On the bookmarks bar, click **Region A > vc-wld01-a Client.**
    c. Select **VCFSSO** as the login method and click **LOGIN**.
        - User name: prod-admin@vcf.lab
        - Password: VMware123!VMware123!
2. Log in to VCF Operations.
    a. Open a **new tab** in Firefox.
    b. On the bookmarks bar, click **Region A > VMware Cloud Foundation Operations**
    c. Select **VCF SSO** as the login method and click **Log In**.
        - User name: prod-admin@vcf.lab
        - Password: VMware123!VMware123!

Task 2: Create a vCenter Group with vCenter Instances

You create a vCenter group and add the vCenter instances (vc-mgmt-a.site-a.vcf.lab and vc-wld01-a.site-a.vcf.lab).

1. Click **Operate** tab.
2. In the navigation pane on the left, select **Administration > Configurations**.
3. Click **vCenter Linking** under **Logical Groupings**.
4. Click **CREATE GROUP**.
5. In the **Group Name** text box, enter **`instance-a-vc-Group`** and click **NEXT**.
6. On the **vCenter Instances** page, select the **vc-mgmt-a.site-a.vcf.lab** and **vc-wld01-a.site-a.vcf.lab** checkboxes and click **NEXT.**
7. On Ready to complete, click **FINISH**.
8. Click **instance-a-vc-Group** to check the status.

   You click the **i** icon next to **In Progress** from the status to read the informational message.
9. The Status changes from **In Progress** to **Active**.

   If the status is still **In progress**, refresh the VCF Operations UI.

   It will take approximately 5 minutes for the status to show as Active in the UI.

Task 3: Verify vCenter Linking

1. In Firefox, open a new tab.
2. Click **Region A > vc-mgmt-a Client**.
3. Select **VCF SSO** as the login method and click **Login**.
   - User name: prod-admin@vcf.lab
   - Password: VMware123!VMware123!
4. Verify that vc-mgmt-a.site-a.vcf.lab and vc-wld01-a.site-a.vcf.lab are linked and shown in the vSphere inventory.

Lab 18 Configuring Distributed Transit Gateway

Objective and Tasks

Create a virtual private cloud in the workload domain with the Distributed gateway connection:

1. Prepare for the Lab
2. Configure Default Transit Gateway for External Connectivity
3. Create an IP Address Block
4. Configure VPC Connectivity Profiles
5. Verify the Default Transit gateway details in vCenter

Task 1: Prepare for the Lab

You log in to Management vCenter.

1. Log in to the vSphere Client with the prod-admin user account.
    a. Open **Firefox**.
    b. On the Firefox bookmarks bar, click **Region A > vc-mgmt-a Client.**
    c. Select **VCF SSO** as the login method and click **Log In**.
        - User name: prod-admin@vcf.lab
        - Password: VMware123!VMware123!

NOTE

Alternatively, you can use the local account **administrator@vsphere.local** as the username and **VMware123!VMware123!** as the password to log in to the Management vCenter.

Task 2: Configure Default Transit Gateway for External Connectivity

1. In Firefox, click the **vSphere Client** tab
2. In the vSphere Client, select **vc-mgmt-a.site-a.vcf.lab** from the inventory
3. Click the **Networks** tab.
4. Click to expand **Transit Gateways.**
5. Click **Default Transit Gateway** with Connection Distributed VLAN**.**
6. Click **Configure tab.**
7. Click **Properties.**
8. Under the General settings, verify that the **Distributed VLAN** is selected under connection
9. Click **EDIT.**
10. On Edit the Transit gateway, complete the details.

Option	Value
Name	Default Transit Gateway
Network Span	Default Span
Connection	Verify that **Distributed VLAN** is selected.

External Connection	Click **X** to clear **Day0 Distributed Vlan Connection.** Click the Dotted line and click **Create New.** On the Add External Connection wizard, complete the details. • Enter Mgmt-**`DTGW01-External-Connection`** in the Name text box. • Enter **`22`** in the VLAN ID text Box • Enter **`10.1.7.129/25`** in the Gateway CIDR IPv4 Address text box. • Click **SAVE.**

a. Click **SAVE**.

Task 3: Create an IP Address Block

You create an IP address block.

1. Click the **vSphere Client** menu bar on the left-hand side of your screen.
2. From the **vSphere Client** menu, select **Inventory**.
3. In the **Inventory** click the **Networking** view**.**
4. Select **Virtual Private Clouds** from the Networking view and click **Configure** tab.
5. Expand **Settings**.
6. Click **IP Blocks**.
7. Click **ADD IP ADDRESS BLOCK**.
8. Enter the configuration to Add IP Address Block for Private network and click **SAVE**.

Option	Value
Name	172.16.100.0/24
Visibility	Private
CIDRs/Ranges	172.16.100.0/24
Excluded IP Ranges	Leave it blank
Reserved for Specific Subnet	No

9. Click **ADD IP ADDRESS BLOCK.**
10. Enter the configuration to Add IP Address Block for External network and click **SAVE**.

Option	Value
Name	10.1.7.128/25
Visibility	Select External
CIDRs/Ranges	10.1.7.128/25
Excluded IP Ranges	Leave it blank
Reserved for Specific Subnet	No

Task 4: Configure VPC Connectivity Profiles

1. Select **Virtual Private Clouds** from the Networking view and click **Configure** tab.
2. Expand **Settings**.
3. Click Connectivity Profiles
4. Click on dotted line next to **Default VPC Connectivity Profile** and select **Edit.**
5. Select the configuration to Edit VPC Connectivity Profile and click **SAVE**.

Option	Value
Name	Default VPC Connectivity Profile
Transit Gateway	Default Transit Gateway
External IP Blocks	10.1.7.128/25
Private Transit Gateway IP Blocks	Click **X** to delete Day0 Private Tgw Ip Block Select **172.16.100.0/24**
Virtual Network Appliance Cluster	Leave it blank
N-S Services	Off
Default Outbound NAT	Off

External IP Block for Default Outbound NAT	Leave it blank

Task 5: Verify the Default Transit Gateway details in vCenter

1. In vSphere Client, Navigate to the Networking View.
2. Expand **vc-mgmt-a.site-a.vcf.lab > dc-a > Virtual Private Clouds**.
3. Click **Default Transit Gateway**.
4. Click **Configure** tab.
5. Expand **Settings.**
6. Click **Properties** to verify the Default Transit Gateway configuration Details.

Option	Value
Name	Default Transit Gateway
Connection	Distributed VLAN
External Connection	MGMT-DTGW01-External-Connection
HA Mode	Distributed
Network Span	Default Span
VPC Connectivity Profile	1
Transit Gateway Edge Cluster	Not Set
VPC Transit Subnets	100.64.0.0/21
Advertise Rules	0

Lab 19 Creating Virtual Private Cloud (Distributed)

Objective and Tasks

Create a virtual private cloud in the workload domain with the Distributed gateway connection:

1. Prepare for the Lab
2. Create Virtual Private Cloud (Distributed)
3. Create Private VPC Subnets in VPC
4. Create Private Transit Gateway Subnets in VPC
5. Create Public Subnets in VPC
6. Deploy Virtual Machines in VPC
7. Test Connectivity Between the Subnets within the VPC
8. Review the VPC in NSX Manager

Task 1: Prepare for the Lab

You log in to SDDC Manager.

1. Log in to the vSphere Client with the prod-admin account.
    a. Open **Firefox**.
    b. On the Firefox bookmarks bar, click **Region A > vc-mgmt-a Client.**
    c. Select **VCF SSO** as the login method and click **Log In**.
        - User name: prod-admin@vcf.lab
        - Password: VMware123!VMware123!

Task 2: Create Virtual Private Cloud (Distributed)

You create a virtual private cloud with Distributed connectivity.

1. In the vSphere Client, select **vc-mgmt-a.site-a.vcf.lab** from the inventory.
2. Click the **Networking** view.
3. Expand **vc-mgmt-a.site-a.vcf.lab** > **dc-a** to see the Virtual private clouds.
4. Right-click **Virtual Private Clouds** and click **New VPC...** with the following configuration.

Option	Value
Name	vpc-mgmt-01a
Description	distributed transit gateway based VPC.
Private VPC IP CIDRs	Enter **172.16.200.0/24** Click **Advanced Settings**
Connectivity Profile	Default VPC Connectivity Profile
Connectivity Policy	None

   a. Click **SAVE**.

Task 3: Create Private VPC Subnets in VPC

You create a private VPC subnet in virtual private cloud.

1. Create a subnet called web-private-subnet-mgmt01.
   a. Expand **Virtual Private Cloud.**
   b. Right-click **vpc-mgmt-01a** and click **New Subnet.**
   c. Create the subnet with the following configuration details and click **NEXT**.

Option	Value
Name	web-private-subnet-mgmt01
Access Mode	Private - VPC
IP Block	default-vpc-mgmt-01a (172.16.200.0/24
Auto allocate Subnet CIDR from IP Blocks	Yes
Subnet Size	16

Advanced Settings	Click to Expand
VPC Gateway Connectivity	Yes
DHCP Config	DHCP Server

d. On DHCP Configuration, click **NEXT.**

e. On Ready to Complete, click **FINISH**.

Task 4: Create Private Transit Gateway Subnets in VPC

You create Private Transit gateway subnets in VPC.

1. Create a subnet called web-private-tgw-subnet-mgmt01.

   a. Right-click **vpc-mgmt-01a** and click **New Subnet**.

   b. Create the subnet with the following configuration details and click **NEXT**.

Option	Value
Name	web-private-tgw-subnet-mgmt01
Access Mode	Private -Transit Gateway
IP Block	Any
Auto allocate Subnet CIDR from IP Blocks	Yes
Subnet Size	16
Advanced Settings	Click to Expand
VPC Gateway Connectivity	Yes
DHCP Config	DHCP Server

   c. On the **DHCP Configuration** wizard, click **NEXT.**

   d. On the **Ready to Complete** wizard, click **FINISH**.

Task 5: Create Public Subnets in VPC

You create a public subnet in VPC.

1. Create a subnet called web-public-subnet-mgmt01.

   a. Right-click **vpc-mgmt-01a** and click **New Subnet**.

b. Create the subnet with the following configuration details and click **NEXT**.

Option	Value
Name	web-public-subnet-mgmt01
Access Mode	Public
IP Block	Any
Auto allocate Subnet CIDR from IP Blocks	Toggle to select **Yes**.
Subnet Size	16
Advanced Settings	Click to Expand
Gateway Connectivity	Yes
DHCP Config	DHCP Server

c. On DHCP Configuration, click **NEXT.**

d. On Ready to Complete, click **FINISH**.

2. Expand **vpc-mgmt-01a** and verify that the three subnets are listed.
3. Select **vpc-mgmt-01a** and click the **Summary** tab to view the VPC summary.

Task 6: Deploy Virtual Machines in VPC

You deploy virtual machines and connect to different type of subnets in VPC.

1. In Firefox, click **vSphere Client (vc-mgmt-a.site-a.vcf.lab)**.
2. In the vSphere Client, click the **VMs and Templates** view.
   a. Expand **vc-mgmt-a.site-a.vcf.lab > dc-a** to locate web-mgmt-vm-template.
3. Deploy a virtual machine from a template.
   a. Right-click **web-mgmt-vm-template** and select **New VM from This Template**.
   b. On the Select a name and folder page, enter **`private-vpc-web-vm`** in the **virtual machine name** text box and click **NEXT**.
   c. On the **select a compute resource** page, click **NEXT**.
   d. On the **Select storage** page, select **vsan-mgmt-01a** and click **NEXT**.
   e. On the Select Clone options, select the **customize this virtual machines hardware** checkbox and click **NEXT**.

f. On the **customize hardware** page, click the **Network adapter 1** drop-down menu, click **Browse**, select **web-private-subnet-mgmt01** and click **OK,** and click **NEXT.**

g. On the **Ready to complete** page, click **FINISH**.

h. In the vSphere Client, monitor the **Clone virtual machine** task from Recent Tasks.

The task takes 2 to 3 minutes, and verify that the task is **Completed** without any errors.

i. In the vSphere Inventory, right-click **private-vpc-web-vm** and click **Power > Power On.**

4. Deploy the second virtual machine from a template.

a. Right-click **Web-mgmt-vm-template** and select **New VM from This Template**.

b. On the Select a name and folder page, enter **`private-tgw-web-vm`** in the **virtual machine name** text box and click **NEXT**.

c. On the **select a compute resource** page, click **NEXT**.

d. On the **Select storage** page, select **vsan-mgmt-01a** and click **NEXT**.

e. On the Select Clone options, select the **customize this virtual machines hardware** checkbox and click **NEXT**.

f. On the **customize hardware** page, click the **Network adapter 1** drop-down menu, click **Browse**, select **web-private-tgw-subnet-mgmt01** and click **OK,** and click **NEXT.**

g. On the **Ready to complete** page, click **FINISH**.

h. In the vSphere Client, monitor the **Clone virtual machine** task from Recent Tasks.

The task takes 2 to 3 minutes, and verify that the task is **Completed** without any errors.

i. In the vSphere Inventory, right-click **private-tgw-web-vm** and click **Power > Power On.**

5. Deploy the third virtual machine from a template.

a. Right-click **Web-mgmt-vm-template** and select **New VM from This Template**.

b. On the Select a name and folder page, enter **`public-web-vm`** in the **virtual machine name** text box and click **NEXT**.

c. On the **select a compute resource** page, click **NEXT**.

d. On Review details, click **NEXT**.

e. On the **Select storage** page, select **vsan-mgmt-01a** and click **NEXT**.

f. On the Select Clone options, select the **customize this virtual machines hardware** checkbox and click **NEXT**.

g. On the **customize hardware** page, click the **Network adapter 1** drop-down menu, click **Browse**, select **web-public-subnet-mgmt01** and click **OK,** and click **NEXT.**

h. On the **Ready to complete** page, click **FINISH**.

i. In the vSphere Client, monitor the **Clone virtual machine** task from Recent Tasks.

The task takes 2 to 3 minutes, and verify that the task is **Completed** without any errors.

j. In the vSphere Inventory, right-click **public-web-vm** and click **Power > Power On.**

Task 7: Test Connectivity Between the Subnets within the VPC

1. In Firefox, click the **vSphere Client (vc-mgmt-a.site-a.vcf.lab)** tab**.**
2. Select **public-web-vm** and click **LAUNCH WEB CONSOLE.**
   a. Log in to public-web-vm.
      - User name: root
      - Password: VMware1!
   b. View the IP address of the machine.

      ```
      ifconfig –a
      ```

      The IP address of public-web-vm is 10.1.7.147
3. Select **private-tgw-web-vm** and click **LAUNCH WEB CONSOLE.**
   a. Log in to private-tgw-web-vm.
      - User name: root
      - Password: VMware1!
   b. View the IP address of the machine.

      ```
      ifconfig –a
      ```

      The IP address of private-tgw-web-vm is 172.16.100.3
4. In Firefox, click the **vSphere Client** tab.
5. Select p**rivate-vpc-web-vm** and **LAUNCH WEB CONSOLE** from Summary.
   a. Log in to private-vpc-web-vm.
      - User name: `root`
      - Password: `VMware1!`
   b. View the IP address of the machine.

```
ifconfig -a
```

The IP address of private-vpc-web-vm is 172.16.200.3.

c. From the command line, run the `ping <public-web-vm-IP Address>` command to test the connectivity between the VMs connected to private-vpc and public subnets.

```
ping -c 2 10.1.7.147
```

The ping command should succeed.

d. From the command line, run the `ping <private-tgw-web-vm-IP Address>` command to test the connectivity between VMs connected to private-vpc and private-tgw subnets.

```
ping -c 2 172.16.100.3
```

The ping command should succeed.

6. In Firefox, click the **public-web-vm web console** tab.

a. From the command line, run the `ping <private-vpc-web-vm-IP Address>` command to test the connectivity between VMs connected to public-VM and private-vpc subnets.

```
ping -c 2 172.16.200.3
```

The ping command should succeed.

b. From the command line, run the `ping <private-tgw-web-vm-IP Address>` command to test the connectivity between the VMs connected to public-VM and private-tgw subnets.

```
ping -c 2 172.16.100.3
```

The ping command should succeed.

7. In Firefox, click **private-tgw-web-vm** web console tab.

a. From the command line, run the `ping <private-vpc-web-vm-IP Address>` command to test the connectivity between the VMs connected to private-tgw and private-vpc subnets.

```
ping -c 2 172.16.200.3
```

The ping command should succeed.

b. From the command line, run the `ping <public-vm-IP Address>` command to test the connectivity between the VMs connected to private-tgw and public subnets.

```
ping -c 2 10.1.7.147
```

The ping command should succeed.

8. Close the **private-vpc-web-vm**, **private-tgw-web-vm**, and **public-web-vm** tabs in Firefox.

Task 8: Review the VPC in NSX Manager

1. In Firefox, click **Region A > nsx-mgmt-01a** from the bookmarks.
2. Click **VCF SSO** as the login method and click **LOG IN**.
3. (Optional) Log in to NSX Manager:
   - User name: prod-admin@vcf.lab
   - Password: VMware123!VMware123!
4. In the NSX UI, click the **VPCs** tab.
5. Click **Topology** to view the topology deployed in the lab.
6. Click **Virtual Private Clouds** from the left side to view the **VPCs** configured in NSX.
7. Expand **vpc-mgmt-01a** and click **3** in the Subnets column to view the subnets.

Lab 20 Creating VNA Cluster and Update Connectivity Profile

Objective and Tasks

You create a vCenter group with the vCenter instances in the Management domain and the Workload domain:

1. Prepare for the Lab
2. Create VNA Cluster

Task 1: Prepare for the Lab

You log in to the vSphere Client and SDDC Manager.

1. Log in to the vSphere Client.
    a. Open Firefox.
    b. On the bookmarks bar, click **Region A > vc-mgmt-a Client**
    c. Select **VCF SSO** as the login method and click **LOGIN**.
        - User name: prod-admin@vcf.lab
        - Password: VMware123!VMware123!

Task 2: Create VNA Cluster

You create a VNA Cluster in the Management Domain vCenter.

1. In vSphere Client, click **Networking** view.
2. Select **vc-mgmt-a.site-a.vcf.lab**.
3. Click **Configure** tab..

4. Click **Networking > VNA Clusters.**
5. Click **ADD CLUSTER.**
6. Configure the options on the **Add Virtual Network Appliance Cluster (VNAC)** page and click **ADD**.

Option	Value
Cluster Name	vnacl-mgmt-a
Node Form Factor	Small

7. Click **ADD** to configure the VNA node.

Option	Action
Node Name (FQDN)	vna-mgmt-01a.site-a.vcf.lab
vSphere Cluster	cluster-mgmt-01a
Resource Pool	Resources
Data Store	vsan-mgmt-01a
IP Address Type	IPV4 Only
IP Assignment	Static
Management CIDR	10.1.1.178/24
Default Gateway	10.1.1.1
Port Group	mgmt-vds01-mgmt-01a

   a. Click **APPLY**.
   b. Select **vna-mgmt-01a.site-a.vcf.lab** and click **CLONE** to deploy the second edge node.
   c. In the **Edge Node Name (FQDN)** text box, enter **`vna-mgmt-02a.site-a.vcf.lab`** and enter the details in the table to configure the second VNA node, and click **APPLY**.

Option	Action

Node Name (FQDN)	vna-mgmt-02a.site-a.vcf.lab
IP Address Type	IPV4 Only
IP Assignment	Static
Management CIDR	10.1.1.179/24
Default Gateway	10.1.1.1

8. Click **SAVE**.

   The VNA Cluster deployments takes approximately 20 mins to complete.

9. Verify that the Node Connectivity status is **2 Up** and Status is **Success**.

Task 3: Update Connectivity Profile with VNA Cluster details.

1. In vSphere Client, navigate to **Networking** view.
2. Expand **vc-mgmt-a.site-a.vcf.lab > dc-a**
3. Click **Virtual Private Clouds**
4. Click **Configure** tab.
5. Expand **Settings**
6. Click **Connectivity Profiles**
7. Click the **vertical dotted line** and click **Edit** to modify the Default VPC Connectivity profile.
8. Click Virtual Network Appliance Cluster drop-down menu and select **vnacl-mgmt-a.**
9. Toggle to **ON** the **N-S Services**
10. Toggle to turn **ON** the **Default Outbound NAT.**
11. From the External IP Block for Default Outbound NAT, select **10.1.7.128/25.**
12. Click **SAVE**.
13. Verify that the status is **Success** for the Default Connectivity profile.

Lab 21 (Simulation) Creating Edge Cluster and Tier-0 Gateway using vCenter

Objective and Tasks

Create a workload domain with the Supervisor cluster:

1. Prepare for the Lab
2. Create Edge Cluster and Set up Tier-0 Gateway
3. Verify the NSX Edge Cluster in NSX Manager

In the web browser in your local desktop, go to https://core-vmware.bravais.com/s/21811/Creating_Edge_Cluster_and_Tier-0_Gateway_using_vCenter to open the simulation.

Perform the following steps in the simulated environment. Refer to these steps as you work through the simulation.

IMPORTANT

Do not refresh, navigate away from, or minimize the browser tab hosting the simulation. Such actions might pause the simulation and prevent it from progressing.

NOTE

If you're stuck anywhere in the simulation, press the **TAB** key to reveal the next clickable area.

Task 1: Prepare for the Lab

You log in to the vSphere Client and VCF Operations.

1. Log in to the vSphere Client with the administrator account.
   a. Open Chrome.
   b. From the Chrome bookmarks bar, click Region A > **vc-wld01-a Client**.
   c. Log in to the vSphere Client.
      - User name: administrator@wld.sso
      - Password: VMware123!VMware123!
2. Log in to NSX Manager.
   a. Open a new tab in Chrome.
   b. On the bookmarks bar, click **Region A > nsx-wld01-a** .
   c. Log in to NSX Manager.
      - Login Method: Local Account
      - User name: admin
      - Password: VMware123!VMware123!
3. Click **vSphere Client** tab.

Task 2: Create Edge Cluster and Setup Tier-0 Gateway

You create an NSX Edge cluster and setup TIer-0 Gateway.

1. In vSphere Client, click **Networking** view.
2. In the vSphere Inventory, click **vc-wld01-a.site-a.vcf.lab**.
3. Click **Configure** tab**.**
4. Click **Edge Clusters.**
5. Click **ADD CLUSTER.**
6. Configure the options on the **Edge Cluster** page and click **ADD**.

Option	Value
Edge Cluster Name	edgecl-wld01-a
Tunnel Endpoint MTU	9000 (default)

Edge Form Factor	Large

7. Click **ADD** to configure the Edge node.

Option	Action
Edge Node Name (FQDN)	edge-wld01-01a.site-a.vcf.lab
vSphere Cluster	cluster-wld01-01a
Resource Pool	Resources
Data Store	cluster-wld01-01a-vsan01
IP Allocation	static
IP Address Type	IPV4
Management IP	10.1.1.182/24
Default Gateway	10.1.1.1
Port Group	cluster-wld01-01a-vds-01-pg-mgmt
Uplinks	**Uncheck** the Use the host overlay network configuration from the selected vSphere Cluster option.
Edge Node Uplink Mapping Verify the Active and Standby PNICs mapping.	• For fp-eth0, Select vmnic0, vmnic1 • For fp-eth1, Select vmnic1, vmnic0
TEP VLAN	18
TEP IP Assignment	IP POOL

IP Pool	Click the ellipsis icon and select **Create New** to create the TEPs for the Edge nodes. • Name: Edge-TEP • From Subnets, click **Set**. • Click **ADD SUBNET** and select **IP Ranges**. • IP Ranges: **`10.1.5.134-10.1.5.150`** • CIDR: **`10.1.5.128/25`** • Gateway: **`10.1.5.129`** • Click **ADD**. • Click **APPLY**. • Click **SAVE**.

a. Click **APPLY**.

b. Select **edge-wld01-01a.site-a.vcf.lab** and click **CLONE** to deploy the second edge node.

c. In the **Edge Node Name (FQDN)** text box, enter **`edge-wld01-02a.site-a.vcf.lab`** and click **APPLY**.

Option	Value
Node Name	edge-wld01-02a.site-a.vcf.lab
IP Address Type	IPV4 Only
IP Assignment	Static
Management CIDR	10.1.1.183/24
Default Gateway	10.1.1.1
TEP IP Address Type	IPv4
IP Pool	Edge TEP

d. Scroll down, turn off the **Auto generate Passwords with VCF to manage Edge nodes** toggle, and click NEXT.

8. Scroll down to the CLI Administrator credentials section to set passwords for admin and root and click **SAVE**.

Option	Action
CLI Username	admin
New CLI Password	VMware123!VMware123!
Confirm CLI Password	VMware123!VMware123!
New System root Password	VMware123!VMware123!
Confirm System Root Password	VMware123!VMware123!

9. Verify that the Node Connectivity status is **2 Up** and Status is **Success**.
10. Click **Set** under Gateway to set Tier-0 gateway.
11. Configure the options under Workload Domain Connectivity and scroll down the page by clicking the bottom arrow of the scroll option.

Option	Action
Gateway Name	wld01-t0-01
High Availability Mode	Active Standby
Gateway Routing Type	BGP
Local Autonomous System Number (ASN)	65002

    a. Click **Set** next to edge-wld01-01a to configure the first Tier-0 gateway with two uplinks.

    b. Configure the options under the first uplink and click **NEXT**.

Option	Value
Gateway Interface VLAN	19
Gateway Interface CIDR	10.1.6.11/25

BGP Peer IP	10.1.6.1
BFD	No (default)
MTU	8000
BGP Peer ASN	65000
BGP Peer Password	VMware123!
Confirm Password	VMware123!

c. Configure the options under Second Uplink (Optional) and click **APPLY**.

Option	Value
Gateway Interface VLAN	20
Gateway Interface CIDR	10.1.6.131/25
BGP Peer IP	10.1.6.129
MTU	8000
BGP Peer ASN	65000
BGP Peer Password	VMware123!
Confirm Password	VMware123!

d. Click **Set** next to edge-wld01-02a to configure the uplinks.

e. Configure the options under First Uplink and click **NEXT**.

Option	Value

Gateway Interface VLAN	19
Gateway Interface CIDR	10.1.6.12/25
BGP Peer IP	10.1.6.1
BFD	No (default)
MTU	8000
BGP Peer ASN	65000
BGP Peer Password	VMware123!
Confirm Password	VMware123!

f. Configure the options under Second Uplink (Optional) and click **SAVE**.

Option	Value
Gateway Interface VLAN	20
Gateway Interface CIDR	10.1.6.132/25
BGP Peer IP	10.1.6.129
BFD	No (default)
MTU	8000
BGP Peer ASN	65000
BGP Peer Password	VMware123!

Confirm Password	VMware123!

Verify that edge cluster is setup with the T0 gateway.

Task 3: Verify the NSX Edge Cluster in NSX Manager

1. In Chrome browser, click the **NSX (nsx-wld01-a.site-a.vcf.lab)** tab.
2. Log in to NSX Manager.
   a. Log in to NSX Manager.
      - Login Method: Local Account
      - User name: admin
      - Password: VMware123!VMware123!
3. Navigate to **System > Fabric > Edge Clusters.**
4. Click **>** to expand **edgecl-wld01-a.**
5. Verify that the Edge node Connectivity status is **Up** and the node status is **Success**.
6. Click **Networking** tab.
7. Click **Tier-0 Gateways** to verify that the wld01-t0-01 is created and status is Success.
8. Click **admin** user drop-down menu from the top right corner and click **Log Out**.

Lab 22 Configuring Centralized Transit Gateway

Objective and Tasks

Configuring Centralized Transit Gateway:

1. Prepare for the Lab
2. Create External Connection
3. Create an IP Address Block
4. Configure VPC Connectivity Profiles
5. Configure Default Transit Gateway

Task 1: Prepare for the Lab

You log in to the vSphere Client and NSX.

1. Log in to the vSphere Client.
    a. Open Firefox.
    b. On the bookmarks bar, click **Region A > vc-wld01-01a Client**.
    c. Select **Local Account/VCF SSO** as the login method and click **LOG IN**.
        - User name: prod-admin@vcf.lab
        - Password: VMware123!VMware123!
2. Launch NSX Manager in a new tab.
    a. On the Firefox bookmarks bar, click **Region > nsx-wld01-a**.
    b. Select **VCF SSO** as the login method and click **LOG IN**.

Task 2: Verify External Connection

You Verify an External Connection in the Workload Domain vCenter **vc-wld01-a.site-a.vcf.lab**.

1. From the networking view, Select **vc-wld01-a.site-a.vcf.lab** > click **Configure** tab > Expand **Networking** > Click **External Connection.**
2. Verify the external connection.
   a. Expand **wld01-ctgw-01-external-connection.**
   b. Verify the external connections configuration details attached to Transit Gateways for external communication.

Option	Value
Name	wld01-ctgw-01-external-connection
Type	Centralized Connection
Tier-0 Gateway	wld01-t0-01
Remote Networks	Not Set
External IP Blocks	Not Set
Private-Transit Gateway IP Blocks	Off
Provider Outbound SNAT	Off
SNAT IP Blocks	Not Set

Task 3: Create and verify the IP Address Block details

You Create and Verify an IP Address Block in the Workload Domain vCenter **vc-wld01-a.site-a.vcf.lab**.

1. Click the **vSphere Client** menu bar on the left-hand side of your screen.
2. In the **Inventory** click the **Networking** tab**.**
3. Expand **vc-wld01-a.site-a.vcf.lab > dc-a > Virtual Private Clouds**.
4. Click **Configure** tab.
5. Expand **Settings**.
6. Click **IP Blocks**.
7. Click **ADD IP ADDRESS BLOCK**.

8. Enter the configuration to Add IP Address Block for Private network and click **SAVE**.

Option	Value
Name	172.16.101.0/24
Visibility	Private
CIDRs/Ranges	172.16.101.0/24
Excluded IPs	Leave it blank
Reserved for Specific Subnet	No

9. The External IP Block **SUP_EXT-1_10.100.2.0/23** is preconfigured in the lab environment.

   Expand **SUP_EXT-1_10.100.2.0/23** to verify the details.

   a. Click **VIEW STATISTICS.**

10. Verify the IP Block statistics for SUP_EXT-1_10.100.2.0/23

Option	Value
IP Block	SUP_EXT-1_10.100.2.0/23
Visibility	External
CIDRs/Ranges	10.100.2.0/23

Task 4: Configure VPC Connectivity Profiles

You configure VPC Connectivity Profiles in Workload Domain vCenter **vc-wld01-a.site-a.vcf.lab.**

1. Select **vc-wld01-a.site-a.vcf.lab > Virtual Private Clouds** from the Networking view.
2. Click **Configure** tab.
3. Expand **Settings**.
4. Click Connectivity Profiles.
5. Click on dotted line next to **Default VPC Connectivity Profile** and select **Edit.**
6. Edit the VPC Connectivity Profile and click **SAVE**.

Option	Value
Name	Default VPC Connectivity Profile
Transit Gateway	Default Transit Gateway
External IP Blocks	Verify that **SUP_EXT-1_10.100.2.0/23** is selected.
Private Transit Gateway IP Blocks	Select **172.16.101.0/24** from drop-down menu. Do not remove Day0 Private Tgw IP Block from the Private-Transit Gateway IP Blocks.
Edge Cluster	edgecl-wld01-a
N-S Services	On
Default Outbound NAT	On
External IP Block for Default Outbound NAT	Leave it blank

Task 5: Configure Default Transit Gateway

You verify the Default Transit Gateway in Workload Domain vCenter **vc-wld01-a.site-a.vcf.lab.**

1. Select **Default Transit Gateway** under **Virtual Private Clouds** from the Networking view.

   Ensure that you have selected the **vc-wld01-a.site-a.vcf.lab** in the navigation menu.
2. Click **Configure** tab.
3. Click **Properties** under Settings.
4. Click **EDIT**.
5. On Edit the Transit gateway, verify the details.

Option	Value
Name	Default Transit Gateway
Network Span	Default Span
Connection	**Centralized**

External Connection	wld01-ctgw-01-external-connection
HA Mode	Select Active Standby
Transit Gateway Edge Cluster	edgecl-wld01-a
VPC Transit Subnets	Leave the defaults

6. Click **CANCEL**.
7. Click **Yes** on the Warning message.

Lab 23 Creating Centralized Virtual Private Cloud

Objective and Tasks

Create a Virtual Private Cloud in the workload domain with the centralized gateway connection:

1. Prepare for the Lab
2. Create a Virtual Private Cloud (Centralized)
3. Create Private VPC Subnets
4. Create Private Transit Gateway Subnets in VPC
5. Create Public Subnets
6. Deploy Virtual Machines in VPC
7. Test Connectivity between the Subnets within the VPC
8. Review the VPC in NSX Manager

Task 1: Prepare for the Lab

You log in to the vSphere Client.

1. Log in to the vSphere Client with the vcfadmin account.
    a. Open Firefox.
    b. On the Firefox bookmarks bar, click **Region A> vc-wld01-a Client.**
    c. Select **VCF SSO** as the login method and click **Log In**.
        - User name: prod-admin@vcf.lab
        - Password: VMware123!VMware123!

Task 2: Create a Virtual Private Cloud (Centralized)

You create a Virtual Private Cloud with Centralized connectivity.

1. From the **vSphere Client** menu, select **vc-wld01-a.site-a.vcf.lab > click Networking view.**
2. Expand **vc-wld01-a.site-a.vcf.lab > dc-a** to see the virtual private clouds.
3. Right-click **Virtual Private Clouds** under dc-a, click **New VPC**, and configure the new VPC.

Option	Value
Name	wld01-vpc01
Description	centralized transit gateway based VPC.
Private VPC IP CIDRs	**172.16.201.0/24**
Advanced Settings	Click to expand Advanced Settings.
Connectivity Profile	Default VPC Connectivity Profile
Connectivity Policy	None

   a. Click **SAVE**.

4. Expand **Virtual Private Clouds** in the vSphere inventory to view wld01-vpc01.

   From the **Summary** of wld01-vpc01, verify VPC DETAILS, SUBNET AND WORKLOADS, CONNECTIVITY AND SERVICES, and STATISTICS.

Task 3: Create Private VPC Subnets

You create a Private VPC subnet in the Virtual Private Cloud.

1. Create a subnet called wld01-private-vpc-web-subnet.
   a. Right-click **wld01-vpc01** and click **New Subnet**.
   b. Create the subnet, configure details, and click **NEXT**.

Option	Value
Name	wld01-private-vpc-web-subnet
VLAN Extension	No

Access Mode	Private - VPC
IP Block	Any
Auto allocate Subnet CIDR from IP Blocks	Yes
Subnet Size	16
Advanced Settings	Click to expand.
VPC Gateway Connectivity	Yes
DHCP Config	DHCP Server

c. On DHCP Configuration, click **NEXT.**

d. On Ready to Complete, click **FINISH**.

Task 4: Create Private Transit Gateway Subnets

You create a Private Transit Gateway subnet in VPC.

1. Create a subnet called wld01-private-tgw-web-subnet.

   a. Right-click **wld01-vpc01** and click **New Subnet**.

   b. Create the subnet, configure details, and click **NEXT**.

Option	Value
Name	wld01-private-tgw-web-subnet
VLAN Extension	No
Access Mode	Private -Transit Gateway
IP Block	Select **172.16.101.0/24**
Auto allocate Subnet CIDR from IP Blocks	Yes
Subnet Size	16
Advanced Settings	Click to expand.
Gateway Connectivity	Yes
DHCP Config	DHCP Server

   c. On DHCP Configuration, click **NEXT.**

   d. On Ready to Complete, click **FINISH**.

Task 5: Create Public Subnets

You create a public subnet Virtual Private Cloud with Centralized connectivity.

1. Create a subnet called wld01-public-web-subnet.
   a. Right-click **wld01-vpc01** and click **New Subnet**.
   b. Create the subnet, configure details, and click **NEXT**.

Option	Value
Name	wld01-public-web-subnet
Access Mode	Public
IP Block	Any
Auto allocate Subnet CIDR from IP Blocks	Yes
Subnet Size	16 IPs (/28 network)
Advanced Settings	Click to expand.
VPC Gateway Connectivity	Yes
DHCP Config	DHCP Server

   c. On DHCP Configuration, click **NEXT**.
   d. On Ready to Complete, click **FINISH**.

Task 6: Deploy Virtual Machines in VPC

You deploy virtual machines and connect to different type of subnets in VPC.

1. In Firefox, Click to select **vc-wld01-a.site-a.vcf.lab**
2. Click the **VMs and Templates** view.
3. Deploy a virtual machine from a template.
   a. Right-click **web-wld-vm-template** and select **New VM from This Template**.
   b. On the Select a name and folder page, enter **`wld01-private-vpc-web-vm`** in the **virtual machine name** text box and click **NEXT**.
   c. On the **select a compute resource** page, click **NEXT**.

      Verify that cluster-wld01-01a is selected.
   d. On the **Select storage** page, select **vsan-wld01-01a** and click **NEXT**.

e. On the Select Clone options, select the **customize this virtual machines hardware** checkbox and click **NEXT**.

f. On the **customize hardware** page, click the **Network adapter 1** drop-down menu, click **Browse**, select **wld01-private-vpc-web-subnet** and click **OK,** and click **NEXT.**

g. On the **Ready to complete** page, click **FINISH**.

h. In the vSphere Client, monitor the **Import OVF Package** and **Deploy OVF Template** tasks from Recent Tasks.

Monitor the **Clone virtual machine** task from Recent Tasks.
The task takes 2 to 3 minutes, and verify that the tasks are **Completed** without any errors.

i. In the vSphere Inventory, right-click **wld01-private-vpc-web-vm** and click **Power > Power On.**

4. Deploy the second virtual machine from a template.

a. Right-click **web-wld-vm-template** and select **New VM from This Template**.

b. On the Select a name and folder page, enter **`wld01-private-tgw-web-vm`** in the **virtual machine name** text box and click **NEXT**.

c. On the **select a compute resource** page, click **NEXT**.

d. On the **Select storage** page, select **vsan-wld01-01a** and click **NEXT**.

e. On the Select Clone options, select the **customize this virtual machines hardware** checkbox and click **NEXT**.

f. On the **customize hardware** page, click the **Network adapter 1** drop-down menu, click **Browse**, select **wld01-private-tgw-web-subnet** and click **OK,** and click **NEXT.**

g. On the **Ready to complete** page, click **FINISH**.

h. In the vSphere Client, monitor the **Import OVF Package** and **Deploy OVF Template** tasks from Recent Tasks.

Monitor the **Clone virtual machine** task from Recent Tasks.
The task takes 2 to 3 minutes, and verify that the tasks are **Completed** without any errors.

i. In the vSphere Inventory, right-click **wld01-private-tgw-web-vm** and click **Power > Power On.**

5. Deploy the third virtual machine from a template.

a. Right-click **web-wld-vm-template** and select **New VM from This Template**.

b. On the Select a name and folder page, enter **`wld01-public-web-vm`** in the **virtual machine name** text box and click **NEXT**.

c. On the **select a compute resource** page, click **NEXT**.

d. On the **Select storage** page, select **vsan-mgmt-01a** and click **NEXT**.

e. On the Select Clone options, select the **customize this virtual machines hardware** checkbox and click **NEXT**.

f. On the **customize hardware** page, click the **Network adapter 1** drop-down menu, click **Browse**, select **wld01-public-web-subnet** and click **OK,** and click **NEXT.**

g. On the **Ready to complete** page, click **FINISH**.

h. In the vSphere Client, monitor the **Import OVF Package** and **Deploy OVF Template** tasks from Recent Tasks.

   Monitor the **Clone virtual machine** task from Recent Tasks.
   The task takes 2 to 3 minutes, and verify that the tasks are **Completed** without any errors.

i. In the vSphere Inventory, right-click **wld01-public-web-vm** and click **Power > Power On.**

Task 7: Test Connectivity between the Subnets within the VPC

1. In the vSphere Client, select **wld01-public-web-vm** and click **LAUNCH WEB CONSOLE.**
   a. Log in to wld01-public-web-vm.
      - User name: root
      - Password: VMware1!
   b. View the IP address of the machine.

      ```
      ifconfig –a
      ```

      The IP address of wld01-public-web-vm is 10.100.2.19
2. Select **wld01-private-tgw-web-vm** and click **LAUNCH WEB CONSOLE.**
   a. Log in to wld01-private-tgw-web-vm.
      - User name: root
      - Password: VMware1!
   b. View the IP address of the machine.

      ```
      ifconfig –a
      ```

      The IP address of wld01-private-tgw-web-vm is 172.16.101.3

3. In Firefox, click the **vSphere Client** tab.
4. Select **wld01-private-vpc-web-vm** and **LAUNCH WEB CONSOLE** from Summary.
   a. Log in to wld01-private-vpc-web-vm.
      - User name: root
      - Password: VMware1!
   b. View the IP address of the machine.

      ```
      ifconfig -a
      ```

      The IP address of sa-wld01-private-vpc-web-vm is 172.16.201.3.
   c. From the command line, run the `ping <wld01-public-web-vm-IP Address>` command to test the connectivity between the VMs connected to private-vpc and public subnets.

      ```
      ping -c 2 10.100.2.19
      ```

      The ping command should succeed.
   d. From the command line, run the `ping <wld01-private-tgw-web-vm-IP Address>` command to test the connectivity between VMs connected to private-vpc and private-tgw subnets.

      ```
      ping -c 2 172.16.101.3
      ```

      The ping command should succeed.
5. In Firefox, click the **sa-wld01-public-web-vm web console** tab.
   a. From the command line, run the `ping <wld01-private-vpc-web-vm-IP Address>` command to test the connectivity between VMs connected to public-VM and private-vpc subnets.

      ```
      ping -c 2 172.16.201.3
      ```

      The ping command should succeed.
   b. From the command line, run the `ping <wld01-private-tgw-web-vm-IP Address>` command to test the connectivity between the VMs connected to public-VM and private-tgw subnets.

      ```
      ping -c 2 172.16.101.3
      ```

      The ping command should succeed.
6. In Firefox, click the **sa-wld01-private-tgw-web-vm** web console tab.
   a. From the command line, run the `ping <wld01-private-vpc-web-vm-IP Address>` command to test the connectivity between the VMs connected to private-tgw and private-vpc subnets.

```
ping -c 2 172.16.201.3
```

The ping command should succeed.

b. From the command line, run the `ping <wld01-public-vm-IP Address>` command to test the connectivity between the VMs connected to private-tgw and public subnets.

```
ping -c 2 10.100.2.19
```

The ping command should succeed.

7. Close the **wld01-private-vpc-web-vm, wld01-private-tgw-web-vm**, and **wld01-public-web-vm** console tabs in Firefox.

Task 8: Review the VPC in NSX Manager

1. In Firefox, click **Region A > nsx-wld01-a** from the bookmarks.
2. Click **VCF SSO** as the login method and click **LOG IN.**
3. Log in to NSX Manager.
   - User name: prod-admin@vcf.lab
   - Password: VMware123!VMware123!
4. In the NSX UI, click the **VPCs** tab.
5. Click **Topology** to view the topology deployed in the lab.
6. Click **Virtual Private Clouds** to view the VPCs configured in NSX.
7. Expand **wld01-vpc01** and click **3** in the Subnets column to view the subnets.

Lab 24 Reviewing the Workload Domain vSAN Cluster Details

Objective and Tasks

Use the vSphere Client to review a vSAN single-site cluster:

1. Prepare for the Lab
2. Review the vSAN Cluster Details in the vSphere Client
3. Verify the vSAN Cluster Configuration Details Using ESXCLI Commands
4. Review Storage Details in VCF Operations

Task 1: Prepare for the Lab

You log in to the vSphere Client and VCF Operations.

1. Log in to the vSphere Client with the vcfadmin account.
    a. Open Firefox.
    b. On the Firefox bookmarks bar, click **Region A > vc-wld01-a Client.**
    c. Select SSO as the login method and click **Log In**.
        - User name: prod-admin@vcf.lab
        - Password: VMware123!VMware123!
2. Log in to VCF Operations.
    a. Open a new tab in Firefox.
    b. On the bookmarks bar, click **Region A > VMware Cloud Foundation Operations**.
    c. Select **VCF SSO** as the login method and click **Log In**.
        - User name: prod-admin@vcf.lab

- Password: VMware123!VMware123!

Task 2: Review the vSAN Cluster Details in the vSphere Client

You use the vSphere Client to review the vSAN cluster details and answer questions about your lab environment.

1. In Firefox, click the **vSphere** browser tab to return to the vSphere Client.
2. In the **Hosts and Clusters** inventory list on the left, expand **vc-wld01-a.site0a.vcf.lab > dc-a.**
3. Select **cluster-wld01-01a** and click **Summary tab.**
4. Perform this step only if you see alarms listed on the Summary tab.

   Under Issues and Alarms on the **Summary** tab of the cluster-wld01-01a page, select **Actions > Reset To Green** next to each alarm.

   You must repeat this step until all the vSAN alarms are gone.
5. Review the details of the workload domain cluster.

   Q1. What is the storage capacity listed for the cluster-wld01-01a-vsan01 vSAN data store?

   NOTE

   Record the capacity and usage values for future labs. The actual usage might be different in your lab environment.

   ---

6. On the **cluster-wld01-01a** page, click the **Hosts** tab.

   Q2. How many hosts are listed under the cluster-wld01-01a cluster?
7. On the **cluster-wld01-01a** page, click the **Configure** tab.
8. In the navigation menu on the left side of the **Configure** tab, select **vSAN > Disk Management**.

   Review the Disk Management details.

Task 3: Verify the vSAN Cluster Configuration Details Using ESXCLI Commands

You use Remmina Remote Desktop Client to verify the vSAN cluster configuration details using ESXCLI commands.

1. Verify the SSH service for the esx-05a.site-a.vcf.lab host.

a. On the **Disk Management** page of the cluster-wld01-01a cluster, click **esx-05a.site-a.vcf.lab**.

b. Click the **Configure** tab.

c. In the navigation menu on the left side of the **Configure** tab, select **System > Services**.

d. In the list of services, select **SSH** and verify that the Daemon is Running.

2. On the Linux taskbar, click the **Remmina** icon.

3. In **Remmina** window, Right-click **esx-05a** and click **Edit** and Update the **User Password** field with the password noted from the previous step.

To find the root user password, navigate to the **Downloads** folder and open `PageOfCredentials.json` using Firefox, then search for **esx-05a.site-a.vcf.lab**.

---

4. In the **Remmina** window, double-click **esx-05a** to open an SSH connection.

5. Run the following commands to debug and troubleshoot the vSAN cluster.

```
esxcli vsan storagepool list
esxcli vsan debug controller list
```

Q1. How many disks are in the storage pool?

Q2. What is the driver name of the controllers that are used by vSAN?

6. Run the `esxcli storage core adapter list` command to query the adapters information.

7. Run the `esxcli network vswitch dvs vmware list` command to list the distributed virtual switches.

Q3. How many distributed virtual switches are listed?

Q4. What are the names of the distributed virtual switches that are listed?

Q5. How many uplinks are connected to each distributed virtual switch?

Q6. How many configured ports are available on sa-wld01-cluster-wld01-01a-vds01?

Q7. What is the MTU size that is configured for each distributed virtual switch?

8. Run the `df -h` command to display the file system information.

    Q8. What is the name of the vSAN data store?

    Q9. What is the size of the vSAN data store?

9. Run the `vdq -qH` command to query the vSAN disks in a human-readable format.
10. Click **X** to close the **esx-05a** SSH session in Remmina Remote Desktop Client.

Task 4: Review Storage Details in VCF Operations

You examine the vSAN data store for the workload domain to compare its storage details with the values that you recorded in an earlier task.

1. In Chrome, click the **VCF Operations** browser tab.
2. In the navigation page on the left, Click **Overview > Storage**
3. Under Storage by type, scroll down to the **vSAN Clusters** sections and verify the capacity usage for **cluster-wld01-01a**.

Lab 25 Analyzing the Placement of VM Objects and Components in the vSAN Cluster

Objective and Tasks

Use the vSphere Client to analyze the placement of VM objects and components in the single-site vSAN cluster:

1. Prepare for the Lab
2. Deploy a Virtual Machine
3. Verify the VM Object and Component Placement

Task 1: Prepare for the Lab

You log in to the vSphere Client.

1. Log in to the vSphere Client with the vcfadmin account.
    a. Open **Firefox**.
    b. On the Firefox bookmarks bar, click **Region A > vc-wld01-a Client**.
    c. Select **VCF SSO** as the login method and click **Log In**.
        - User name: prod-admin@vcf.lab
        - Password: VMware123!VMware123!

Task 2: Deploy a Virtual Machine

You deploy a VM in the vSphere Client with the default vSAN storage policy.

1. Add a new virtual machine to the cluster-wld01-01a cluster.
    a. In the **Hosts and Clusters** inventory on the left, right-click **cluster-wld01-01a** and select **New Virtual Machine**.

b. On the **Select a creation type** page, verify that **Create a new virtual machine** is selected and click **NEXT**.

c. Enter **`tempvm-wld01-01a`** in the **Virtual machine name** text box and click **NEXT**.

d. On the **Select a compute resource** page, verify that **cluster-wld01-01a** is selected and click NEXT.

e. On the **Select storage** page, select **vsan-wld01-01a** and click NEXT.

f. On the **Select compatibility** page, leave the default settings and click **NEXT**.

g. On the **Select a guest OS page**, leave the default settings and click **NEXT**.

h. On the **Customize hardware** page, expand **New Hard disk** *.

i. From the **Sector Size** drop-down menu, verify that **512n** is selected and click **NEXT**.

j. On the **Assign Tag** page, leave the default settings and click **NEXT**.

k. On the **Ready to complete** page, review the details and click **FINISH**.

2. In the **Recent Tasks** pane, monitor the `Create virtual machine` task until it is completed successfully.
3. Power on the new VM.
   a. In the **Hosts and Clusters** inventory on the left, expand **cluster-wld01-01a**.
   b. Right-click **tempvm-wld01-01a** and select **Power > Power On**.

Task 3: Verify the VM Object and Component Placement

You use the vSphere Client to verify the placement of the VM objects and components.

1. In the **Hosts and Clusters** inventory on the left, select **tempvm-wld01-01a**.
2. On the **Summary** tab, review the details under Storage Policies.

   Q1. Which VM storage policy is applied to the VM?

   Q2. What is the VM Storage Policy Compliance status?
3. Click the **Monitor** tab.
4. In the navigation menu on the left, select **vSAN > Physical disk placement**.
5. Review the vSAN component placement created for the VM.

   Q3. How many virtual objects Components are created?

   Q4. How many components are created for each object?

Lab 26 Running vSAN Tests

Objective and Tasks

Run the proactive tests and data migration tests to verify the vSAN performance and health:

1. Prepare for the Lab
2. Run vSAN Proactive Tests
3. Run a Data Migration Precheck

Task 1: Prepare for the Lab

You log in to the vSphere Client.

1. Log in to the vSphere Client with the vcfadmin account.
    a. Open Firefox.
    b. On the Firefox bookmarks bar, click **Region A > vc-wld01-a Client**.
    c. Select **VCF SSO** as the login method and click **Log In**.
        - User name: prod-admin@vcf.lab
        - Password: VMware123!VMware123!

Task 2: Run vSAN Proactive Tests

You run the vSAN proactive tests to verify that the vSAN cluster components are working as expected. Run these tests outside business hours because they can often be network intensive.

1. In Firefox, click the **vc-wld01-a.site-a.vcf.lab** vSphere Client.
2. In vSphere Inventory, select **cluster-wld01-01a** and click the **Monitor** tab.
3. Click **vSAN > Proactive Tests**.
4. Expand **VM Creation Test**.

5. Click **RUN TEST**.
6. On the **Run VM Creation Test** dialog box, click **RUN**.

   This test creates a simple tiny VM on every host and deletes the VM on successful testing.

   The VM Creation test status is Success on all the four ESX hosts (esx-05a, esx-06a, esx-07a, and esx-08a).

   The success status concludes that the vSAN is operational.
7. Expand **Network Performance Test**.

   The network performance test is designed to assess if a connectivity issue exists and if the network bandwidth between the hosts can satisfy the requirements of vSAN.
8. Click **RUN TEST**.
9. Select the **Enable network diagnostic mode** checkbox and click **RUN**.

   a. Verify that the network performance test status is passed.

Task 3: Run a Data Migration Precheck

1. In Firefox, click the **vc-wld01-a.site-a.vcf.lab** vSphere Client.
2. In vSphere Inventory, select **cluster-wld01-01a** and click the **Monitor** tab.
3. Under **vSAN > Data Migration Pre-check**.
4. From the **Pre-check data migration for** drop-down menu, select **esx-05a.site-a.vcf.lab**.
5. Verify that vSAN data migration has given **Ensure accessibility** and click **PRE-CHECK.**

   From the Latest test result, verify that the host can enter maintenance mode.

   You can ignore the other `Non-Compliant` messages that appear on the **Object State** and **Predicted Health** tabs.

Lab 27 Creating and Applying a vSAN Storage Policy

Objective and Tasks

Create a VM Storage policy and apply it to the virtual machine:

1. Prepare for the Lab
2. Verify the VM Storage Policy Applied to a vSAN Data Store
3. Create a VM Storage Policy
4. Create a Virtual Machine with a Custom VM Storage Policy
5. Check Compliance with the VM Storage Policy

Task 1: Prepare for the Lab

You log in to the vSphere Client.

1. Log in to the vSphere Client with the prod-admin account.
    a. Open **Firefox**.
    b. On the Firefox bookmarks bar, click **Region A** > **vc-wld01-a Client**.
    c. Select **VCF SSO** as the login method and click **LOG IN**.
        - User name: prod-admin@vcf.lab
        - Password: VMware123!VMware123!

Task 2: Verify the VM Storage Policy Applied to a vSAN Data Store

You create a VM Storage policy and application to the data store.

1. In Firefox, click the **vc-wld01-a.site-a.vcf.lab** vSphere Client.

2. From the **vSphere** menu, select **Policies and Profiles.**
3. Select **VM Storage Policies**.
4. Select the **cluster-wld01-01a - Optimal Datastore Default Policy - AutoRAID** checkbox to verify the details of the vSAN storage policy applied to the cluster-wld01-01a cluster.
5. Click the **Rules** tab to verify the rules for the vSAN data store.

Option	Value
Name	cluster-wld01-01a - Optimal Datastore Default Policy - AutoRAID
Description	vSAN ESA Default Storage - AutoRAID
Storage Type	vSAN
Failures to tolerate	vSAN managed (based on cluster configuration)
IOPS limit for Object	0
Object space reservation	Thin provisioning

6. Click the **VM Compliance** tab to verify that the virtual machines running on the data store are compliant with the VM storage policy defined on the data store.
7. Click the **VM Template** tab to check the VM templates stored on the vSAN data store.
8. Click **Storage Compatibility** to view the data store compatibility against the VM Storage policy.

Task 3: Create a VM Storage Policy

1. Click **CREATE** from VM Storage Policies.
2. Verify that **vc-wld01-a.site-a.vcf.lab** is selected.
3. On the **Name and description** wizard, enter the options and click **NEXT**.

Option	Value
Name	WLD01-VM-Storage-Policy
Description	VM Storage policy for workload VMs

4. On the **policy structure** wizard, select the **Enable rules for "vSAN" storage** checkbox and click **NEXT**.

   Leave the default values (cleared) for all the other options.

5. On the **vSAN** wizard, Click the Toggle to **turn off** vSAN ESA Auto-RAID (9.1 Onwards).

Option	Value
Site disaster tolerance	none-standard cluster
Failure to tolerate	1 failure - RAID-1 (mirroring)

   a. Click the **Storage rules** tab to verify the storage rules.

Option	Value
Encryption services	No preference
Data reduction	No preference
Storage tier	No preference

   b. Click the **Advanced Policy Rules** tab and verify the options.

Option	Value
Number of disk stripes per object	1
IOPS limit for object	0
Object space reservation	Thin provisioning
Flash read cache reservation	0
Disable object checksum	turned off
Force provisioning	turned off

   c. Click **NEXT**.

6. On Storage compatibility, check the listed data store compatible with the policy and click **NEXT**.

7. Click **FINISH** on the review page.

Task 4: Create a Virtual Machine with a Custom VM Storage Policy

1. From the **vSphere Client** menu, select the **Inventory** view.
2. Select and right-click **cluster-wld01-01a** and click **New Virtual Machine**.
3. Select **Create a new virtual machine** and click **NEXT**.
4. In the **virtual machine name** text box, enter **`wld01-vm01`** and click **NEXT**.
5. Click **NEXT** on the **select a compute resource** wizard.
6. On the **Select storage** wizard, select the options and click **NEXT**
   a. From the **VM Storage Policy** drop-down menu, select **WLD01-VM-Storage-Policy.**
   b. Select the **vsan-wld01-01a** data store and click **NEXT**.
7. Click **NEXT** on the **select compatibility** wizard.
8. Click **NEXT** on the **select a guest OS** wizard.
9. Click **NEXT** on the **customize hardware** wizard.
10. Click **NEXT** on the **Assign Tag** wizard.
11. Click **FINISH** on the **Ready to complete** wizard.

Task 5: Check Compliance with the VM Storage Policy

1. In vSphere Client, select **wld01-vm01** and right-click **VM Policies > Check VM Storage Policy Compliance.**
2. On the **Summary** tab, verify that the virtual machine appears Compliant under Storage Policies.
3. On the **Summary** tab, verify the details listed under related objects.

Option	Value
Cluster	cluster-wld01-01a
Host	esx-05.site-a.vcf.lab
Networks	edge-wld01-01a-External-1
Storage	vsan-wld-01a

NOTE

The ESX host might be different in your lab.

Lab 28 Implementing Certificates

Objective and Tasks

Use an OpenSSL certificate authority to rotate the certificate on the SDDC Manager server:

1. Prepare for the Lab
2. Configure the Certificate Authority in SDDC Manager
3. Rotate the Certificate for SDDC Manager
4. Manage Certificates from VCF Operations

Task 1: Prepare for the Lab

You log in to the vSphere Client, VCF Operations, and SDDC Manager.

1. Log in to the vSphere Client with the vcfadmin account.
    a. Open Firefox.
    b. On the Firefox bookmarks bar, click **Region A > vc-mgmt-a Client**.
    c. Select **VCF SSO** as the login method and click **Log In**.
        - User name: prod-admin@vcf.lab
        - Password: VMware123!VMware123!
2. Launch SDDC Manager in a new tab.
    a. On the Firefox bookmarks bar, right-click **SDDC Manager** from **Region A** folder and select **Open in new tab**.
    b. Click the **SDDC Manager** browser tab.
        - User name: administrator@vsphere.local
        - Password: VMware123!VMware123!

3. Log in to VCF Operations.
    a. Open a new tab in Firefox.
    b. On the bookmarks bar, click **Region A > VMware Cloud Foundation Operations**.
    c. Select VCF SSO as the login method to log in to VCF Operations.
        - User name: prod-admin@vcf.lab
        - Password: VMware123!VMware123!

Task 2: Configure the Certificate Authority in SDDC Manager

You configure SDDC Manager to connect to the internal OpenSSL CA.

1. In the navigation pane on the left, select **Security > Certificate Authority**.
2. On the **Configure Certificate Authority** page, click **EDIT**.
3. From the **Certificate Authority Type** drop-down menu, select **OpenSSL**.
4. Configure the OpenSSL CA and click **SAVE**.

Option	Action
Common Name	Enter **`sddcmanager-a.site-a.vcf.lab`** in the text box.
Organizational Unit	Enter **`VCF`** in the text box.
Organization	Enter **`Learning`** in the text box.
Locality	Enter **`Palo Alto`** in the text box.
State	Enter **`CA`** in the text box.

Country	Select **US - United States** from the drop-down menu.

The `CA Configuration saved successfully` message appears at the top of the page.

Task 3: Rotate the Certificate for SDDC Manager

You use SDDC Manager to deploy CA-signed certificates to VMware Cloud Foundation components.

1. In the navigation pane on the left, select **Inventory > Workload Domains**.
2. In the Domain column of the workload domains table, click **mgmt-a**.
3. Click the **Certificates** tab.
4. Generate a new CSR for SDDC Manager.
   a. From the list of resource types, select the **SDDC MANAGER** checkbox and click **GENERATE CSRS**.
   b. Configure the options on the **Details** page and click **NEXT**.

Option	Action
Algorithm	Select **RSA** from the drop-down menu.
Key Size	Select **3072** from the drop-down menu.
Email	Leave blank.
Organizational Unit	Enter **`VCF`** in the text box.
Organization Name	Enter **`Learning`** in the text box.
Locality	Enter **`Palo Alto`** in the text box.
State	Enter **`CA`** in the text box.
Country	Select **US - United States** from the drop-down menu.

   c. On the **Subject Alternative Name** window, leave the default setting and click **NEXT**.
   d. On the **Summary** page, review the configuration and click **GENERATE CSRS**.

e. Wait for the `CSR Generation is successful` message to appear before continuing.



You might need to refresh the browser periodically for the message to appear.

5. Generate the signed certificate from the CA.
   a. Select the **SDDC MANAGER** checkbox and click **GENERATE SIGNED CERTIFICATES**.
   b. In the **Generate Certificates** dialog box, select **OpenSSL** from the **Select Certificate Authority** drop-down menu.
   c. Click **GENERATE CERTIFICATES**.
   d. Wait until the `Certificate Generation is successful` message appears before continuing.
6. Verify that `Certificate Generation – SUCCESSFUL` appears in the Certificate Operation Status column.
7. Install the certificate on SDDC Manager.
   a. Select the **SDDC MANAGER** checkbox and click **INSTALL CERTIFICATES**.
   b. Refresh the browser every 2 minutes.

      The Failed to load domain details HTTP failure response… error appears because of the certificate change.
   c. Click X to close SDDC Manager tab in Firefox.
   d. In Firefox, open the Application menu and select New Private Window.
   e. From the bookmarks tool bar, click **Region A > SDDC Manager.**
   f. On the Be careful, Something doesnt look right page, click **Advanced > Proceed to sddcmanager-a.site-a.vcf.lab (Risky)** to continue.
   g. After the browser refreshes, verify that `Certificate Installation – SUCCESSFUL` appears in the Certificate Operation Status column.
   h. Log in to SDDC Manager:
      - Username: administrator@vsphere.local
      - Password: VMware123!VMware123!
   i. From the Task bar, verify that the **Certificate Operation: Replace Certificate** task status is Successful

8. (Optional) If the `Certificate Installation – SUCCESSFUL` status does not appear in the browser, reboot SDDC Manager.
    a. On the Linux taskbar, click the **Remmina** icon.
    b. Double-click **SDDCManager** to open an SSH connection to SDDC Manager.
    c. Run the `su` command to switch to the root user account.
    d. Enter **`VMware123!VMware123!`** at the password prompt.
    e. Run the `reboot now` command to reboot SDDC Manager.

       Remmina Remote Desktop Client automatically ends the SSH session and closes the window.
    f. Return to Firefox and wait for the reboot to complete.

       The process takes 5 to 10 minutes to finish. You must refresh the page every minute until SDDC Manager reappears.
9. Verify that SDDC Manager has the new certificate installed.
    a. In the navigation pane on the left, select **Inventory > Workload Domains**.
    b. In the Domains column of the workload domains table, click **Mgmt-a**.
    c. Click the **Certificates** tab.
    d. In the SDDC MANAGER row, verify that the certificate issuer is sddcmanager-a.site-a.vcf.lab and the certificate issue date is today.

Task 4: Manage Certificates from VCF Operations

1. In Firefox, click the **ops-a.site-a.vcf.lab** tab.
2. Click the **Manage** tab.
3. In the navigation pane, click **Fleet Management > Certificates.**
4. Under VCF Management, select **VCF Automation (**where **Component IP/FQDN** is **auto-a.site-a.vcf.lab** and **Category** is **TLS Certificate)** and click **RENEW CERTIFICATE.**
5. On the **Renew Certificate** dialog box, select the checkbox **I understand that I must wait for outgoing operations to complete before starting a new rotation** and click **PROCEED**.
6. From the top-right corner of the **VCF Operations** page, turn on the **Activate Auto-renewal** toggle.

NOTE

After enabling **Activate Auto-renewal**, the autorenewal status for VCF Automation and VCF Operations appears Activated after approximately 5 minutes.

7. Click **CONFIRM** in the **Activate Auto-renewal** dialog box.

Lab 29 Configuring Backup

Objective and Tasks

Configure the backup for SDDC Manager, NSX Manager, and VCF Management components:

1. Prepare for the Lab
2. Configure Backups for VCF Instances
3. Configure SFTP Settings for VCF Fleet Management
4. Configure Backups for VCF Management

Task 1: Prepare for the Lab

You log in to the vSphere Client, VCF Operations, and SDDC Manager.

1. Log in to VCF Operations.
    a. Open Firefox.
    b. On the bookmarks bar, click **Region A > VMware Cloud Foundation Operations**.
    c. Select **VCF SSO** as Login Method and Log in to VCF Operations.
        - User name: prod-admin@vcf.lab
        - Password: VMware123!VMware123!

Task 2: Configure Backups for VCF Instances

You use the VMware Cloud Foundation Operations UI to configure backup for VCF instances, which include SDDC Manager and NSX Manager backups.

1. In Firefox, click the **VMware Cloud Foundation Operations** tab.
2. Register a backup server for SDDC Manager and NSX Manager backups.

a. In the VMware Cloud Foundation Operations UI, click **Operate** tab > **Administration > SDDC Manager**.

b. Click the **Backup Settings** tab.

c. Click **Site Settings**.

d. On Backup Settings, configure the backup server for SDDC Manager and NSX Manager.

Option	Action
HOST FQDN or IP	Enter **10.1.10.205** in the text box.
Port	Enter **22** in the text box.
Transfer Protocol	Select **SFTP** from the drop-down menu.
Username	Enter **nsxbackup** in the text box.
Password	Enter **VMware123!** in the text box.
Backup Directory	Enter /**backups** in the text box.
SSH Fingerprint	The text box is filled automatically.
Confirm Fingerprint	Select the checkbox**.**
Encryption Passphrase	Enter **VMware123!VMware123!** in the text box.

e. Click **SAVE**.

f. In the **Confirm your changes to backup settings** dialog box, click **CONFIRM**.

The following message appears after configuring an external SFTP server:
`When you configure an external SFTP server, NSX Manager backups are automatically scheduled at regular intervals. Only IPV4 is supported for this purpose.`

3. Configure automatic backups for SDDC Manager and NSX Manager.

a. Click **SDDC Manager Configurations** under Backup Settings.

b. Click **EDIT** near **Backup Schedule**.

You have to wait for 1 to 2 minutes to be able to click **Edit** while the system is working in the background

c. Configure the options under Backup Schedule.

Option	Action
Automatic Backup	Turn on the toggle to enable.
Backup Frequency	Select **Weekly** from the drop-down menu.
Days of the Week	Select **Sun**.
Schedule Time	Enter **12 : 30** in the text boxes and select **AM** from the **AM/PM** drop-down menu.
Take Backup on State Change	Turn on the toggle to enable.
Retain Last Backup	Enter **7** in the text box.
Retain Hourly Backup for Days	Enter **1** in the text box.
Retain Daily Backup for Days	Enter **7** in the text box.

   d. Click **SAVE**.

      Wait for a minute for VCF Operations to save the Backup Schedule settings for SDDC Manager.

4. Under Last Backup Status, click **BACKUP NOW**.
5. Monitor **Status** under Last Backup Status and verify that the Last Backup Status is successful.

Task 3: Configure SFTP Settings for VCF Fleet Management

You use the VMware Cloud Foundation Operations UI to configure SFTP settings for VCF components backup.

1. In Firefox, click the **VMware Cloud Foundation Operations** tab.
2. Click the **Build** tab.
3. Click **Lifecycle > VCF Management**
   a. Click **Backup & Restore** tab.
   b. Click **ADD BACKUP LOCATION**.
   c. On Add Backup Location, configure the SFTP server for VCF Management components.

Option	Action
VCF Instance	instance-a
SFTP host FQDN or IP address	Enter **10.1.10.205** in the text box.
Port	Enter **22** in the text box.
Username	Enter **nsxbackup** in the text box.
Password	**VMware123!**
Backup Directory	Enter **/backups** in the text box.
Encryption Passphrase	**VMware123!VMware123!** • Click **ADD**
SSH Fingerprint	Select **Confirm fingerprint** checkbox.

d. Click **ADD**.

e. You navigate to **Tasks > Configure backup location for vcf instance: instance-a** to monitor the status.

This task should take about 15 to 20 minutes to complete.

Wait for the task to complete before you move to the next task.

Task 4: Create Backup Schedule for VCF Management

You use the VMware Cloud Foundation Operations UI to back up settings for each of the VCF Management components.

1. Click **Lifecycle** > **VCF Management Backup & Restore**.
2. Click **CREATE SCHEDULE.**
3. Configure VCF Management Services backup schedule.
   a. Click **VCF Instance** drop down menu to select instance-a
   b. Select **Frequency** drop down menu to select **Daily**.
   c. Select Start time as **11:00 PM** from the drop-down menu.
   d. Toggle to turn ON **Incremental backup**.
   e. Select **Retention** as **Retain all backups.**
   f. Click **NEXT**.

g. Click **FINISH**.

h. Click **Tasks** tab to monitor the **Configure backup schedule for VCF Instance: instance-a** status.

   Wait for the task to complete before moving to the next step.

   This task should take about 15 to 20 minutes to complete.

4. Click **Backup & Restore** tab.
5. Click **BACKUP NOW** to take the backup.
6. Select **instance-a f**rom the VCF Instance drop-down menu.
7. Click **Management Component** checkbox to select all the components**.**
8. Click **BACK UP NOW.**
9. Click **Tasks** tab to verify that **Back up components of VCF instance: instance-a** is completed.

   Wait for the task to complete before moving to the next step.

   This task should take about 30 to 35 minutes to complete.

Lab 30 (Optional) Using APIs

Objective and Tasks

Use the VMware Cloud Foundation APIs to perform administrative tasks:

1. Prepare for the Lab
2. Use the API to Determine vSAN Storage Usage
3. Use the API to Obtain NSX Cluster Details

Task 1: Prepare for the Lab

You log in to the vSphere Client, VCF Operations, and SDDC Manager.

1. Log in to the vSphere Client with the vcfadmin account.
    a. Open Firefox.
    b. On the Firefox bookmarks bar, click **Region A > vc-mgmt-a Client**.
    c. Select **VCF SSO** as the login method and click **Log In**.
        - User name: prod-admin@vcf.lab
        - Password: VMware123!VMware123!
2. Log in to VCF Operations.
    a. Open a new tab in Firefox.
    b. On the bookmarks bar, click **VMware Cloud Foundation Operations**.
    c. Select **VCF SSO** as the login method and click **Log In**.
        - User name: prod-admin@vcf.lab
        - Password: VMware123!VMware123!
3. Launch SDDC Manager in a new tab.

a. On the Firefox bookmarks bar, right-click **SDDC Manager** from **Region A** folder and select **Open in new tab**.

b. Click the **SDDC Manager** browser tab.

- User name: admin
- Password: VMware123!VMware123!

Task 2: Use the API to Determine vSAN Storage Usage

You use the API to determine the storage space consumption in the management domain.

1. In Firefox, click the VCF Operations tab.
2. Click the Build tab.
3. In the navigation pane on the left, select **Developer Center > APIs & SDKs**.
4. In the right pane, click **API Explorer** from the SDDC Manager API tile.
5. Under API Categories, expand **Domains > GET /v1/domains**.
6. Scroll down to the **Try it out** section and click **EXECUTE**.
7. Record the values for the total and used storage space for the management domain.
   a. Under Response, expand **PageOfDomain > Domain (mgmt-a) > Capacity > Storageinfo**.
   b. Under `"total"`, expand **DataMetric** and record the total storage capacity in Terabytes. _____
   c. Under `"used"`, expand **DataMetric** and record the used storage capacity in Terabytes. _____
8. Verify that the storage values retrieved with the API match the values that appear on the vSAN cluster in VCF Operations.
   a. Click **Operate** tab.
   b. In the navigation pane on the left, click **Overview > Storage**.
   c. Scroll down to the the vSAN Clusters and verify **cluster-mgmt-01a** Storage information.
   d. Verify that the total and used storage values that you recorded match the values that appear under Capacity Summary for the vSAN cluster (cluster-mgmt-01a).

Task 3: Use the API to Obtain NSX Cluster Details

You gather information about NSX Edge nodes using the API.

1. Click the **Build** tab.
2. In the navigation pane on the left, select **Developer Center > APIs & SDKs**.
3. In the right pane, click **API Explorer** from the SDDC Manager API tile.
4. Under API Categories, expand **NSX-T Clusters > GET /v1/nsxt-clusters**.
5. Scroll down to the **Try it out** section and click **EXECUTE**.
6. Expand **PageOfNsxTCluster**.

   If the API does not return a result after 3 minutes, refresh the browser and rerun the API.
7. Obtain the details about the first NSX cluster.
   a. Under `"elements"`, expand the **NsxTCluster** instance.
   b. Under `"domains"`, expand **DomainReference (mgmt-a)** and verify that the NSX cluster is associated with the mgmt-a management domain.
   c. Under `"nodes"`, expand **NsxTManager (nsxt-mgmt-01a)**.
   d. Verify that the FQDN of the NSX Manager node is nsx-mgmt-01a.site-a.vcf.lab.
   e. Find the `"version"` line and review the version of the NSX Manager appliance associated with the cluster.

      Q1. What is the NSX Manager version associated with the cluster?
   f. Find the `"vipFqdn"` line and review the FQDN for the NSX cluster.

      Q2. What is the VIP FQDN for the NSX cluster that is associated with the EDU management domain?
8. Obtain the details about the second NSX cluster.
   a. Under `"elements"`, expand the second **NsxTCluster** instance and verify that the **DomainReference (wld01-a)** is listed. If not, click on the third **NSXTCluster**.
   b. Under `"domains"`, verify that **DomainReference (wld01-a)** and **NSXTManager (nsx-wld01-01a)** are listed.
   c. Under `"nodes"`, expand **NsxTManager (nsx-wld01-a)**.
   d. Verify that the FQDN of the NSX Manager node is **nsx-wld01-01a.site-a.vcf.lab.**
   e. Find the `"version"` line and review the version of the NSX Manager appliance associated with the cluster.

      Q3. What is the NSX Manager version associated with the cluster?
   f. Find the `"vip"Fqdn` line and review the FQDN for the NSX cluster.

      Q4. What is the VIP address for the NSX cluster that is associated with the sa-wld01 domain?

Lab 31 (Optional) Using SoS Commands

Objective and Tasks

Use SoS commands to verify the cluster health and generate a log bundle:

1. Use SoS Commands to Verify Cluster Health

Task 1: Use SoS Commands to Verify Cluster Health

You use commands from the Supportability and Serviceability (SoS) utility to verify cluster health and generate a log bundle.

1. On the Linux taskbar, click the **Remmina** icon.
2. Double-click **SDDCMANAGER**.
3. Run the `su -` command to switch to the root user account.
4. Enter **`VMware123!VMware123!`** at the password prompt.
5. Run the `/opt/vmware/sddc-support/sos -h` command to see the help options.

   Q1. Which health checks are skipped when you run the `--health-check` option?

   Q2. Which SoS option do you run to verify the validity and expiration of certificates?

6. Run the `/opt/vmware/sddc-support/sos --password-health` command to verify the health of the passwords.

   Q3. What is the vCenter management domain password expiration set to?

7. Run the `/opt/vmware/sddc-support/sos --services-health` command to verify the health of the SDDC Manager services.

   Q4. Which services on SDDC Manager does this health check examine?

8. Generate a log bundle with ESX host logs for the workload domain and save it to the `/tmp` directory.

   ```
   /opt/vmware/sddc-support/sos --esx-logs --domain-name wld01-a --log-dir /tmp
   ```

   The ESX log bundle collection takes about 50 to 60 minutes to complete.

9. Type exit to logout as root.

   ```
   exit
   ```

10. Type exit to close the SSH session.

    ```
    exit
    ```

Answer Key

Lab 8 Assigning Permissions to VCF SSO Users

Q1. Can you access SDDC Manager with the dev-user account?

A1. No. The dev-user account is not authorized to access SDDC Manager, and the Invalid credential message appears.

Lab 24 Reviewing the Workload Domain vSAN Cluster Details

Q1. What is the storage capacity listed for the cluster-wld01-01a-vsan01 vSAN data store?

A1. 1.89 TB.

Q2. How many hosts are listed under the cluster-wld01-01a cluster?

A2. Four.

Q1. How many disks are in the storage pool?

A1. 3

Q2. What is the driver name of the controllers that are used by vSAN?

A2. VMware NVMe SSD Controller.

Q3. How many distributed virtual switches are listed?

A3. Two.

Q4. What are the names of the distributed virtual switches that are listed?

A4. vds01-wld01-01a and vds02-wld01-01a

Q5. How many uplinks are connected to each distributed virtual switch?

A5. Two.

Q6. How many configured ports are available on sa-wld01-cluster-wld01-01a-vds01?

A6. 64.

Q7. What is the MTU size that is configured for each distributed virtual switch?

A7. 8000.

Q8. What is the name of the vSAN data store?

A8. vsan-wld01-01a

Q9. What is the size of the vSAN data store?

A9. 3.5 TB

Lab 25 Analyzing the Placement of VM Objects and Components in the vSAN Cluster

Q1. Which VM storage policy is applied to the VM?

A1. cluster-wld01-01a -Optimal Datastore Default Policy - AutoRAID

Q2. What is the VM Storage Policy Compliance status?

A2. Compliant.

Q3. How many virtual objects Components are created?

A3. 24 vSAN components

Q4. How many components are created for each object?

A4. 8

Lab 30 (Optional) Using APIs

Q1. What is the NSX Manager version associated with the cluster?

A1. 9.1.0.0.xxxxxxxx

Q2. What is the VIP FQDN for the NSX cluster that is associated with the EDU management domain?

A2. nsx-mgmt-a.site-a.vcf.lab

Q3. What is the NSX Manager version associated with the cluster?

A3. 9.1.0.0.xxxxxxxx

Q4. What is the VIP address for the NSX cluster that is associated with the sa-wld01 domain?

A4. nsx-wld01-a.site-a.vcf.lab

Lab 31 (Optional) Using SoS Commands

Q1. Which health checks are skipped when you run the `--health-check` option?

A1. None. The basic health check runs all available checks.

Q2. Which SoS option do you run to verify the validity and expiration of certificates?

A2. The `--certificate-health` option.

Q3. What is the vCenter management domain password expiration set to?

A3. Sep 07, 2053.

Q4. Which services on SDDC Manager does this health check examine?

A4. NTP, SOSREST, POSTGRES, COMMON_SERVICES, SDDC_MANAGER_UI, DOMAIN_MANAGER, OPERATIONS_MANAGER, VIP-MANAGER-i18n, LCM, LCM_DB, APPLIANCE_DISK_USAGE.